

**SMS I: AI-ASSISTED DIAGNOSIS AND QR CODE
INTEGRATION FOR STREAMLINED CLINIC
MANAGEMENT SYSTEM AT BESTLINK
COLLEGE OF THE PHILIPPINES**

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In Partial Fulfilment
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Bachelor of Science in Information Technology

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DEDICATION

We would like to express our sincere and profound gratitude to all individuals who have significantly contributed to our journey. This work is dedicated to our families, whose unwavering support and understanding have provided a nurturing environment for our growth. We also extend our appreciation to our relatives and friends, whose encouragement and enthusiasm have inspired us to persevere through challenges.

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ABSTRACT

Title SMS I: AI-Assisted Diagnosis and QR Code
Integration for a Streamlined Clinic Management
System at Bestlink College Of the Philippines

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Introduction:

The traditional way of keeping patient records and generating reports does not improve the processes involved in education, which leads to inefficiencies, delays, and poor patient care. This capstone project entitled " SMS I: An AI-Assisted Diagnosis and QR Code Integration Streamlined Clinic Management System for Bestlink College of the Philippines," is designed to provide accurate, efficient, and highly accurate healthcare service by integrating AI-assisted diagnosis and QR code technology.

Method:

The project adopts an Agile Scrum methodology, allowing for flexibility and incremental progress. The system was created with three principal aims: (1) to establish an all-inclusive /go-to clinic management system, (2) to enhance diagnostic accuracy with AI, and (3) to use QR codes to make patient data efficient. Positions like project manager, developers, and UI/UX designers were well-defined during the whole development process.

Results:

The system combines AI technology, which performs detailed analysis of patients' data that supports clinicians, with its QR code technology to help retrieve information that is also easy to point to, thus preventing mistakes. The usability and functionality tests confirm the system is doing what it was designed to do, which ultimately leads to improved operational efficiencies and better patient care at Bestlink College of the Philippines.

Discussion:

The system solves the problem of outdated healthcare management processes by using modern technologies. AI enhances diagnosis accuracy, and QR codes enable real-time and secure access to patients' data. This system was implemented successfully at a local hospital, showcasing the power of digital transformation in the healthcare scene and perhaps initiating more innovations in educational facilities in times to come.

Keyword:

AI-assisted diagnosis, QR code integration, clinic management system, healthcare technology, Agile Scrum, patient data efficiency.

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Chapter 1

Introduction

In the current time, people cannot imagine their lives without technology. The technology sector has changed and developed many products. A computer is an example of technology that helps people in many ways. People use computers to finish office work, for business, and for entertainment. Computers help us make manual processes easier and more efficient.

Most of us already take an active interest in how our environment is changing. As we continue our research, we ask the Head Physician, Dr. Jane Ann F. Fernandez MD, who provided valuable insights into how modern technology is influencing healthcare delivery. Our research focuses on how staff members at BESTLINK COLLEGE OF THE PHILIPPINES CLINIC can provide individuals with effective care in a reasonable amount of time.

AI-assisted diagnosis is emerging as a powerful tool in healthcare, with the potential to revolutionize how medical professionals diagnose and treat patients. This technology leverages machine learning algorithms to analyze vast amounts of medical data, including patient records, medical

images, and genetic information, to identify patterns and predict potential diagnoses.

By incorporating AI-assisted diagnosis into our clinic management system, we aim to enhance the accuracy and efficiency of patient care at BCP.

A key feature of our system is the integration of QR codes. Nurses and Dr. Jane Ann F. Fernandez can obtain their details instantly by scanning a QR code, which minimizes errors and eliminates the need for humans to provide information. This feature guarantees that the appropriate data is constantly available to the clinic personnel, helping them manage patient information more effectively. By using QR codes, the process is made simple and easy.

1.1 Background of the Capstone Project

In today's rapidly evolving healthcare landscape, the demand for accurate and timely information has never been more critical. We developed this system to make the BCP-CLINIC acquire the right information and make their reports easily, to consume time works, and easily deliver monthly reports and records. The BCP Clinic plays a crucial role in safeguarding the health and well-being of students, faculty, and staff. However, the traditional methods of record-keeping and report generation often lead to inefficiencies, inaccuracies, and delays in decision-making.

To address these challenges, we have developed an advanced clinic management system that integrates QR code technology. This system is designed to streamline the workflow within the BCP Clinic by enabling quick and easy access to patient records and other essential information. By simply scanning a QR code, clinic staff can instantly retrieve accurate data, reducing the time spent on administrative tasks and minimizing the risk of errors.

AI-assisted diagnosis can play a significant role in supporting this system by providing valuable insights and potential diagnoses based on patient data. This can help clinicians make more informed decisions and improve the overall quality of care.

In addition, the use of QR code enhances the security of patient information by limiting access to authorized personnel only. This ensures that sensitive health data is protected while still being easily accessible to those who need it. This method can also help health services researchers evaluate health outcomes for people and populations, as well as effectively examine care access and quality.

The purpose of this project is to greatly increase the effectiveness and efficiency of healthcare service delivery at BCP by using

modern technology, including QR codes, in clinic operations. As a result, the clinic is more flexible and capable of meeting the demands of the local community, which enhances the quality of life and the efficiency of the school environment.

1.2 Context and Scope Context

The Bestlink College of the Philippines (BCP) Clinic serves as a vital health service provider within the institution, catering to the needs of students, faculty, and staff. The clinic is faced with issues in maintaining patient records, providing timely reports, and guaranteeing the authenticity of health-related data as the demand for effective healthcare services continues increasing. These challenges are compounded by the need for reliable, secure, and easily accessible technology that can support clinic operations while securing private patient information.

AI-assisted diagnosis can be integrated into the clinic management system to address some of these challenges by providing a more efficient and accurate way to analyze patient data and generate potential diagnoses. This can help reduce the burden on clinicians and improve the overall quality of care.

In response to these challenges, the capstone project focuses on the development and implementation of an advanced clinic management

system that integrates QR code technology. The system is intended to improve data administration, streamline clinic operations, and improve the quality of BCP's medical services overall.

Scope:

The scope of this project encompasses the following key areas:

1. Development of a QR Code-Integrated System: Design and implementation of a system that uses QR codes for easy access to patient records and other essential data. Integration of QR code scanning functionality to facilitate quick retrieval of information, reducing administrative burden and improving accuracy.
2. Efficient Data Management: Creation of a centralized database that securely stores patient information, medical records, and other clinic-related data. Implementation of user access controls to ensure that only authorized personnel can access sensitive information.
3. Streamlined Reporting Process: Development of features that automate the generation of monthly and annual reports, ensuring they are accurate and delivered on time. Provision of tools that allow clinic staff to analyze health trends and outcomes within the BCP community.

4. Stakeholder Engagement: Collaboration with clinic staff, IT professionals, and relevant stakeholders to ensure the system meets the operational needs of the BCP Clinic.

Consideration of feedback from users to refine and enhance the system's functionality.

Limitations:

The system will be designed primarily for use within the BCP Clinic, and its implementation is limited to the specific needs and environment of the college.

The QR code system is focused on internal clinic operations and is not intended for use in external healthcare facilities.

Anticipated Outcomes

The successful deployment of a clinic management system enhances data security and contributes to better health outcomes within the BCP community.

A significant reduction in the time required for administrative tasks, allowing clinic staff to focus more on patient care.

AI-assisted diagnosis can contribute to these outcomes by:

Improving diagnostic accuracy: AI algorithms can analyze patient data more comprehensively and identify potential diagnoses that might be missed by human clinicians.

Reducing diagnostic errors: AI can help identify and flag potential errors in diagnosis, leading to safer patient care.

Streamlining the diagnostic process: AI can help automate some of the tasks involved in diagnosis, freeing up clinicians' time for more complex patient interactions.

1.3 Problem Statement

Bestlink College of the Philippines currently faces challenges in managing clinical services efficiently, leading to potential errors, delays, and suboptimal patient care. These challenges include:

Manual record keeping: Paper-based records are time-consuming, prone to errors, and difficult to access.

Communication gaps: Delays in communication between healthcare providers can impact patient outcomes.

The problem statements are:

- 1. Inefficient clinical management systems:** Current systems may not be fully integrated or optimized for efficient operations.
- 2. Manual processes:** Reliance on manual processes can lead to errors, delays, and inefficiencies.

AI-assisted diagnosis can help address these challenges by:

- 1. Automating data analysis:** AI can analyze patient data more efficiently and accurately than manual methods, reducing the risk of errors.
- 2. Improving communication:** AI can help facilitate better communication between healthcare providers by providing them with more comprehensive and accurate patient information.

Implementing QR code technology can address these challenges by:

- 1. Digitizing patient records:** QR codes can store and access patient information electronically, reducing errors and improving accessibility.

By integrating QR code technology into the clinical management system, Bestlink College of the Philippines can:

- 1. Improve efficiency:** can reduce errors and save time.

2. Enhance patient safety: Reduce medication errors and improve communication between healthcare providers.

1.4 Objective and Goal

Objective	Goals
Digitize patient records: Transition from paper-based records to a digital format using QR codes.	Improve patient safety: Reduce medication errors, improve communication, and enhance overall patient care.
	Enhance data management: Create a centralized, accessible database of patient information using QR codes.
	Improve patient satisfaction: Provide more efficient, effective, and personalized care.

Table 1: Objective and Goal

AI-assisted diagnosis can contribute to these objectives and goals by:

1. Improving the accuracy and efficiency of data analysis: AI

can help identify patterns and insights in patient data that might be missed by human analysts.

2. Providing clinicians with more comprehensive and accurate

information: This can help them make more informed decisions about patient care

1.5 Significance and Relevance

Improved Efficiency: QR codes can streamline various processes, such as Student Information, and appointment scheduling. This can reduce administrative burdens and improve overall efficiency.

Enhanced Patient Experience: QR codes can provide a more convenient and accessible way for Nurses to interact with the Students For example, students can easily access their medical records.

Data-driven decision-making: QR codes can capture valuable data, such as Student Information, appointment history, and treatment outcomes. This data can be used to inform evidence-based decision-making and improve the quality of care.

Cost Reduction: QR code integration can help reduce operational costs and improve resource allocation.

AI-assisted diagnosis can further enhance these aspects by:

Improving the efficiency of data analysis: AI can analyze large amounts of data quickly and accurately, saving time and resources.

Providing clinicians with more actionable insights: AI can help identify trends and patterns in patient data that can be used to improve care delivery.

Relevance:

Nurse's Needs: Many nurses are comfortable using technology and appreciate the convenience and efficiency that QR codes can offer because one can of QR code they get the full information of the student

Industry Trends: Some Healthcare organizations worldwide are increasingly adopting Technology and most of them are using QR codes to improve patient care and streamline operations.

Institutional Goals: The integration of QR codes aligns with Bestlink College's commitment to providing high-quality healthcare services and utilizing innovative technologies.

AI-assisted diagnosis is also becoming increasingly relevant in the healthcare industry, with many hospitals and clinics adopting this technology to improve patient care.

1.6 Structure of the Document

In this document, Bestlink College of the Philippines' clinic management system with QR code implementation is presented in detail.

A clear and thorough summary of the project is provided via the document's multiple sections.

I. Title

A brief and informative project title that summarizes the project intention.

II. Table of Content

A compilation of all categories and subcategories along with their respective page numbers.

III. Introduction

Gives a brief overview of the subject, states the problem and outline the goals of the capstone project.

IV. Related Studies & Literature Review

Reviews relevant literature and studies related to the topic.

V. Methodology

Explain the project's design, strategies for gathering data, and the tools implemented.

VI. System Development and Implementation

The process involves creating software or IT systems to meet business needs. It includes planning, analysis, design, development, testing, deployment, and maintenance.

VII. Summary, Conclusion, and Recommendations

A summary provides a brief overview of key points, the conclusion interprets findings, and recommendations suggest actions for improvements.

Chapter 2

Related Studies and Literature Review

2.1 Agile Scrum Methodology Overview

Agile Scrum, integrated with QR code technology, is key in developing a Clinic Management System. This approach focuses on customer collaboration, involving healthcare providers, clinic staff, patients, and administrators throughout the process. Short sprints allow continuous feature refinement based on feedback, ensuring the system meets real-world needs. QR codes boost adaptability, offering quick access to patient info and accommodating evolving clinic requirements. This flexibility enables the team to respond effectively to feedback and healthcare trends, resulting in a more efficient and user-friendly management system.

Agile Scrum Methodology Related Studies and Research

According to Kokol, P. (2022). "Agile software development in Healthcare: A Synthetic scoping review". It is focused on improving the software process, quality of healthcare software, reduction of development resources, and general improvement of healthcare delivery. More research must be done on scaling agile approaches to large-scale healthcare software development projects.

According to Thiagarajan, S., Saldanha, P. R. M., Govindan, R., Leena, K. C., & Prathyusha, P. V. (2024). "Development of Agile Scrum Perception Tool to Evaluate Students' Opinions on Agile Methodology in Nursing Education Exploring the student's opinions of AM in teaching helps understand how agile can be utilized in higher education to positively improve learning ability, leadership, and teamwork. Hence, the study aimed to develop the agile scrum perception tool (ASPT) to evaluate the opinions on AM among nursing students.

According to Alotaibi, F., & Almudhi, R. (2023). "Application of Agile Methodology in Managing the Healthcare Sector". This study significantly contributes to the healthcare sector by providing insights into the application of agile methodology, potentially enhancing healthcare management and patient outcomes.

According to Zierock, B., Angar, S., Rimmler, M., (2023). "Strategic Transformation and Agile thinking in Healthcare Projects". This article delves into the pivotal role that prototyping and testing, as integral components of user-centered design, can play in revolutionizing the healthcare domain. It explores how these methods can address the unique considerations of healthcare and highlights their potential benefits and challenges

According to Desai, M., Tardif-Douglin, M., Miller, I., Blitzer, S., Gardner, D. L., Thompson, T., Edmondson, L., & Levine, D. M. (2024).

“Implementation of Agile in healthcare: methodology for a multisite home hospital accelerator”. They believe the healthcare community could deliver innovation to the bedside faster if it followed other sectors by employing an organizational framework for efficiently accomplishing work.

2.2 Enterprise Architecture Concepts

This study focuses on implementing a QR code-based clinic management system within the Enterprise Architecture (EA) framework. EA delivers a systematic method to business process and technology alignment, ensuring that the technological components and operational requirements of the clinic are consistent. The basic concepts of EA include a holistic view of an organization's structure, covering four key areas: business processes, applications, data, and technology.

AI-assisted diagnosis represents a significant technological advancement that can be seamlessly integrated into this EA framework. It aligns with the principles of EA by enhancing the overall efficiency and effectiveness of the clinic's operations, improving patient care, and facilitating the integration of new technologies.

By leveraging an EA framework like TOGAF (The Open Group Architecture Framework), this system will follow established standards to ensure that it fits within the larger organizational architecture. By dividing the architecture into layers, such as business architecture, which defines clinic procedures, and application architecture, which covers the functionality of the QR code-enabled system, TOGAF directs the system's design and implementation.

AI-assisted diagnosis can be incorporated into each layer of the EA framework, enhancing its capabilities and contributing to the overall goals of the clinic management system.

This alignment helps achieve the clinic's organizational goals, improves service efficiency, and supports the integration of new technologies as healthcare demands evolve

Business Architecture

Defines the alignment of the iSMS with the clinic's operational processes, focusing on streamlining patient data management and improving healthcare services. It ensures that QR code technology is effectively integrated into workflows, optimizing the efficiency of clinic staff and enhancing patient care.

AI-assisted diagnosis plays a crucial role in business architecture by:

Streamlining diagnostic processes: AI algorithms can analyze patient data quickly and accurately, reducing the time required for diagnosis and allowing clinicians to focus on other aspects of patient care.

Improving diagnostic accuracy: AI can help identify potential diagnoses that might be missed by human clinicians, leading to more effective treatment plans.

Facilitating better communication: AI can help clinicians communicate more effectively with patients by providing them with clear and concise information about their diagnoses and treatment options.

Data Architecture

Structures the management and flow of sensitive healthcare information within the iSMS, ensuring data is securely stored and accessed through encrypted QR codes. The system supports timely and accurate health reporting, complying with data privacy regulations to protect patient information.

Ai-assisted diagnosis enhances data architecture by:

Providing a robust platform for data analysis: AI algorithms can analyze vast amounts of patient data, including medical images, lab results, and electronic health records, to identify patterns and insights that can

improve diagnosis and treatment.

Improving data security: AI can help protect patient data by identifying and mitigating potential security risks.

Facilitating data-driven decision-making: AI can help clinicians make more informed decisions about patient care by providing them with real-time insights based on patient data.

Application Architecture

Represents the software components of the iSMS, particularly the integration of QR code functionality. It emphasizes modular development, where core features such as patient registration, record management, and reporting are incrementally built and enhanced.

AI-assisted diagnosis can be integrated into the application architecture as a separate module or integrated into existing modules:

AI-powered diagnostic module: This module would allow clinicians to upload patient data, such as medical images or lab results, and receive AI-generated diagnoses and insights.

AI-enhanced record management: AI can be used to automate tasks such as patient record entry and data extraction, improving the efficiency of record management.

AI-driven reporting: AI can be used to generate comprehensive reports that provide insights into patient outcomes and trends in diagnosis and treatment.

Technology Architecture

Focuses on the underlying infrastructure that supports the clinic management system, ensuring reliable server and database functionality. The system leverages secure cloud storage and QR code scanning technology to facilitate fast, accurate access to patient data.

AI-assisted diagnosis requires a robust technology infrastructure to support its computational demands:

High-performance computing: AI algorithms require significant processing power to analyze large datasets.

Secure cloud storage: Patient data must be stored securely in the cloud to ensure privacy and confidentiality.

Advanced data analytics tools: AI algorithms require specialized tools for data analysis and visualization.

Enterprise Architecture Concepts Related Studies and Research

According to Hidalgo, Q., Camacho, S., Reyes, B., & Fuentes, M. (2023). "Implementation of QR codes on medical equipment as a tool for hospital management." These QR codes were affixed to the medical equipment to

facilitate easy tracking and organization. The implementation of the QR code and database-driven approach significantly improved the medical device management system's organization, enabling efficient maintenance and precise tracking of equipment locations.

According to Neiles, F., Boulton, S., & De Burgh-Thomas, A. (2023). Their aim was to introduce the use of QR codes to direct patients to specific PILs on the British Association for Sexual Health and HIV or Family Planning Association websites. Methods QR codes were created using an online tool for PILs relating to 16 sexual health related conditions and 16 contraceptive methods.

According to Jung, E., Jung, J., & Yun, H. (2024). This study was to compare and analyze the impact of video practice education using beam technology and self-learning through QR code videos on the acquisition of emergency patient management skills in relation to self-efficacy.

According to Zhou, Y. (2021). In order to improve the quality of medical services, an indoor navigation system based on QR code has been designed for hospitals, which realizes indoor navigation by scanning a QR code. This low-cost technology has accurate positioning, which brings convenience in locating and navigating departments.

2.3 Microservice Architecture

The challenge lies in enhancing the efficiency and accessibility of healthcare services at Bestlink College of the Philippines. QR codes offer a promising solution by digitizing various processes and reducing manual tasks.

Proposed Microservices Architecture

Microservices architecture can effectively handle the complexity and scalability of integrating QR codes into healthcare services. This approach divides the system into smaller, independent services that can be developed, deployed, and scaled independently.

Services	Description
QR Code Generation Service	Generates unique QR codes for various use cases, such as student information, medical date schedule and medical results. Integrates with a database to store and manage QR code data.
Student Management Service	Handles student information, medical history, section, and contact number of parents.

	Processes QR code scans to retrieve student data and update records.
Medical Results Management Service	Processes and stores lab results. Generates QR codes for medical results that can be accessed by patients and healthcare providers.
Inventory Management Service	Tracking and management of medical supplies, equipment, and medications.
Admission Management Service	Ensure effective patient admission processes in clinics.
Report and Analytics	Includes customizable reports, real-time analytics, data visualization, compliance reporting, and data exporting.

Table2: Proposed Microservice Architecture

Integrating AI-Assisted Diagnosis into Microservices

The AI-assisted diagnosis service can be implemented as a separate microservice, interacting with other services through APIs. This allows for:

Independent Scalability: The AI service can be scaled up or down based on demand, without impacting other services.

Flexibility: The AI service can be updated or replaced with newer technologies without disrupting the entire system.

Improved Security: Patient data is only shared with the AI service when necessary, enhancing data security.

Benefits and Challenges of Microservices

Benefits	Challenges
Scalability: Each microservice can be grow independently, which helps manage resources better, and scaled independently based on demand, ensuring optimal performance.	Scalability: While microservices are designed to be scalable, managing the scaling process and ensuring all services scale appropriately can be challenging.
Flexibility: New features and services can be added without affecting the entire system.	Complexity: Microservices add more difficulty because the services need to communicate and work together.
Scalability: Each microservice can be scaled independently based on	Data Consistency: Ensuring data consistency across distributed

demand, ensuring optimal performance.	services can be challenging, as each microservice may have its own database.
Resilience: If one microservice fails, the others can continue to operate, minimizing downtime.	Security Concerns: Microservices often span multiple cloud environments, increasing the risk of data breaches and loss of control over application components.
Maintainability: The code should be easy to modify and maintain. Microservices are smaller and more focused, making them easier to maintain and update.	Inter-service Communication: Managing communication between microservices can be complex, especially when dealing with high volumes of data and requests.
Continuous Delivery: Microservices can be developed, tested, and deployed independently, enabling faster development cycles.	Monitoring and Debugging: Monitoring the health of individual microservices and debugging issues can be more complicated compared to monolithic architectures.

Table 3: Benefits and Challenges of Microservice

2.4 DevOps and CI/CD: A Strategic Approach

DevOps, a cultural and technical approach, bridges the gap between development and operations teams to enable faster and more reliable software delivery. Continuous Integration (CI) and Continuous Delivery (CD) are key practices within DevOps that automate the building, testing, and deployment of applications.

Key Principles of DevOps and CI/CD:

Automation: Automate repetitive tasks like building, testing, and deployment to reduce errors and improve efficiency.

Collaboration: Foster collaboration between development and operations teams to break down silos and improve communication.

Continuous Improvement: Continuously seek ways to optimize processes and improve the quality of software delivery.

Version Control: Use a version control system to track changes to the codebase and facilitate collaboration.

Applying DevOps and CI/CD to QR Code Implementation:

1. Infrastructure as Code (IaC):

Use IaC tools like Terraform or Ansible to define and manage the infrastructure (e.g., servers, networks, storage) required for the QR code system. This ensures consistency, reproducibility, and

efficient provisioning of resources.

2. Source Code Management:

Choose a version control system like Git to manage the codebase for the QR code system. This enables collaboration, tracking of changes, and easy rollback if necessary.

3. Continuous Integration (CI):

Set up a CI pipeline to automate the building, testing, and packaging of the QR code system. This ensures that code changes are integrated and tested frequently, catching errors early.

4. Continuous Delivery (CD):

Implement a CD pipeline to automate the deployment of the QR code system to various environments (e.g., development, testing production). This reduces manual intervention and enables faster delivery of new features.

5. Containerization:

Consider using containers (e.g., Docker) to package the QR code system and its dependencies into a portable unit. This ensures consistent behavior across different environments and simplifies deployment.

6. Monitoring and Logging:

Implement monitoring tools to track the performance and health of the QR code system. Use logging to capture system events and troubleshoot issues.

7. Feedback Loops:

Establish feedback loops to gather information about the QR code system's performance and user experience. Use this feedback to continuously improve the system and address any issues.

Benefits of DevOps and CI/CD for QR Code Implementation

Faster Time to Market: Automated processes and streamlined workflows accelerate the delivery of new features and improvements.

Improved Reliability: CI/CD practices help identify and address issues early, reducing the risk of production failures.

Enhanced Scalability: The infrastructure can be easily scaled up or down to meet changing demands.

Greater Efficiency: Automation and collaboration reduce manual effort and improve overall efficiency.

Increased Quality: Continuous testing and feedback lead to higher-quality software.

2.5 Relevant Studies and Research Foreign:

“Patient Care Management using QR Code: Embracing Blockchain Technology” (2021)

The rise of QR codes has revolutionized technology, providing quick, reliable, and secure information access. With mobile devices, QR codes offer seamless connectivity, further secured by blockchain integration to ensure authorized access. Introducing MediQuick, a patient management app that combines QR code and blockchain technologies to create secure hospital databases. MediQuick enables patients to schedule appointments, access medical records, and retrieve prescribed medications efficiently and safely.

“During the Summer Outbreak of SARS-CoV-2 Delta Variant in Chengdu, China: An Observational Study” (2021)

During the SARS-CoV-2 Delta outbreak, Chengdu implemented effective health QR code measures, including circle-layer confinement, efficient hospital triage, and specialized care for quarantined pregnant women and children. These strategies prevented further transmission, enhanced patient triage efficiency, and provided satisfactory healthcare services. QR codes notably reduced the time needed to obtain epidemiological histories compared to traditional methods.

“Using QR Codes as a Form of eHealth to Promote Health Among Women in a Pandemic: Cross-sectional Study” (2022)

QR codes, vital during the pandemic, remain underused in healthcare by patients. Limited research focuses on their health promotion, especially for women, and user perspective assessments are scarce. COVID-19, declared a global pandemic on March 11, 2020, underscored the importance of health promotion. QR codes in gynecology clinics can enhance women's health during pandemics. The developed questionnaire provides significant insights for clinical practice, with potential applications in other areas and countries.

“Collection of patient-generated health data with a mobile application and transfer to hospital information system via QR codes” (2023)

Efficient collection of patient-generated health data (PGHD) is essential for patient care but integrating it into electronic medical records (EMRs) is often challenging. QR codes offer a two-way data interface between EMR forms and patient mobile apps, enhancing data exchange. The system serializes PGHD into QR codes, scans, and decodes them into EMR templates for accurate data integration. Laboratory results are processed similarly, allowing patients to access and save data via mobile

apps. This method streamlines PGHD integration into EMRs, improving care efficiency and data accessibility.

“Improving health clinic services through the application of QR code-based digital technology”. (2023)

This study focuses on improving the Hasna Medika Group's service system using Android-based QR Code technology to enhance efficiency and mobile access to health information. The approach aims to speed up service processes and improve overall service quality. Employing mixed-methods research, the study integrates quantitative and qualitative data analysis to develop an effective QR Code-based public service system.

“Enhancing Congenital Cardiac Postprocedural Care: A QR Code Enabled Medical Bracelet Pilot Study (2024)”

The study explored the feasibility of using QR code-enabled medical bracelets for postoperative CHD patients to ensure quick communication of vital information during emergencies. Pediatric cardiac patients' families were given QR bracelets linking to secure medical information sheets. Surveys indicated 65% usage in medical care, 55% found them useful, 70% had a positive experience, and 80% recommended them for other cardiac patients. High satisfaction levels suggest QR code bracelets can reduce medical errors and treatment delays.

**“QR Codes in Medical Education - An Analog and Digital Bridge
(2021)”**

This two-part series explores the use of QR codes in health education. The first article provides an overview of QR codes, including their structure, symbology, and usage requirements, to help educators develop the necessary skills for implementation. The second article will focus on custom QR codes, tools for generation and decoding, and their application in Augmented Reality. The series aims to ensure educators can effectively integrate QR code technology into their teaching materials, enhancing both traditional and online education.

Local:

**“Proposed Innovations on Contact Tracing System using
Contactless Smart Cards in the Philippines” (2021)**

Manual contact tracing in the Philippines faces challenges due to low smartphone penetration. This study explores using RFID-based contactless smart cards for a reliable contact tracing system. A survey indicated high user acceptance and preference for smart cards over current methods, suggesting their potential adoption for effective contact tracing in the Philippines.

“Development and Pilot Testing of GoTrace: A Web Application for Contact Tracing Using Quick Response (QR) Codes” (2021)

Amid COVID-19 challenges, strict health protocols and contact tracing are essential, particularly in the Philippines where vaccine distribution is slow. This study introduces GoTrace, a QR code-based web application for contact tracing, integrating daily health surveys and location data to create real-time, accurate reports. Positive pilot testing feedback suggests GoTrace is a promising, adaptable, and user-friendly solution for various environments.

“Effectivity of digitized health information card” (2022)

As the COVID-19 pandemic continues with new variants, ensuring efficiency and safety is vital. Researchers developed Digitized Health Information Cards using QR codes and Google Forms to support paperless transactions. This study assessed the system's effectiveness in terms of accessibility, convenience, and user satisfaction through surveys and weighted mean analysis. Positive feedback highlighted the system's convenience and accessibility, with students showing a strong preference for digitized forms over manual writing, suggesting QR codes significantly benefit student welfare.

“Development of employee attendance and management system using quick response (QR) code in Sorsogon state university, Castilla campus, Philippines” (2022)

The development of a QR code-based Attendance System aimed to streamline the management of employee attendance at Sorsogon State University, Castilla Campus. Using the Rational Unified Process (RUP) and descriptive evaluation methods, the system assigned QR codes to employees for attendance confirmation. The findings highlighted improved accessibility, cost-effectiveness, and reduced errors. The system demonstrated high functionality, reliability, efficiency, accuracy, usability, maintainability, and portability, making it an effective, user-friendly solution for contactless attendance tracking.

“Contact tracing application using quick response code and thermal scanner” (2022)

In response to the COVID-19 pandemic, a Contact Tracing Application using QR Codes and Thermal Scanners was developed to enhance health safety at All for Christ Church, Tanyag, Taguig City. The system streamlines the identification of persons under investigation (PUIs) by using thermal scanners for temperature checks, which are sent directly to a database. It generates unique QR codes for registrants, aiding contactless identification. This initiative aims to mitigate virus spread

improve health awareness, and reduce manual labor, proving vital in pandemic management efforts.

2.6 Integration of Information Systems in Enterprise Environments

Integrating information systems into the business environment of a Clinic Management System using QR codes presents both challenges and significant benefits, particularly when dealing with various modules like Student Information, Inventory, Medical, Admission, and Reports and Analytics.

AI-assisted diagnosis is a powerful technology that can significantly enhance the effectiveness and efficiency of this integrated system. It can be integrated into various modules, providing valuable insights and improving decision-making across the clinic.

Challenges in Integrating Information Systems

1. Complexity in Synchronizing Modules: Integrating multiple modules—such as Student Information, Medical, and Admission—can be challenging, especially when they have different workflows and data requirements. Ensuring seamless interaction between these modules demands precise coordination and technical expertise.

AI-assisted diagnosis can help address this challenge by:

Standardizing data formats: AI can be used to translate data between different modules, ensuring consistency and reducing the need for complex data mapping.

Automating data exchange: AI can facilitate seamless data flow between modules, simplifying integration and reducing the need for manual intervention.

2. Data Security and Privacy Concerns: Handling sensitive student health records requires strict adherence to security standards, especially when integrating multiple systems. Managing secure access to personal data while complying with data protection regulations can be a challenge.

AI-assisted diagnosis can enhance data security by:

Enhancing data encryption: AI can be used to develop more robust encryption methods, protecting sensitive patient data.

Identifying and mitigating security risks: AI can analyze data patterns to identify potential security vulnerabilities and recommend solutions.

3. User Adaptability and Training: Clinic staff and students may face difficulties in adapting to the new system, especially with the use of QR codes for health records and services. Continuous training is necessary to ensure that all users are proficient in the system's functionalities.

AI-assisted diagnosis can improve user adaptability by:

Providing intuitive user interfaces: AI can help design user-friendly interfaces that are easy to learn and use, reducing the need for extensive training.

Offering personalized guidance: AI can provide tailored guidance and support to users based on their individual needs and experience levels.

4. Cost and Time for Implementation: Integrating different modules into a cohesive system requires significant resources—both in terms of time and financial investment. The initial setup, testing, and troubleshooting phases can be resource intensive.

AI-assisted diagnosis can help reduce implementation costs by:

Automating tasks: AI can automate many tasks involved in system integration, such as data migration and testing, reducing the need for manual effort.

Streamlining development processes: AI can assist in code generation and testing, accelerating the development cycle.

5. Maintenance and Scalability: Ensuring that the integrated system remains scalable to accommodate future enhancements, such as expanding to other departments or adding new functionalities, can pose a long-term challenge.

AI-assisted diagnosis can improve scalability by:

Adapting to changing data volumes: AI can handle increasing data volumes without compromising performance, ensuring the system remains scalable.

Supporting new functionalities: AI can be easily integrated with new modules and functionalities, enabling the system to evolve and adapt to changing needs.

Benefits of Integrating Information Systems were as follows:

1. Improved Efficiency: By integrating all the relevant modules, the clinic's workflows become more streamlined. QR codes allow for fast, automatic access to student health records, eliminating manual processes like data entry, which significantly reduces wait times and errors in service delivery.

AI-assisted diagnosis further enhances efficiency by:

Automating diagnostic processes: AI can analyze patient data and generate potential diagnoses, reducing the time and effort required for manual diagnosis.

Providing real-time insights: AI can provide clinicians with immediate insights based on patient data, enabling faster and more accurate decision-making.

2. Enhanced Data Consistency: Integration ensures that all modules access and update the same data set. For example, when a student's medical record is updated in the medical module, the information is simultaneously reflected in the Student Information module. This eliminates data discrepancies and ensures that all systems are working with consistent, up-to-date information.

AI-assisted diagnosis contributes to data consistency by:

Centralizing data access: AI can provide a single source of truth for patient data, ensuring that all modules access the same information.

Maintaining data integrity: AI can monitor data for inconsistencies and errors, ensuring data accuracy and reliability.

3. Better Decision Making: The Reports and Analytics module, fed by integrated data from other modules, can generate real-time insights into clinic operations, resource use, and patient care trends.

With accurate and comprehensive data, administrators can make informed decisions, improve resource allocation, and identify areas that require attention.

AI-assisted diagnosis enhances decision-making by:

Providing predictive analytics: AI can analyze data to identify potential health risks and predict future trends, enabling proactive decision-making.

Generating personalized recommendations: AI can offer tailored recommendations based on individual patient data, improving the quality of care and resource allocation.

4. Improved Patient Experience: Students can easily access services by scanning their QR codes, which simplifies appointment scheduling and information retrieval. This enhances the overall patient experience, leading to greater satisfaction and streamlined operations for the clinic.

AI-assisted diagnosis can further improve the patient experience by:

Providing personalized information: AI can provide patients with tailored information about their conditions, treatment options, and health management strategies.

Enabling self-service options: AI can empower patients to access information and manage their health records independently, enhancing convenience and autonomy.

5. Cost Savings in the Long Run: While initial setup costs may be high, integrating the information systems leads to long-term savings by reducing administrative overhead, preventing redundancies, and eliminating the need for multiple standalone systems.

AI-assisted diagnosis contributes to cost savings by:

Reducing diagnostic costs: AI can help reduce the need for expensive and time-consuming diagnostic tests.

Optimizing resource allocation: AI can provide insights into resource utilization, enabling better resource allocation and reducing unnecessary expenses.

Common Integration Techniques and Technologies

1. Application Programming Interfaces (APIs): APIs enable different systems, such as the Student Information and Medical modules, to communicate seamlessly. APIs allow data to flow in real time, ensuring that information remains synchronized across systems.

AI-assisted diagnosis can leverage APIs to:

Access patient data: AI can access patient data from various modules through APIs to perform analysis.

Share insights: AI can share its insights and recommendations with other modules through APIs, facilitating integrated decision-making.

2. Enterprise Service Bus (ESB): ESBs function as a central data highway for the various systems, helping to route data between modules like Admission and Clinic Staff. This facilitates a smooth exchange of data and reduces the complexity of integrating multiple systems.

AI-assisted diagnosis can integrate with ESBs to:

Access data from multiple sources: AI can access data from different modules through the ESB, enabling comprehensive analysis.

Share insights across the system: AI can share its insights with other modules through the ESB, facilitating data-driven decision-making.

3. Middleware: Middleware acts as a bridge between different systems, allowing them to exchange data even if they use different technologies or formats. Middleware is especially useful when

integrating legacy systems with newer technologies.

AI-assisted diagnosis can be integrated with middleware to:

Connect to legacy systems: AI can access data from legacy systems through middleware, enabling comprehensive analysis.

Ensure data compatibility: AI can help translate data between different formats, ensuring compatibility across the integrated system.

4. Database Integration and ETL (Extract, Transform, Load):

This process ensures that data from various modules (like Medical and Admission) is stored in a central database, making it easier to access, update, and analyze information. ETL tools help in consolidating data into the Reports and Analytics module, ensuring that data-driven insights are accurate.

AI-assisted diagnosis can enhance database integration by:

Automating data extraction and transformation: AI can automate the ETL process, reducing manual effort and improving efficiency.

Enhancing data quality: AI can identify and correct data errors, improving the accuracy and reliability of the database.

Impact of Information Systems Integration on Decision Making

Information systems integration plays a pivotal role in decision-making by providing accurate, real-time data across all clinic operations. By consolidating data from various modules, administrators can assess patient care quality, clinic staff performance, and resource utilization through reports generated in the Reports and Analytics module. This allows for better planning, more strategic decision-making, and proactive problem-solving.

AI-assisted diagnosis significantly enhances decision-making by:

Providing predictive insights: AI can analyze data to identify potential health risks and predict future trends, enabling proactive interventions.

Generating personalized recommendations: AI can offer tailored recommendations based on individual patient data, improving the quality of care and resource allocation.

Examples of Successful Information Systems Integration

1. Cleveland Clinic: Cleveland Clinic implemented an integrated health management system that consolidated patient records, clinical workflows, and operational data. This integration resulted in improved patient outcomes, enhanced staff collaboration, and optimized resource management.

2. Mount Sinai Health System: This system integrated patient management, billing, and analytics platforms to provide a holistic view of clinic operations. It resulted in reduced administrative workload and better patient care delivery.

Chapter 3

Methodology

3.1 Agile Scrum Methodology in the Project

Using Agile Scrum in the building of the Clinic Management system at Bestlink College of the Philippines can significantly increase adaptability, teamwork, and responsiveness to changing details. The following gives a high-level summary of Agile Scrum. For this project, I applied:

3.2 Roles and Responsibilities

Name	Role	Responsibilities
Joseph Oliver P. Banac	Project Manager	Develop project plans, schedules, and budgets. Facilitate team meetings and stakeholder updates. Manage risk assessments and mitigation strategies. Ensure project deliverables align with goals.
Angelo B. Abargos	Developer	Front-End: Create responsive

		UI components, implement design mockups, and optimize performance. Back-End: Build and maintain server-side logic, APIs, and database interactions. Full Stack: Integrate front-end and back-end systems, ensuring seamless functionality.
Ariel A. Forcado	UI/UX Designer	Conduct user research to gather insights into user needs. Design wireframes, prototypes, and high-fidelity mockups. Test designs through usability testing and gather feedback for iterations. Collaborate with developers to ensure design fidelity in implementation.

Lister John L. Evangelista	Security Analyst	Develop and enforce security policies and best practices. Monitor systems for security breaches and respond accordingly. Stay updated on cybersecurity trends and compliance regulations.
Andiella Mhae B. Gabad	Documentation Specialist	Develop user manuals, installation guides, and API documentation. Maintain a knowledge base for project artifacts and resources. Ensure all documentation is up-to-date and easily accessible. Assist in creating training materials for users.

Table 4: Roles and Responsibilities

Roles

Scrum Master: The Scrum Master helps the team follow Scrum practices, removes obstacles, manages communication, and promotes collaboration to meet project goals on time.

Product Owner: The Product Owner is the main person in charge of the project. They define the project's vision and goals, prioritize features, make sure the product meets user needs, and help the development team and stakeholders communicate. This helps guide the project to successful completion.

Development Team: The development team, which includes individuals with different backgrounds and specialties, is responsible for carrying out the actual development work.

Project Manager: The project manager in IT oversees the planning, execution, and delivery of IT projects. This involves coordinating teams, and managing resources, timelines, and budgets. The project manager ensures that project objectives align with business goals and works to mitigate risks to successfully deliver technical solutions.

Developer: The developer's role is to create, code, and put into place practical solutions to ensure that the technical aspects of the project match the defined objectives. This involves combining user requirements and system functions while solving problems and using new ideas.

UI/UX Designer: The UI/UX Designer's role is to create easy-to-use interfaces and smooth experiences by researching user needs, designing wireframes, and ensuring that the system is easy to use and accessible to meet project goals.

Security Analyst: The security analyst's job is to check, plan, and set up security measures that keep data and systems safe during the project. The analyst makes sure to find and fix any weaknesses in the project's development and launch phases

Documentation Specialist: The documentation specialist creates, organizes, and maintains project records. This includes technical reports, user manuals, design specifications, and progress documentation. The specialist ensures that the documents are clear, accurate, and compliant with project standards.

3.3 Sprint Cycles

Regular check-ins and progress evaluations are part of sprint cycles. This could entail holding more frequent feedback meetings between managers and staff in performance management. This check-in offers the chance to talk about accomplishments, difficulties, and prospective goal revisions.

Sprint Cycle	Duration	User Stories / Tasks	Key Features / Tasks	Deliverables	Responsible Team Member
Sprint Cycle 1: Inventory Management					
Sprint Planning	5 days	Define key process of inventory. Define what necessary data need in inventory.	Assess inventory functions.	Functionalities	Angelo Abargos, Ariel Forcado
Daily Standups	5 days	Monitor / maintain alignment	Sync team in medical processes, inventory management and reports	Progress on system	Team
Sprint Review	5 days	Review inventory functionalities	Present Inventory management. Gather feedback from	Functional inventory	Angelo Abargos, Ariel Forcado

			stakeholder s		
Sprint Cycle 2: Medical Management					
Sprint Planning	5 days	Define key medical processes. Define what necessary data need in medical.	Assess medical module functions.	Medical function.	Angelo Abargos, Joseph Oliver Banac
Daily Standups	5 days	Track medical module progress.	Medical module and testing.	Medical management	Team
Sprint Review	5 days	Present medical and features.	Present medical manageme nt to stakeholde rs. Gather their feedback.	Functional medical management	Andiella Mhae Gabad, Lister John Evangelista.
Sprint Cycle 3: Admission and Student					

Sprint Planning	5 days	Plan Student information and admission of students.	Assess student module and admission module.	Student module function And Admission module	Angelo Abargos, Lister John Evangelista
Daily Standups	5 days	Monitor Student module and Admission module progress	Student module and admission testing.	Student module and Admission module	Team
Sprint Review	5 days	Present Student module and Admission module.	Gather feedback on student and admission module from stakeholders.	Functional Student and admission module.	Joseph Oliver Banac, Ariel Forcado
Sprint Cycle 4: Reporting and Reminder					
Sprint Planning	5 days	Plan reporting features.	Generate Reports	Reporting features and export data.	Angelo Abargos,

					Andiella Mhae Gabad
Daily Standups	5 days	Track reporting and data integration.	Develop filters and export of reports	Generating Report data.	Team
Sprint Review	5 days	Present the report module to the stakeholder s.	Feedback on report module form stakeholder s	Functional Report module	Ariel Forcado, Angelo Abargos
Sprint Cycle 5: Data Management					
Sprint Planning	5 days	Plan and manage data.	Data handling	Data handling	Angelo Abargos, Lister John Evangelista
Daily Standups	5 days	Monitor the progress in data manageme nt.	Track data manageme nt	Data Management	Team
Sprint Review	5 days	Review and improve	Reviewing and improving	Data management	Joseph Oliver Banac,

		data manageme nt	data manageme nt		Andiella Mhae Gabad
Sprint Cycle 6: Security Features					
Sprint Planning	5 days	Plan and identify system security features.	Role based access control, session token, authenticati on.	Security	Joseph Oliver Banac, Lister John Evangelista
Daily Standups	5 days	Research security applicable in system.	Security System features	RBAC authenticatio n session token	Team
Sprint Review	5 days	Review and Improve security.	Test security features	Functional Security.	Angelo Abargos, Joseph Oliver Banac
Sprint Cycle 7: AI-Assisted Diagnosis					
Sprint Planning	5 days	Plan AI features in the system.	AI-Assisted diagnosis feature	Planning AI feature	Angelo Abargos, Andiella

					Mhae Gabad
Daily Standups	5 days	Monitor changes in AI, track progress.	Testing AI model capabilities	AI-assisted diagnosis	Team
Sprint Review	5 days	Review and Improve AI, present to the stakeholders.	Feedback on AI feature.	Improve AI gradually.	Andiella Mhae Gabad, Ariel Forcado
Sprint Cycle 8: QR Code Integration					
Sprint Planning	5 days	Plan QR code features.	QR functionalities	Scannable QR	Angelo Abargos, Ariel Forcado
Daily Standups	5 days	Research QR code integration.	Data retrieval using QR	QR functionalities	Team
Sprint Review	5 days	Improve QR Code and present it to the stakeholders.	Feedback in QR code module.	QR code more functionalities	Angelo Abargos, Andiella Mhae Gabad

Sprint Cycle 9: Documentation					
Sprint Planning	5 days	Review all documentations.	Checking the document.	Documentation	Team
Daily Standups	5 days	Consult documents and system.	Consultation	Document Enhance	Team
Sprint Review	5 days	Refining Documents and requirements.	Revision of Document.	Document Quality	Team
Sprint Cycle 10: System Testing for Bugs and Integration					
Sprint Planning	5 days	Test system functionalities.	Findings of bugs and testing.	Integration.	Angelo Abargos, Ariel Forcado
Daily Standups	5 days	Cluster integration, feedback and research.	Performance checking and penetration test.	System enhancement	Team

		Error handling.			
Sprint Review	5 days	Review and analyze system.	Review and Improving system.	System features.	Angelo Abargos, Lister John Evangelista
Sprint Cycle 11: Backup, Monitoring and Deployment					
Sprint Planning	1 week	Planning and monitoring of the system. Deployment of system	Backup and monitoring of system. Deploying system online.	Backup and Deployment	Angelo Abargos, Ariel Forcado
Daily Standups	1 week	Improving data management and backups. Tracking and Quality testing for deployment	Data storage, Data management, System Deployment	Data management and backup, deployment.	Team

Sprint Review	1 week	Checking the system and always monitoring and backup. Deployment of system and presenting to stakeholders	Monitoring system. Deployed Cloud / Online	Data storage and system monitoring. Access the system Online	Team
Sprint Cycle 12: Final Review of Features and Refinement					
Sprint Planning	1 week	Final Review of features and system and documentation.	Documentation and System Review	Review	Team
Daily Standups	1 week	Integration Testing,	System Testing	Testing	Team

		System Testing			
Sprint Review	1 week	Refinement of System, Improving the system overall.	System Refinement, Quality testing system overall.	System Refinement, Quality testing	Team

Table 5: Sprint Cycles

3.4 Scrum Artifacts

Product Backlog: A product backlog in performance management could be an employee's prioritized list of skills, competencies, or areas for development. It is possible to periodically review and modify this backlog.

Product Backlog (User Stories)

User Story No.	User Stories	User Story Priorities	Status
1	As a clinic physician, I can log in clinic system. <i>Can log in</i>	5	Done
2	As a clinic physician, I want to view the dashboard. <i>Overview Dashboard</i>	5	Done
3	As a clinic physician, I want to view and manage	5	Done

	student information. <i>Manage Student Data</i>		
4	As a clinic physician, I want to view, manage medical results, medical history, and health form. <i>Manage Medical Affiliation's</i>	4	Done
5	As a clinic physician, I want to track and manage the inventory of medical supplies <i>Inventory Management</i>	5	Done
6	As a clinic nurse, I want to update student medical records in real- time so that other staff members and authorized users always see the most current data. <i>Real-Time Data Accuracy for Student Medical Records</i>	1	Done
7	As a clinic nurse, I want to view a summarized version of a student's medical history so that I can make informed decisions quickly during a clinic visit. <i>Medical History Summary for Clinic Staff</i>	1	Done
8	As a system administrator, I want to track and audit user access to medical records to ensure data security and identify any unauthorized access. <i>Access Logs for Data Auditing</i>	1	Done
9	As a system administrator, I want daily backups of the clinic's data to ensure that no critical health information is lost in case of a system failure or data corruption. <i>Data Backup and Recovery</i>	1	Done

10	As a school administrator, I want access to real-time health reports (e.g., clinic visits, prevalent illnesses) so I can make informed decisions regarding school policies or closures. <i>Real-Time Health Reports</i>	1	Done
11	As a clinic administrator, I want to analyze student health trends (e.g., seasonal illnesses) to plan proactive healthcare strategies and reduce the spread of infections. <i>Health Monitoring and Trend Analysis</i>	2	Done
12	As a school administrator, I want to ensure that student health data is stored and shared securely in compliance with privacy laws (e.g., HIPAA), so the clinic meets all regulatory standards. <i>Data Privacy and Student Health Confidentiality</i>	1	Done
13	As a clinic nurse, I want to track and manage the inventory of medical supplies so I can ensure we have sufficient stock of essential items (e.g., bandages, medications) without running out. <i>Inventory Management</i>	2	Done

Table 6: Product Backlog (User Stories)

Product Backlog for EIS Information Security

User Story No.	User Stories	User Story Priorities	Status
1	As a system administrator, I want to define user roles and assign permissions so that users can only access the data and functionalities necessary for their role, minimizing security risks. <i>Role-Based Access Control (RBAC)</i>	1	Done
2	As a system administrator, I want to require users to authenticate using both their password and a secondary method (e.g., OTP or biometrics) to prevent unauthorized access. <i>Multi-Factor Authentication (MFA) for User Access</i>	1	Done
3	As a data security officer, I want to ensure that all sensitive student health data is encrypted both when stored and transmitted to prevent unauthorized access and data breaches. <i>Data Encryption</i>	1	Done
4	As a compliance officer, I want a detailed log of all user activities related to student health data, so I can track and investigate any unauthorized access or changes. <i>Audit Trails and Activity Logging</i>	1	Done

5	As a network security officer, I want to be alerted to any suspicious activities or unauthorized access attempts to the system so that I can take immediate action to prevent breaches. <i>Intrusion Detection and Prevention System (IDPS)</i>	2	Done
6	As a system administrator, I want automatic daily backups of the EIS data so that the clinic can quickly recover in case of an incident or data loss. <i>Data Backup and Recovery</i>	1	Done
7	As a system administrator, I want to securely transmit data between the EIS and external systems (e.g., the registrar) to prevent unauthorized interception or data tampering. <i>Secure Data Sharing Protocols (External Integrations)</i>	1	Done
8	As a security officer, I want to conduct regular vulnerability assessments and penetration tests on the system to detect and fix security flaws, ensuring the clinic's data is protected. <i>Periodic Vulnerability Assessments and Penetration Testing</i>	1	Done
9	As a security officer, I want to enforce a policy requiring users to create complex passwords (including letters, numbers, symbols, and special character) and reset them periodically to enhance security.	1	Done

	<i>Secure Password Policies</i>		
10	As a compliance officer, I want the EIS system to comply with regulations like HIPAA to ensure that all student health data is protected and handled in accordance with legal standards. <i>Compliance with Data Protection Regulations</i>	1	Done
11	As a security officer, I want the system to automatically log out users after a certain period of inactivity to reduce the risk of unauthorized access if a session is left open. <i>Session Timeouts and Automatic Logouts</i>	2	Done
12	As a clinic administrator, I want the clinic staff to receive regular training on data security to prevent human errors that could lead to security breaches. <i>Security Awareness Training for Clinic Staff</i>	4	Done

Product Backlog for EIS Standards

User Story No.	User Stories	User Story Priorities	Status
1	As an admin, I want to customize reports.	1	Done
2	As an admin, I want to filter the list of medical.	1	Done
3	As an admin, I want to automate reminder for nurses related to inventory tracking.	1	Done

4	As an admin, I want to monitor the medical examination.	1	Done
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Table 7: Product Backlog (EIS)

Product Backlog for Integration

User Story No.	User Stories	User Story Priorities	Status
1	As a clinic administrator, I want the system to integrate with the school's registrar / student portal so that I can access and retrieve student information (e.g., contact details, grade level, enrollment status) in real- time without manually entering data. <i>Registrar Integration for Student Data</i>	1	Done
2	As a clinic administrator, I want the system to integrate with human resources so that they can assign role, access, and permission of accounts of the employee of the clinic. <i>Human Resource Integration for Account Management</i>	1	Done

Table 8: Product Backlog (Integration)

Sprint Backlog

User Story No.	User Stories	User Story Priorities	Revision Priority	Status
1	Log and admit student medical history,	2	2	Done

	examinations			
2	Track and manage inventory.	2	2	Done
3	Legal compliance as GDPR and HIPAA.	2	2	Done
4	Generate reporting and reminders to accounts.	3	3	Done
5	AI and QR integration	1	1	Done
6	Security integration	1	1	Done
7	Human resource integration	1	1	Done
8	Registrar integration	1	1	Done
9	Medical records management	1	1	Done
10	Scheduling and appointment	3	3	Done

Table 9: Sprint Backlog

Scrum Board

Done
Developed Inventory Management
Developed Medical Management
Developed Admission and Student Management
Developed Reports, Reminder, and Notification
Developed Security Features
Developed AI-assisted Diagnosis
Developed QR Code Integration
Manage the Documentation, Technical Documentation, and Manual

System Testing, Debugging, and Vulnerability Test
Data Storage, Backup, Monitoring, Deployment
Registrar Integration and Human Resource Integration
Authentication and Verification

Table 10: Scrum Board

Burndown Chart

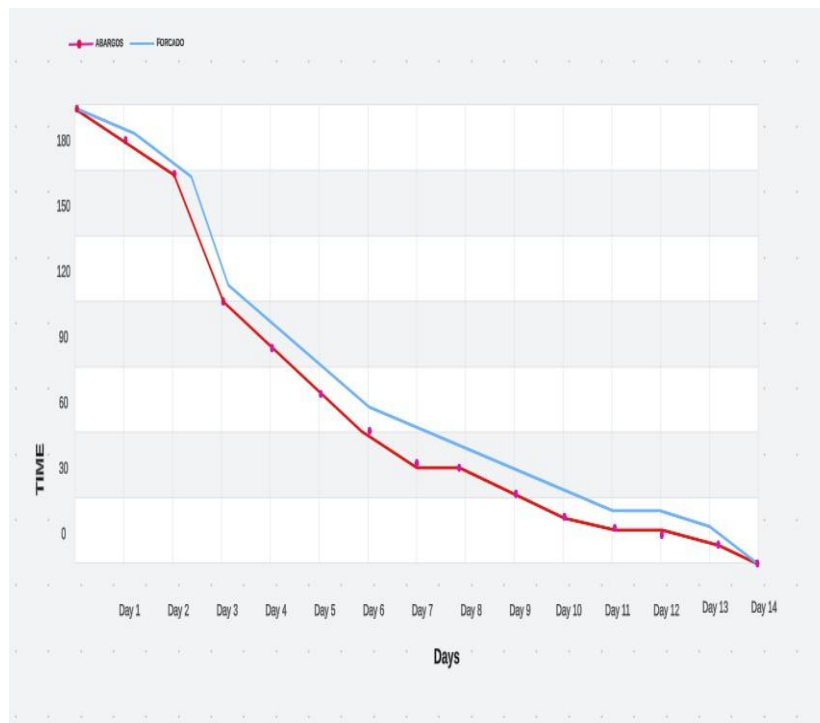


Figure 1: Burndown chart: Programmer / UI /UX Designer

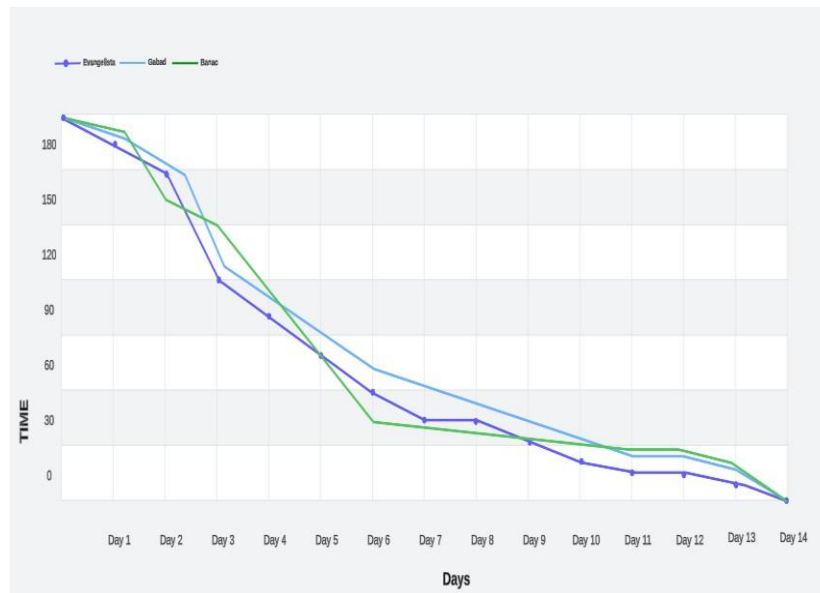


Figure 2: Burndown chart: Documents

3.5 Microservice Architecture

Our Clinic Management System's microservices design is made up of several essential components that work together to deliver a simplified and efficient experience for both healthcare practitioners and patients.

Account Management and Authentication Service, this service will manage secure user sign-ups, logins, and role-based access for patients, physicians, and staff. It will protect important health data and allow only authorized individuals access to the system. Patient Records Service: This service will store and manage patient medical histories, diagnoses, treatments, and prescriptions. With AI-powered, it will assist healthcare providers in offering better decision support, predicting potential health risks, and recommending preventative measures based on historical data.

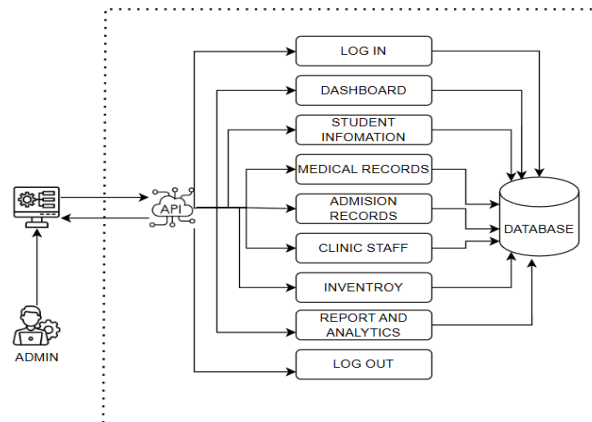


Figure 3: Microservices Architecture Diagram (Admin)

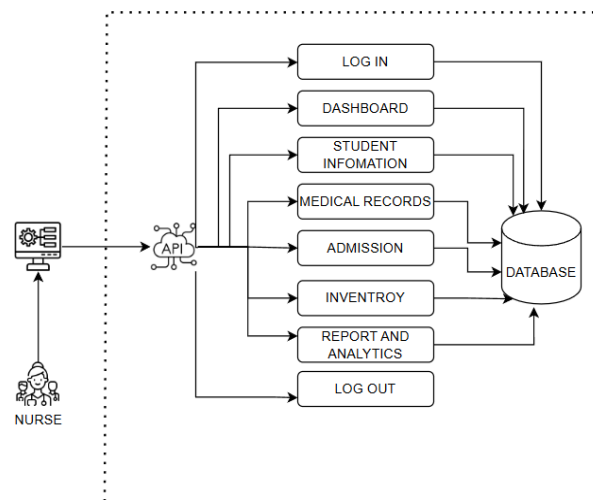


Figure 4: Microservices Architecture Diagram (User)

This Microservice diagram shows an application's architecture, illustrating individual services and their relationships. It focuses on communication mechanisms and data flows, which aid in understanding system structure and scalability. By adopting this microservices architecture, each component of the CLINIC MANAGEMENT SYSTEM

operates independently. This design not only improves the overall patient and staff experience but also ensures the system is scalable and adaptable to future needs, whether it be incorporating new technologies, expanding service offerings, or improving operational efficiencies.

Data Flow Diagram

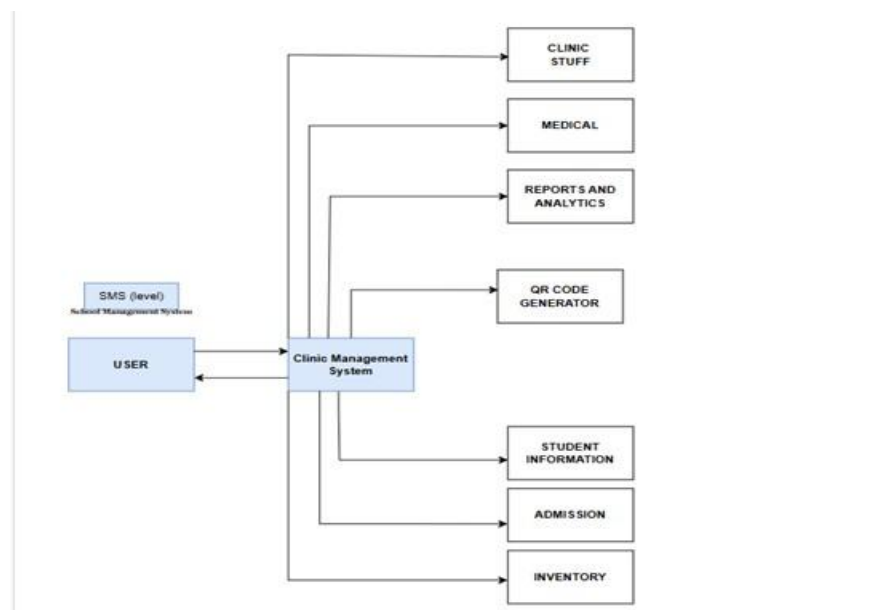


Figure 5: Data Flow Diagram

Data Flow Diagram (DFD) Level 1 shows how data moves within a system. It breaks down the main processes into sub-processes, showing inputs and outputs for each. This helps to understand how data is processed and where it goes, providing a clearer picture of the system's operations.

3.6 DevOps Implementation

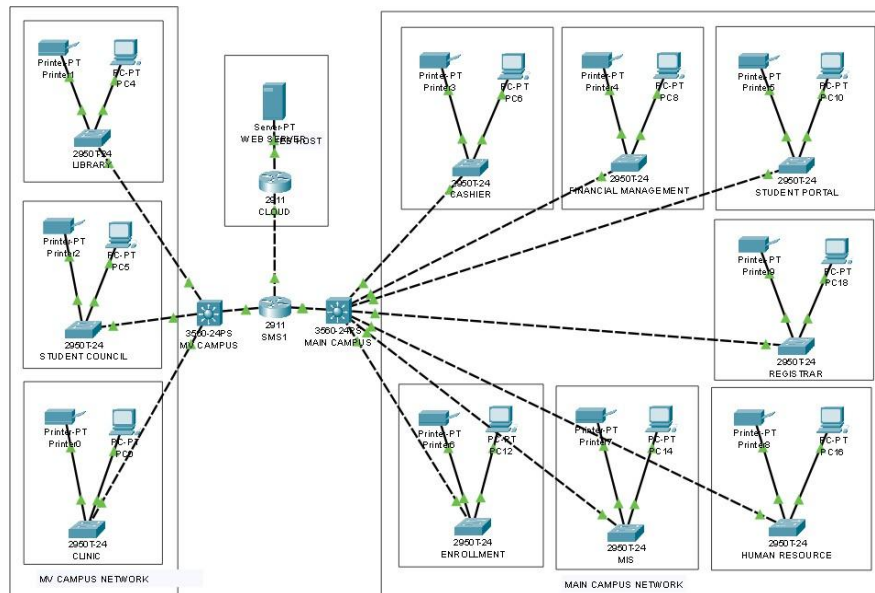


Figure 6: Network Topology

A simple network configuration comprising a router, switch, and other connected devices is depicted in this diagram. Every device in this network, which has a star topology, is linked to a single switch. The basic parts of this diagram consists of the following:

Internet: Stands for network.

Router: Links the internet and the local network together.

Switch: A gadget that permits several devices to connect to the same network.

PC: A personal Computer

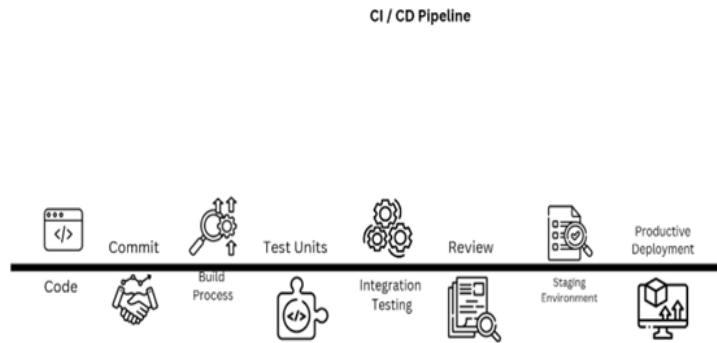


Figure 7: CI/CD Pipeline Code

This figure shows a standard CI/CD pipeline. The first step is writing and committing to a plan. Under continuous integration, the code subsequently undergoes a build process, unit testing, Integration Testing, Review, and Staging Environment and Product Deployment. The code is reviewed, staged, and then put into production after completing these steps, which are all part of continuous delivery.

This pipeline ensures the code is built, tested, and deployed effectively and efficiently, and helps to enhance the software development process.

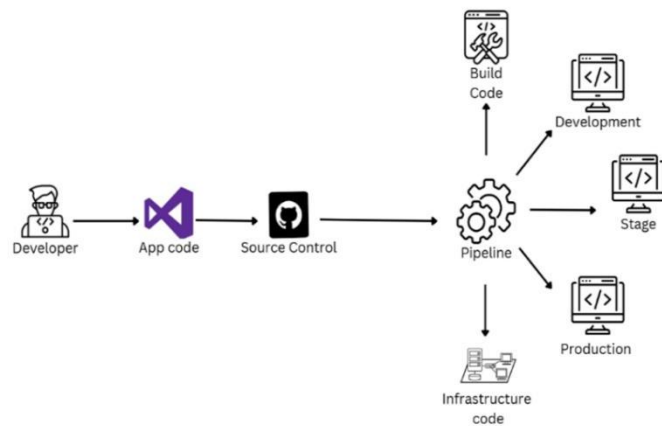


Figure 8: Infrastructure as Code

With this diagram, we can see the software development process's Infrastructure represented as code. Presented below is an outline of the procedures:

1. **Developers:** developers write the application code using (VS Code).
2. **App Code:** The code is committed to the Git Hub repository.
3. **Sources Control:** Git is a repository management system that keeps track of all the changes made to code.
4. **Pipeline** making modifications to the code in Git, a CI/CD pipeline is started.
5. **Build Code:** The pipeline generates the application code automatically.

6. Development: For testing and debugging, the developed application is placed in a development environment.

7. Stage: For testing and debugging, the developed application is placed in a development environment. To replicate production settings, the code is moved to a staging environment once it has been tested in the development environment.

8. Production: Lastly, the application is made available to users in production if everything in the staging environment goes as planned.

3.7 Innovation Integration

The Business Process Architecture (BPA) is depicted in the following figures at three levels of hierarchy. The integration of all subsystems is demonstrated in Top Level 1, which offers a summary of the primary system. In particular, the proponents' subsystem and related characteristics are described in Top Level 2. The data flow is finally depicted in Top Level 3, which offers a more thorough view of the interactions taking place within the proponents' subsystem.

Business Process Architecture

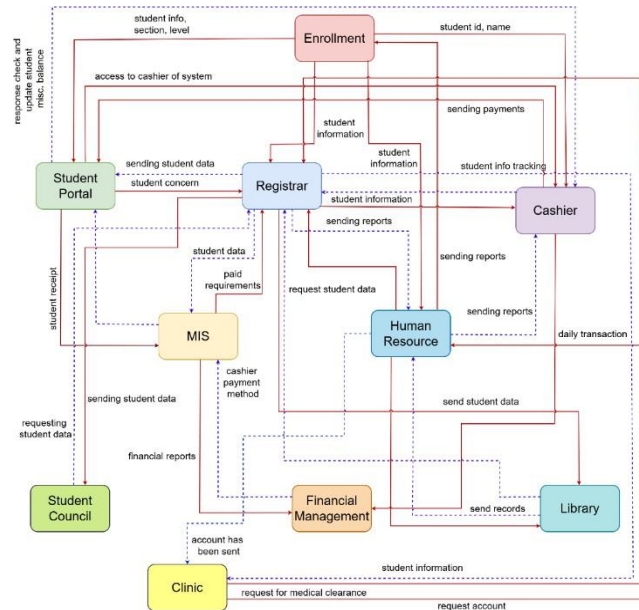


Figure 9: BPA 1

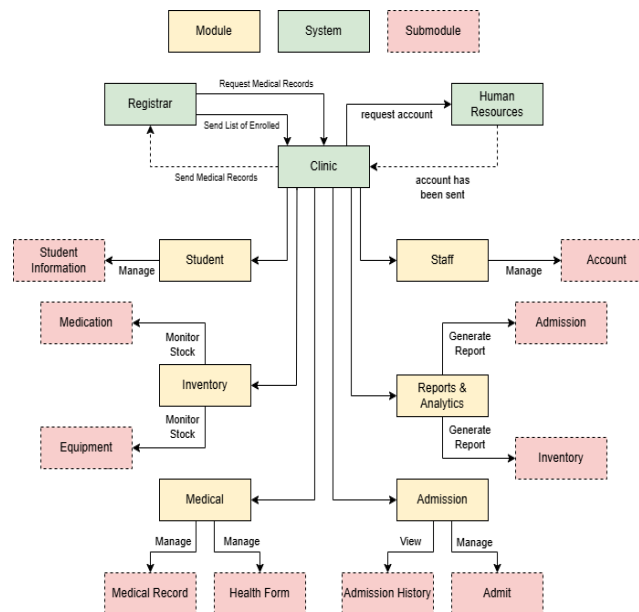


Figure 10: BPA 2

Data Flow Diagram Level 3 – Admin End

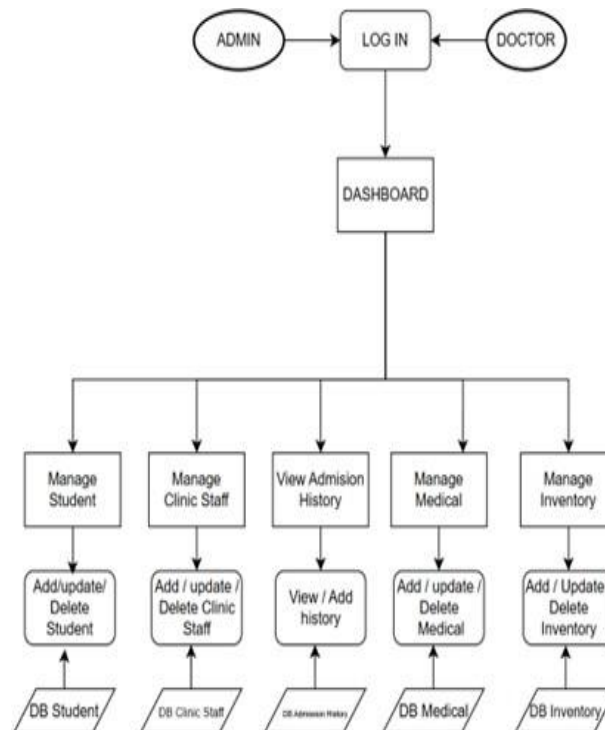


Figure 11: DFD 3- Admin End

This diagram shows how to manage a clinic. The system's key characteristics are:

Admin: The admin user has access to the dashboard, where they can modify Clinic, (add, update, or delete) Student, (add, edit, or delete) Clinic Staff (add, edit, or delete) Medical (add, edit, or delete) Inventory and Can also View and Add Admissions History.

Doctor: The doctor can log in and access all the modules in clinic management system.

Evaluation: The system allows you to add, update, and delete evaluations, which are designed to be used in the clinical process.

Data Flow Diagram Level 3 – User End

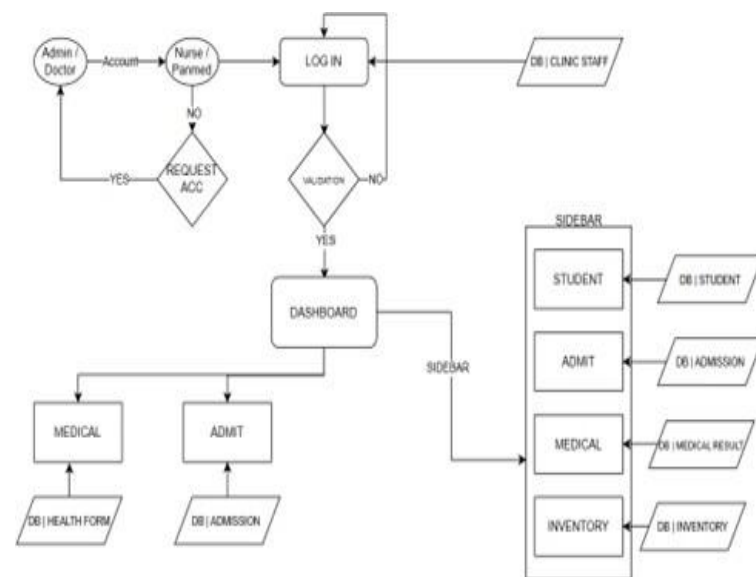


Figure 12: DFD 3 – User – End

This illustration shows simple, straightforward clinical processes. It emphasizes the importance of user authentication and data storage for outcome tracking.

Use Case Diagram

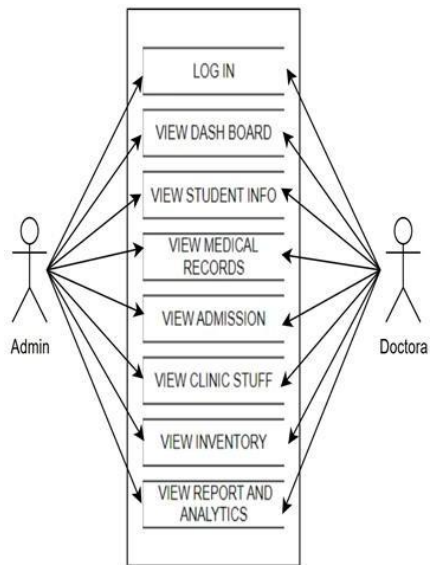


Figure 13: Use – Case Diagram – Doctor

Nurse / PanMed

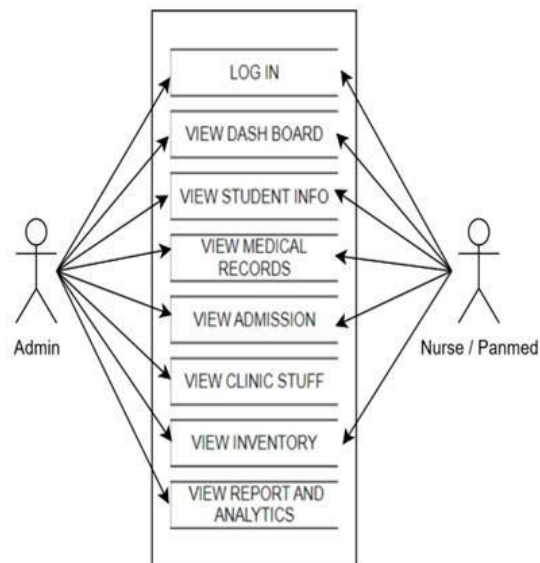


Figure 14: Use Case Diagram – Nurse

Sequence Diagram

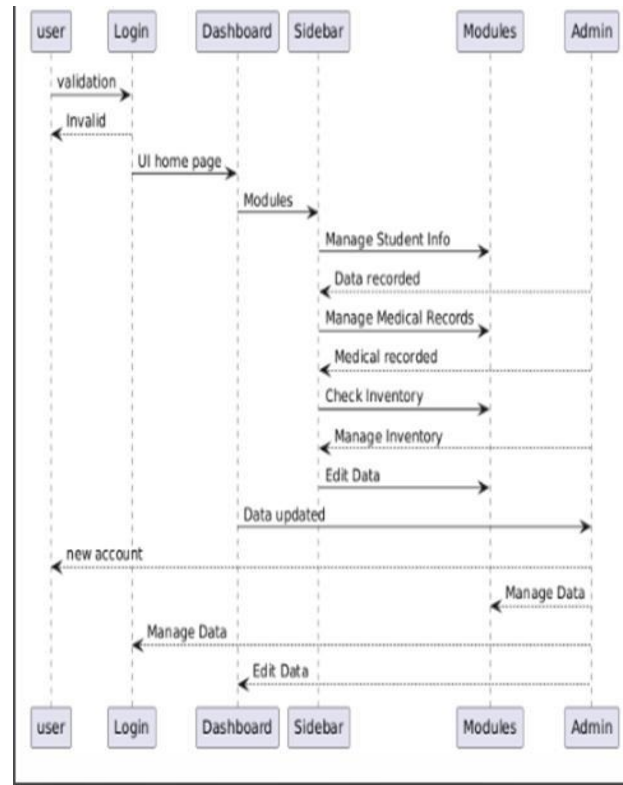


Figure 15: Sequence Diagram

Chapter 4

System Development and Implementation

4.1 Requirement Analysis

The requirement analysis for the Clinic Management System involved identifying key stakeholders and evaluating their levels of power and interest to guide effective system development. Stakeholders were assessed based on their ability to influence the project (power) and their level of concern or involvement (interest). The School Administrator, Project Manager, and Project Sponsor exhibited high power, with the Administrator and Project Manager also showing high interest, indicating the need for close management. Healthcare Providers and the Project Team held medium power but high interest, warranting active engagement and regular updates. Compliance and Finance Officers possessed medium-to-high power with moderate interest, suggesting they should be kept satisfied through periodic communication. Meanwhile, Students demonstrated high interest but low power, requiring accessible and user-friendly system features. Parents had moderate interest and low power, indicating a need for occasional updates. This power-interest assessment provided a foundation for prioritizing stakeholder input and

aligning the system requirements with user needs and organizational goals.

The chart below will be used to establish stakeholders and their levels of power and interest for use on the power/interest chart as part of the stakeholder analysis.

Key	Stakeholders	Power	Interest	Description
A	School Administrator	5	4	Oversees clinic operations and policy alignment; key decision-maker.
B	Finance Officer	4	4	Concerned with budgeting, funding, and financial reporting.
C	Compliance Officer	4	4	Ensures system meets regulatory and legal standards.
D	Healthcare Provider	3	3	Key users; system must support clinical workflow and patient care.
E	Project Sponsor	5	5	Provides funding and strategic support; oversees project value alignment.

F	Project Manager	5	5	Responsible for planning, execution, and delivery of the project.
G	Project Team	3	4	Responsible for system development and implementation.
H	Parents	1	2	Indirect users; interested in student health services and communication.
I	Students	1	3	Primary users of the clinic services; directly impacted by system usability.

Table 11: Stakeholder Identification (Power & Interest)

The Power/Interest Chart for the Clinic Management System is shown below. Each letter represents a distinct stakeholder with a different level of influence and interest, based on the stakeholder identification table.

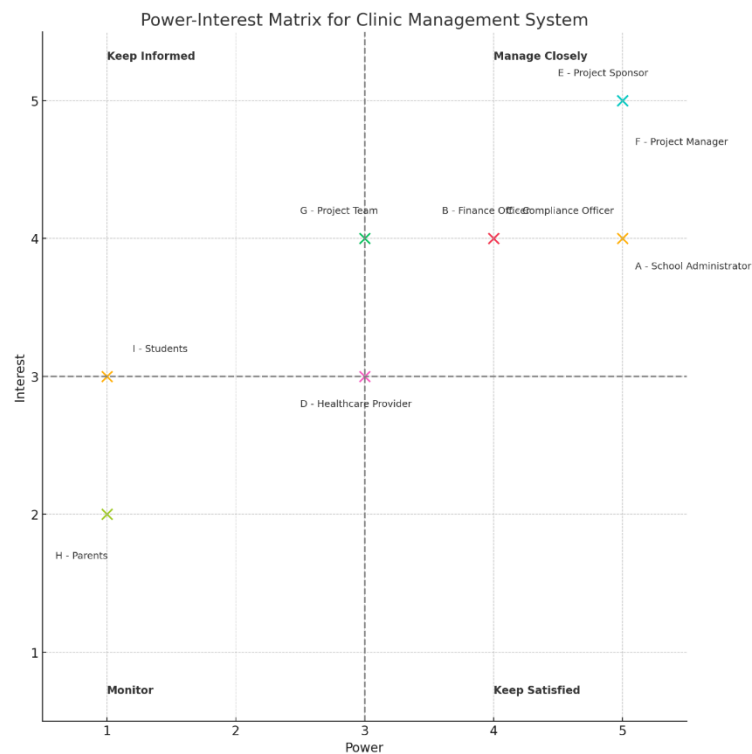


Figure 16: Power Interest Matrix

4.1.1 Stakeholder Identification

This section delineates and examines the principal stakeholders involved in the SMS-I-CLINIC project. Stakeholders comprise individuals or groups who hold a vested interest in, or are influenced by, the outcomes of the project. Their roles encompass a spectrum—from direct system users to those indirectly impacted by the success or shortcomings of the system. The process of stakeholder identification and analysis is not a perfunctory task; rather, it is a critical step in eliciting and aligning stakeholder needs, ensuring the system ultimately delivers value across a variety of perspectives, some of which may inherently conflict.

An inadequate stakeholder analysis can result in considerable project impediments, such as delays, unmet expectations, and eventual project failure. A system designed without sufficient input from relevant stakeholders risks low adoption rates, user dissatisfaction, and failure to meet its intended goals in healthcare delivery.

Hence, comprehensive stakeholder engagement and analysis form the foundational pillars of project success. By recognizing and involving stakeholders early, project teams are better positioned to foresee challenges, reduce risks, and promote collective ownership and collaboration throughout the system development lifecycle.

Key Stakeholders and Their Interests:

Stakeholder Category	Specific Stakeholder Examples	Interests & Expectations	Potential Conflicts
Executive Management	School Administrator	Seeks alignment with institutional policy and operational efficiency; desires	May conflict with clinical or technical teams over priorities or implementation methods.

		influence in decision-making.	
Financial Oversight	Finance Officer	Focused on budgeting, funding allocation, and financial transparency.	May resist funding priorities that lack immediate financial justification.
Regulatory Compliance	Compliance Officer	Ensures legal and regulatory adherence within system functionalities.	May enforce constraints that delay system deployment or conflict with user preferences.
Clinical Users	Healthcare Provider	Requires support for clinical workflows and patient care delivery.	May find system design inadequate if clinical usability is deprioritized.
Strategic Leadership	Project Sponsor	Aims to align project with	Conflicts may arise with project

		organizational strategy and deliver value; provides funding.	team if timelines or deliverables deviate from strategic goals.
Project Leadership	Project Manager	Accountable for planning, execution, and successful project delivery.	Balances competing interests; may face pressure from stakeholders with divergent expectations.
Technical Implementation	Project Team	Focused on system development and implementation fidelity.	May experience friction with stakeholders demanding rapid delivery or additional features.
Indirect Users	Parents	Interested in health services	May have concerns overlooked due to

		and effective communication.	indirect role in system use.
Primary Users	Students	Directly impacted by system usability and service quality.	Their usability needs may be underestimated during technical or policy-focused decisions.

Table 12: Stakeholder Identification

Importance of Stakeholder Analysis:

Stakeholder analysis is crucial because it helps identify and understand the interests, needs, and influence of individuals or groups involved in a project or decision. By recognizing key stakeholders, their expectations, and potential impact, organizations can better manage relationships, mitigate risks, and ensure the project's success. It enables more effective communication, resource allocation, and decision-making, ultimately fostering collaboration and reducing the likelihood of conflicts.

Ensuring the system meets everyone's needs:

Each stakeholder group has different priorities. Patients want a user-friendly system that provides accurate information and efficient communication. Doctors need reliable tools for managing patient records and accessing relevant information. Hospital administrators are concerned with cost-effectiveness, security, and compliance. By understanding these diverse needs, developers can design a system that effectively serves all stakeholders.

Preventing project delays and conflicts: Failing to consider stakeholder requirements can lead to disagreements and conflict during development. For example, if developers prioritize ease of use for patients without considering the needs of doctors, the system might lack features essential for their workflow, causing frustration and delays.

Achieving project goals and objectives: Stakeholder analysis helps align project goals with the expectations of the stakeholders involved. By understanding their interests and concerns, the project team can ensure that the final product meets their needs and achieves the desired outcomes. For example, hospital administrators might be concerned about the cost of implementing the system. Understanding this concern allows the project team to prioritize cost-effective solutions and ensure the project aligns with the hospital's budget.

Improving communication and collaboration: Stakeholder analysis facilitates open and effective communication throughout the project lifecycle. By understanding each stakeholder's perspective, the project team can tailor communication to their specific needs and concerns, ensuring everyone is informed and involved in the decision-making process.

Managing risk and mitigating potential issues: Stakeholder analysis helps identify potential risks and challenges associated with each stakeholder group. For example, if a regulatory body has specific requirements for data security, understanding these requirements early on allows the project team to design the system, accordingly, minimizing the risk of non-compliance

4.1.2 Requirements Gathering Techniques

To capture the diverse and sometimes conflicting needs of stakeholders for the SMS-I-CLINIC project, a multi-faceted approach to requirements gathering was employed. This ensured a comprehensive understanding of the requirements and minimized the risk of overlooking critical details. The techniques used included surveys, interviews, workshops, observations, and document analysis.

Each technique played a unique role in gathering different types of information and addressing various aspects of the project.

1. Surveys: Online surveys were distributed to a broad range of stakeholders, including patients, doctors, nurses, and administrative staff. These surveys collected both quantitative data (e.g., frequency of SMS usage, satisfaction ratings) and qualitative data (e.g., open ended feedback on system features and usability). The surveys were designed to be concise and easy to complete, ensuring a high response rate. The data collected was analyzed using statistical methods to identify trends and patterns.

2. Interviews: Semi-structured interviews were conducted with key stakeholders, including representatives from each stakeholder group. These interviews provided a deeper understanding of individual needs, workflows, and priorities. Open-ended questions were used to encourage detailed responses and uncover unanticipated requirements. Interview transcripts were meticulously documented and analyzed to identify recurring themes and concerns.

3. Observation: Direct observation of existing workflows within the clinic provided valuable insights into current processes and potential areas for improvement. This involved observing staff interactions, document flows, and communication patterns. The observations helped identify inefficiencies and pain points that could be addressed by the new SMS-based system. Detailed notes were taken during the observations, and these notes were subsequently analyzed to identify recurring issues and opportunities for improvement.

4. Document Analysis: Existing clinic policies, procedures, and communication protocols were reviewed to ensure that the new system aligned with existing regulations and standards. This involved analyzing relevant documents to identify any constraints or requirements that might impact the system's design. The analysis helped ensure that the system would be compliant with all relevant regulations and would integrate seamlessly with existing clinic processes.

4.1.3 User Stories and Use Cases

A user story is a brief, straightforward account of a feature or functionality written from the viewpoint of the end user. A use case is a thorough explanation of how a system communicates with an outside party (a user or another system) in order to achieve a particular objective.

4.1.4 Functional Requirements for Integration

I. Functional Requirements for Integration:

These requirements specify the features and behaviors necessary for successful integration.

A. Data Exchange:

Appointment Scheduling: The system must allow for seamless integration with the clinic's appointment scheduling system. This includes the ability to send appointment reminders, confirmations, and updates via SMS. The data exchange must be bi-directional, allowing updates from the SMS system to be reflected in the appointment scheduling system and vice versa.

Patient Data Exchange: The system must securely exchange relevant patient data (e.g., name, contact information, medical history) with the clinic's Electronic Health Record (EHR) system. This exchange must adhere to relevant data privacy regulations. The specific data fields to be exchanged must be clearly defined.

Test Result Integration: The system must integrate with the laboratory information system (LIS) to allow for the secure delivery of test results to patients via SMS. This requires secure transfer of results data, including appropriate error handling for failed transmissions.

Prescription Management: The system should integrate with the pharmacy system (if applicable) to allow for the secure transmission of prescription information to patients. This requires secure data transfer and confirmation of successful delivery.

B. System Interfaces:

API Integration: The SMS-I-CLINIC system must provide well-documented APIs for integration with other systems. These APIs should be secure and support various communication protocols. **Data Mapping:** A clear data mapping strategy must be defined to ensure consistent data representation across different systems. This includes handling data format conversions and ensuring data integrity.

Error Handling: The system must provide robust error handling mechanisms to manage data transmission failures, system errors, and other exceptions. Error messages must be informative and actionable.

II. Non-functional Requirements for Integration

These requirements define the performance, security, and usability characteristics of the integrated system.

A. Performance:

Response Time: The system must respond to requests within an acceptable timeframe (e.g., under 2 seconds).

Scalability: The system must be scalable to accommodate an increasing number of users and messages.

Reliability: The system must be highly reliable and available, minimizing downtime and ensuring data integrity.

B. Security:

Data Encryption: All data exchanged between systems must be encrypted using industry-standard encryption protocols.

Authentication and Authorization: Secure authentication and authorization mechanisms must be implemented to control access to sensitive data.

Data Integrity: Mechanisms must be in place to ensure the integrity of data exchanged between systems.

C. Usability:

User Interface: The system's user interface must be intuitive and easy to use for all stakeholders.

Documentation: Comprehensive documentation must be provided for system administrators and users.

4.2 Business Process Architecture

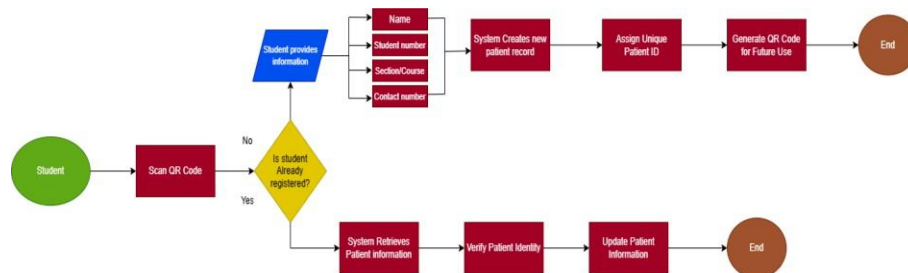
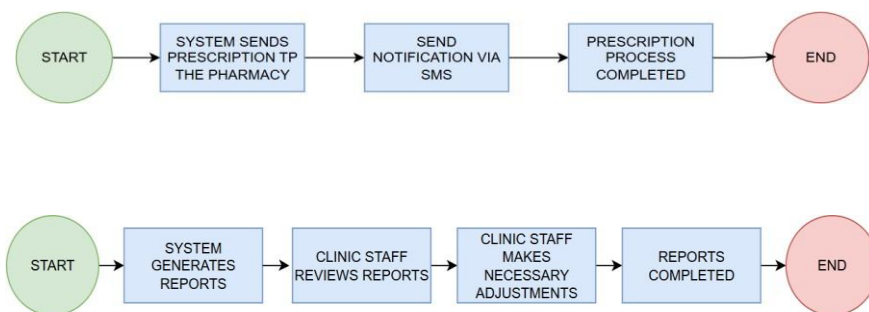


Figure 17 : Student Business Process Architecture



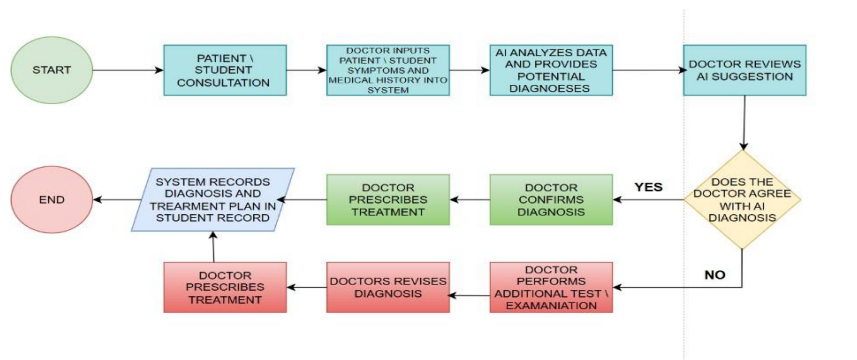


Figure 18: Admin Business Process Architecture

4.2.3 Alignment of Integrated Systems with Business Processes

This document details the alignment of the SMS I (AI-Assisted Diagnosis and QR Code Integration for Streamlined Clinic Management System) with business processes at Bestlink College of the Philippines, specifically focusing on the integration points with the Registrar and HR systems.

Alignment with Business Processes

The SMS I system aligns with Bestlink's business processes by acting as a central hub for student and employee health information. This integration reduces data redundancy, improves data accuracy, and streamlines workflows across various departments.

Specific Integration Points –

Registrar Integration:

A. Student Information: The system integrates with the Registrar's database to access student information (ID, name, program, contact details).

This eliminates the need for manual data entry in the clinic system, ensuring data accuracy and consistency. Integration points include:

Student enrollment: Automatic updates to the SMS I system when a new student enrolls.

Student record updates: Real-time updates of student contact information and program details.

Graduation / withdrawal: Automatic removal of students from the SMS I system upon graduation or withdrawal.

B. Appointment Scheduling: The system allows appointment scheduling through an online portal linked to the Registrar's system, enabling students to easily book appointments.

HR Integration:

A. Employee Information: The system integrates with the HR database to access employee information (ID, name, department, contact details).

This ensures accurate record- keeping and facilitates efficient health management for staff. Integration points include:

Employee onboarding: Automatic addition of new employees to the SMS I system.

Employee record updates: Real-time updates of employee contact information and departmental changes.

Employee termination: Automatic removal of employees from the SMS I system upon termination.

B. Health Records Management: The system allows HR to access aggregated health data for reporting and planning purposes (e.g., tracking common illnesses among staff).

Support, Improvement, and Automation of Business Processes:

Support: SMS I provides a centralized platform for managing student and employee health records, supporting the Registrar and HR departments in their respective functions.

Improvement: The system improves data accuracy by eliminating manual data entry and reducing the risk of human error. It also enhances efficiency by automating tasks such as appointment scheduling and record updates.

Automation: The system automates the transfer of student and employee information between the clinic system and the Registrar/HR databases, reducing manual effort and improving data consistency. The QR code integration further automates patient check-in and data capture.

Benefits to the Organization:

Improved Efficiency: Streamlined workflows reduce the administrative burden on the Registrar and HR departments.

Reduced Errors: Automated data transfer minimizes data entry errors and ensures data consistency across systems.

Enhanced Decision-Making: Aggregated health data allows for better informed decisions regarding health resource allocation and preventative measures.

Improved Service Delivery: Easier appointment scheduling and efficient record management improve the overall patient experience.

Cost Savings: Automation of tasks reduces labor costs and improves resource utilization.

Enhanced Data Security: Centralized data management improves data security and privacy.

4.2.4 Business Process Improvement

This document outlines opportunities for business process improvement within Bestlink College of the Philippines' clinic management system (SMS I), focusing on streamlining processes, reducing manual effort, and enhancing overall workflow. These improvements directly contribute to the project's goals and align with the organization's objectives for efficient healthcare delivery.

1. Streamlining Patient Information Management -

Before:

Manual Record Keeping: Clinic staff relied on paper-based records, leading to time-consuming data entry, potential errors, and difficulties in accessing information.

Data Redundancy: Information was often duplicated across different forms and systems, increasing the risk of inconsistencies and errors.

Limited Accessibility: Accessing patient records required physical presence at the clinic, hindering timely access for healthcare providers and administrators.

After:

QR Code Integration: The SMS I system utilizes QR codes to digitize patient records, enabling quick and easy access to information through mobile devices.

Centralized Database: Patient data is stored in a centralized database, eliminating redundancy and ensuring data consistency across the system.

Real-Time Access: Healthcare providers and administrators can access patient records remotely, enabling faster decision-making and improved coordination of care.

Impact:

Reduced Manual Effort: Data entry is automated, freeing up clinic staff to focus on patient care.

Improved Accuracy: Centralized data storage and real-time updates minimize errors and inconsistencies.

Enhanced Efficiency: Faster access to information streamlines patient care, appointment scheduling, and reporting processes.

Contributed to Organizational Goals:

Enhanced Healthcare Services: The improved efficiency and accuracy of patient information management contribute to delivering high-quality healthcare services to students, faculty, and staff.

Data-Driven Decision Making: The centralized database enables the collection of valuable data for analyzing health trends and making informed decisions regarding resource allocation and preventative measures.

2. Automating Appointment Scheduling -**Before:**

Manual Scheduling: Appointments were scheduled through phone calls or in-person visits, leading to potential scheduling conflicts and delays.

Limited Flexibility: Students had limited options for scheduling appointments, often needing to adjust their schedules to accommodate clinic hours.

After:

Online Appointment Scheduling: The SMS I system integrates with the Registrar's system, enabling students to schedule appointments online through a user-friendly portal.

Real-Time Availability: The system displays real-time availability, allowing students to choose convenient appointment slots.

Impact:

Reduced Manual Effort: Clinic staff are relieved of the burden of managing appointments through phone calls and manual scheduling.

Increased Flexibility: Students can schedule appointments at their convenience, improving accessibility and reducing disruptions to their academic schedules.

Improved Efficiency: The system automates scheduling processes, minimizing conflicts and ensuring efficient utilization of clinic resources.

Contribution to Project Goals:

Streamlined Clinic Management: The automated appointment scheduling system contributes to the project's goal of improving efficiency and effectiveness within the clinic.

Enhanced Patient Experience: The online appointment scheduling system provides a more convenient and user- friendly experience for students.

Contribution to Organizational Goals:

Improved Student Satisfaction: The increased flexibility and convenience of online appointment scheduling contribute to a positive student experience.

Enhanced Operational Efficiency: The automated scheduling system improves the overall efficiency of the clinic, freeing up staff time and resources.

3. AI-Assisted Diagnosis -**Before:**

Manual Diagnosis: Diagnoses relied heavily on the physician's experience and knowledge, potentially leading to missed diagnoses or delays in treatment.

Limited Data Analysis: Physicians had limited access to comprehensive patient data, making it challenging to identify patterns and trends.

After:

AI-Powered Insights: The SMS I system incorporates AI algorithms to analyze patient data, providing physicians with insights and potential diagnoses based on medical history, symptoms, and relevant research.

Enhanced Accuracy: The AI system can identify patterns and potential diagnoses that might be overlooked through manual analysis, improving diagnostic accuracy and reducing the risk of misdiagnosis.

Impact:

Improved Diagnostic Accuracy: The AI-assisted diagnosis system enhances the accuracy and effectiveness of medical diagnoses.

Faster Diagnosis: The system speeds up the diagnostic process, allowing for quicker treatment and improved patient outcomes.

Enhanced Physician Support: The AI system provides valuable support to physicians, empowering them to make more informed decisions and improve patient care.

Contribution to Project Goals:

Improved Diagnostic Accuracy: The AI-assisted diagnosis system directly contributes to the project's goal of enhancing diagnostic accuracy within the clinic.

Streamlined Clinic Management: The system automates aspects of the diagnostic process, improving efficiency and freeing up physician time for patient care.

Contribution to Organizational Goals:

Enhanced Healthcare Quality: The improved accuracy and efficiency of diagnosis contributes to delivering high-quality healthcare services to students, faculty, and staff.

Cost Reduction: The AI-assisted diagnosis system can potentially reduce the need for expensive and time-consuming diagnostic tests, leading to cost savings for the institution.

4. Data-Driven Reporting and Analytics -**Before:**

Manual Reporting: Clinic staff manually generated reports, which was time-consuming, prone to errors, and limited in terms of insights.

Limited Data Analysis: The lack of comprehensive data analysis hindered the ability to identify health trends and make informed decisions regarding resource allocation and preventive measures.

After:

Automated Reporting: The SMS I system automates the generation of reports, providing real-time insights into clinic operations, patient demographics, health trends, and resource utilization.

Advanced Analytics: The system incorporates data analytics tools to identify patterns, trends, and potential areas for improvement, enabling data-driven decision-making.

Impact:

Improved Efficiency: Automated reporting saves clinic staff time and effort, allowing them to focus on patient care.

Enhanced Insights: The system provides valuable insights into clinic operations, enabling administrators to make informed decisions regarding resource allocation, preventative measures, and overall healthcare strategy.

Data-Driven Decision Making: The system empowers administrators to make evidence-based decisions based on real-time data analysis.

Contribution to Project Goals:

Streamlined Clinic Management: The automated reporting system contributes to the project's goal of improving efficiency and effectiveness within the clinic.

Improved Healthcare Quality: The data-driven insights provided by the system enable administrators to make informed decisions that enhance the quality of healthcare services.

Contribution to Organizational Goals:

Cost Reduction: The system's insight into resource utilization can help optimize resource allocation and reduce unnecessary expenses.

Improved Public Health: The ability to identify health trends and patterns allows the institution to implement proactive measures to prevent the spread of illness and promote overall student health.

4.3 Application Architecture

The Application Architecture component of our SMS I project delves into the design and structure of software applications that make up the integrated clinic management system. This section outlines the key components, communication patterns, and design principles that guide the development of the application architecture.

4.3.1 Components of Application Architecture

The application architecture of our SMS I system comprises several key components, each serving a specific purpose in the overall functionality:

1. User Interface (UI): The UI component is the front-end interface that allows users to interact with the system.

Including:

Web Interface: A web-based interface accessible through a browser, allowing students, staff, and administrators to access the system from various devices.

Mobile Application: A dedicated mobile app for students to access their medical records, schedule appointments, and view health information.

User Dashboards: Personalized dashboards for different user roles (students, doctors, nurses, administrators) providing relevant information and functionalities.

2. Business Logic: This component houses the core logic of the system.

Including:

Patient Management: Managing patient records, medical history, allergies, and contact information.

Appointment Scheduling: Managing appointment bookings, cancellations, and reminders.

AI-Assisted Diagnosis: Processing patient data and providing AI-powered insights and potential diagnoses.

Prescription Management: Generating, managing, and tracking prescriptions.

Inventory Management: Tracking medical supplies and equipment.

Reporting and Analytics: Generating reports on clinical activity, patient demographics, health trends, and resource utilization.

3. Database Management System (DBMS): DBMS is responsible for storing and managing all data related to the clinic, including: Patient Data: Student and staff medical records, health history, allergies, and contact information.

Appointment Data: Appointment schedules, cancellations, and reminders.

Medical Data: Diagnostic results, prescriptions, and treatment plans.

Inventory Data: Medical supplies, equipment, and their stock levels.

User Data: User accounts, roles, and permissions.

4. Application Programming Interface (API): APIs act as the communication bridge between different components of the system, enabling data exchange and seamless integration.

This includes:

Registrar Integration API: Connecting with the Registrar system to access and update student information.

HR Integration API: Connecting with the HR system to access and update employee information.

AI Service API: Connecting with the AI-powered diagnosis service to process patient data and receive insights.

External API Integration: Potentially connected with other external services, such as pharmacy systems for prescription management.

5. QR Code Integration: This component handles the generation, storage, and scanning of QR codes for:

Patient Record Access: Students can access their medical records by scanning their unique QR code.

Appointment Scheduling: Students can schedule appointments by scanning a QR code displayed on the appointment calendar.

Data Capture: Clinic staff can scan QR codes to quickly capture patient information and update medical records.

6. Security and Authentication: This component ensures secure access to the system and protects sensitive medical data:

User Authentication: Secure logins with password management and multi-factor authentication.

Role-Based Access Control (RBAC): Restricting user access to specific functionalities based on their roles (student, doctor, nurse, administrator).

Data Encryption: Encrypting sensitive data both at rest and in transit to protect it from unauthorized access.

Audit Trails: Tracking user activity to identify and investigate any unauthorized access or data manipulation.

Communication Patterns

The SMS I system employs a combination of communication patterns:

Client-Server Architecture: The UI (web interface, mobile app) interacts with the server-side logic, which processes requests and interacts with the database and other services.

API-Based Communication: Different components of the system communicate through APIs, enabling data exchange and integration.

Message Queues: For asynchronous communication, message queues can be used to handle tasks like sending notifications or processing large data sets.

Design Principles

The following design principles guide the development of the SMS I application architecture:

Modularity: The system is designed as a collection of independent modules (services) that can be developed, deployed, and scaled independently. This improves maintainability, scalability, and flexibility.

Scalability: The system is designed to handle growing data volumes and user traffic, ensuring performance and availability.

Security: Security is paramount, with robust authentication, authorization, and data encryption mechanisms.

Usability: The user interface is designed for ease of use and accessibility for all user roles.

Maintainability: The system is designed for easy maintenance and updates, allowing for continuous improvement and adaptation.

4.3.2 Application Architecture Diagram

We develop architecture diagrams that show the connections and interactions between the components to clearly represent the application architecture. These diagrams help in understanding how data flows, how user interactions are processed, and how different modules communicate with each other.

Key Features:

Modular Design: Each service handles a distinct function.

QR Code Integration: Services like Student Information and Medical Records support QR code generation and scanning for streamlined access.

AI Module: Integrated into the diagnostic process as a service within the architecture to enhance clinical decision-making.

Cloud Deployment: Uses CI/CD pipelines, containers (Docker), and infrastructure-as-code for deployment.

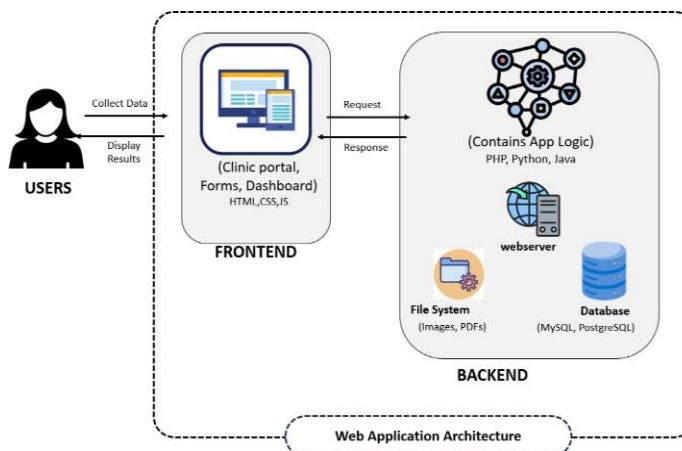


Figure 19: Web Application Architecture

4.3.3 Integration of Software Modules

In the SMS I clinic management system, various software modules need to work together seamlessly to provide a cohesive and efficient healthcare solution. Integration is achieved through a combination of well-defined Application Programming Interfaces (APIs), data synchronization mechanisms, and established communication protocols.

The core modules Authentication, Patient Management, Appointment Scheduling, AI Diagnosis, Prescription Management, Inventory Management, Reporting & Analytics, QR Code Management, and Registrar/HR Integration interact in the following ways:

1. Authentication Module:

This module acts as the central access point, verifying user credentials and granting access to other modules based on user roles (student, doctor, nurse, administrator).

2. Data Flow and Integration Points:

Patient Management Module: This module interacts with the Authentication Module to verify user access before allowing updates to patient records.

Appointment Scheduling Module: This module interacts with the Authentication Module for user authentication, the Patient Management Module to access patient details, and the Reporting & Analytics Module to provide appointment data for reporting and analysis.

AI Diagnosis Module: This module receives patient data from the Patient Management Module and communicates with an external AI service (via API) to generate diagnostic suggestions. The results are then sent back to the Patient Management Module for integration into the patient's record.

Prescription Management Module: This module interacts with the Authentication Module for access control, the Patient Management Module to access patient allergies and medical history, and the Inventory Management Module to check medication availability.

Inventory Management Module: This module interacts with the Authentication Module for access control, the Prescription Management Module to track medication usage, and the Reporting & Analytics Module to generate inventory reports.

Reporting & Analytics Module: This module collects data from various modules (Appointment Scheduling, Patient Management, Inventory Management) to generate reports and perform data analysis.

QR Code Management Module: This module generates and manages QR codes linked to patient records and appointments. It interacts with the Authentication Module for security, the Patient Management Module to link QR codes to patient data, and the Appointment Scheduling Module to link QR codes to appointments.

Registrar/HR Integration Module: This module interacts with the Registrar and HR systems (via APIs) to exchange data, such as student and staff information. It uses secure API communication protocols (REST) and data exchange formats (JSON, XML) to ensure data consistency between the SMS I system and the external systems.

3. Mechanisms and Patterns for Cohesive Operation:

APIs: Well-defined APIs facilitate communication and data exchange between modules.

Data Synchronization: Mechanisms like database triggers and scheduled tasks ensure data consistency across modules.

Message Queues: Asynchronous communication patterns (using message queues or websockets) are used for tasks like appointment updates and notifications.

Event-Driven Architecture: Modules can trigger events to notify other modules of changes.

Data Validation: Input validation and data integrity checks are implemented at multiple points to ensure data quality.

Error Handling: Robust error handling and logging mechanisms are implemented to manage and resolve issues.

4.3.4 Communication and Interaction Patterns

Efficient communication and interaction patterns are essential for the SMS I clinic management system's performance and user experience.

The system employs several key patterns and strategies to ensure seamless data exchange and effective response handling:

1. Synchronous Request-Response: Many interactions within the system follow a synchronous request-response pattern. For example:

Appointment Scheduling: A user (via the web or mobile interface) requests an appointment. The frontend sends a request to the Appointment Scheduling module.

AI-Assisted Diagnosis: Patient Management module sends patient data to the AI Diagnosis module, which in turn sends a request to the external AI service.

Data Retrieval: When a doctor needs to access a patient's record, a request is sent.

Management module: which retrieves the data from the database and returns it to the requesting module (or directly to the user interface).

2. Asynchronous Communication (Message Queues): For tasks that don't require immediate responses, asynchronous communication using message queues is employed:

Appointment Notifications: When an appointment is scheduled or changed, a message is placed in a queue.

Background Tasks: Data synchronization between the system and external systems (Registrar, HR) can be handled asynchronously.

3. API-Based Communication: The system uses well-defined APIs for communication between different modules and external systems:

Module-to-Module Communication: Modules interact primarily via APIs, ensuring loose coupling and promoting modularity.

External System Integration: APIs are used to communicate with the AI service, Registrar system, and HR system.

4. Event-Driven Architecture: The system incorporates aspects of an event-driven architecture:

Appointment Updates: When an appointment is scheduled, canceled, or modified, an event is triggered.

Inventory Alerts: When inventory levels fall below a certain threshold, an event is triggered, alerting relevant personnel.

5. Real-Time Communication (WebSockets): For scenarios requiring real-time updates, WebSockets can be used:

Live Monitoring: Doctors or administrators might use a dashboard that receives real-time updates on patient status or appointment changes via Web Sockets.

By combining these communication and interaction patterns, the SMS I system ensures efficient data exchange, robust error handling, and responsive user experience. The choice of synchronous or asynchronous communication depends on the specific needs of the interaction, balancing the need for immediate responses with the importance of system responsiveness and resource efficiency.

4.4 Data Architecture

4.4.1 Data Sources and Types

The SMS I clinic management system utilizes several data sources to store and manage information. Identifying these sources is crucial for understanding the system's data flow, ensuring data consistency, and facilitating integration with external systems.

1. Clinic Database (MySQL):

Type: Structured (relational database) Data Stored:

Patient Information: Student and staff details (ID, name, contact information, allergies, medical history).

Appointment Data: Appointment schedules, cancellations, reminders.

Medical Data: Diagnostic results, prescriptions, treatment plans.

Inventory Data: Medical supplies, equipment, and their stock levels.

User Accounts: User information, roles, and permissions.

2. External APIs:

Type: Semi-structured (JSON or XML) Data Sources:

AI Service: Used for AI-assisted diagnosis. Exchanges data in JSON format, including patient data and AI-generated insights.

Registrar System: Used to access and update student information. Exchanges data in JSON or XML format, including student IDs, names, contact details, enrollment status, etc.

HR System: Used to access and update staff information. Exchanges data in JSON or XML format, including employee IDs, names, contact details, department, etc.

3. File System:

Type: Unstructured (documents, images)

Medical Documents: Scans of medical records, reports, images.

System Files: Configuration files, logs, and other system-related documents.

Importance of Identifying Data Sources:

Data Flow Understanding: Knowing the data sources helps understand how data moves through the system, identifying potential bottlenecks or redundancies.

Data Consistency: Ensuring data consistency across different sources is crucial for accurate reporting and decision-making.

Integration with External Systems: Identifying the data sources and their formats allows for seamless integration with external systems like the Registrar and HR.

Security and Privacy: Understanding data sources is essential for implementing robust security measures and adhering to data privacy regulations.

4.4.2 Data Flow Diagram

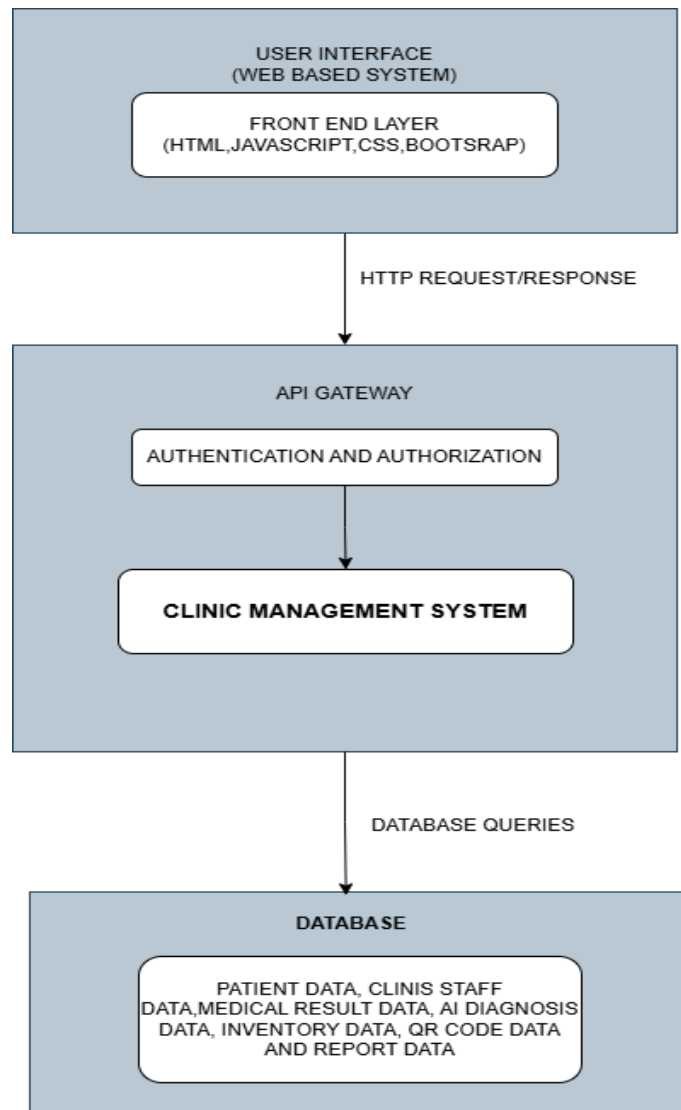


Figure 20: Process Diagram

4.4.3 Data Storage Management

The SMS I clinic management system employs a robust data storage and management architecture to ensure data integrity, consistency, and security. This section details the database system, data organization, indexing, security measures, CRUD operations, and data synchronization mechanisms.

Data Organization and Indexing

Data is organized into relational tables, normalized to minimize redundancy and improve data integrity. Each table represents a specific entity (e.g., Patients, Appointments, Prescriptions, Inventory). Relationships between tables are defined using foreign keys to ensure data consistency.

Data Security

Data security is paramount. The system implements the following security measures:

Access Control: Role-based access control (RBAC) restricts access to data based on user roles.

Data Encryption: Sensitive data (e.g., patient medical records) is encrypted both at rest.

Regular Backups: Regular database backups are performed to protect against data loss.

Data Validation: Input validation is implemented to prevent invalid or malicious data from entering the database.

Auditing: Database activities are audited to track changes and identify potential security breaches.

CRUD Operations

The system supports standard CRUD (Create, Read, Update, Delete) operations for all data entities. These operations are typically performed through APIs and are subject to access control mechanisms.

Create: New records are created using appropriate API calls, with data validation ensuring data integrity.

Read: Data is retrieved using queries optimized by indexing. Access control ensures that only authorized users can access specific data.

Update: Existing records are updated using API calls, with data validation and auditing to track changes.

Delete: Records are deleted using API calls, with appropriate safeguards to prevent accidental data loss.

Data Synchronization between Systems

The SMS I system synchronizes data with external systems (Registrar, HR) to maintain data consistency and integrity. The synchronization process involves the following steps:

1. Data Mapping: Data fields from the SMS I database are mapped to corresponding fields in the external systems.

2. Data Transformation: Data may need to be transformed to ensure compatibility between systems (e.g., data type conversions, format changes).

3. Error Handling: The synchronization process includes robust error handling to manage and resolve any issues that may arise during data exchange.

4. Conflict Resolution: Mechanisms are in place to resolve data conflicts that may occur if changes are made in multiple systems simultaneously.

4.4.4. Data Synchronization Across Systems

Data synchronization is critical to maintaining data consistency across the various modules of the SMS I clinic management system. To achieve this, we have implemented a robust data synchronization strategy that employs a combination of real-time and scheduled batch processes.

Real-Time Synchronization

For modules requiring immediate data consistency, real-time synchronization is employed. This is achieved using (WebSockets) to propagate data changes instantly. Examples of modules using real-time synchronization include:

Appointment Scheduling Module: When an appointment is scheduled, canceled, or modified, the change is immediately reflected in the database and propagated to other relevant modules (Patient Management, Reporting & Analytics) via real-time messaging.

Inventory Management Module: Real-time updates to inventory levels are crucial for efficient management of medical supplies.

Patient Status Updates: In cases where patient status changes (e.g., vital signs, test results), real-time updates are sent to relevant personnel (doctors, nurses) through notifications or dashboards.

Scheduled Batch Processes

For data synchronization that does not require immediate updates, scheduled batch processes are used. These processes run at regular intervals (e.g., nightly, hourly) to update data between modules and external systems. This approach is particularly useful for:

Synchronization with External Systems: Data synchronization with the Registrar system (for student information) and the HR system (for staff information) is typically done using scheduled batch processes.

Reporting and Analytics Data Aggregation: The Reporting & Analytics module might aggregate data from various modules on a scheduled basis to generate reports and perform analysis.

Large Data Transfers: If large datasets need to be transferred or processed, scheduled batch processes are more efficient than real-time synchronization.

Data Synchronization Mechanisms:

The specific mechanisms used for data synchronization include:

Database Triggers: Database triggers are used to automatically update related tables when changes are made in one table.

Message Queues: Message queues are used for asynchronous communication between modules.

APIs: APIs are used to communicate with external systems (Registrar, HR).

ETL Processes: Extract, Transform, Load (ETL) processes are used for scheduled batch synchronization with external systems.

Data Consistency and Integrity:

To maintain data consistency and integrity, the following measures are implemented:

Data Validation: Data validation is performed at multiple points in the system to ensure that only valid data is stored.

Error Handling: Robust error handling and logging mechanisms are implemented to identify and resolve synchronization issues.

4.5 Technology Architecture

4.5.1 Technology Stack and Infrastructure

The technology architecture of our HRM system is built on a well-defined technology stack and infrastructure. We utilized a combination of open-source and industry-standard technologies to ensure scalability, security, and performance. Our technology stack includes:

1. Operating System: We deployed the system on Linux- based servers to ensure stability and security.

2. Programming Languages: Python and JavaScript were chosen as the primary programming language for backend development, while html and CSS were used for frontend development.

3. Database Management: PhpMyadmin, a powerful and open-source relational database, is used to store and manage data.

4. Web Server: Gunicorn and Nginx are used to serve the application and handle HTTP requests.

4.5.2 Network Topology and Configuration

The SMS I clinic management system's network architecture is designed to ensure reliable and secure communication between clients (users accessing the system via web or mobile interfaces) and the system's servers. The chosen topology is a client-server architecture, offering several advantages for a healthcare system.

Network Topology

The system employs a client-server architecture. Clients (users' devices) connect to central servers hosting the application and database.

This topology is chosen for its:

Centralized Management: Easier administration and control of system resources and security.

Scalability: The system can be scaled by adding more servers to handle increased load.

Security: Centralized security measures can be implemented more effectively.

Data Management: Centralized database management simplifies data consistency and integrity.

Network Protocols and Communication Models:

HTTP/HTTPS: The primary communication protocol used between clients and the servers is HTTP/HTTPS.

TCP/IP: The underlying network protocol is TCP/IP, providing reliable and ordered data transmission.

Communication Model: The system primarily uses a request- response.

Security Measures:

Security is a critical aspect of network architecture. The following security measures are implemented:

Firewall: A firewall protects the network from unauthorized access.

Intrusion Detection/Prevention System (IDS/IPS): Monitors network traffic for malicious activity.

Virtual Private Network (VPN): For remote access, a VPN provides secure connections.

Secure Socket Layer/Transport Layer Security (SSL/TLS):

HTTPS encrypts communication between clients and servers.

4.5.3. Software Technologies

We employed various software technologies to enhance the functionality and user experience of the CMS system:

1. Django REST framework: This powerful framework was used for building robust APIs that allowed seamless data interaction between the frontend and backend.

2. React.js: A popular JavaScript library, React.js has made creating dynamic and responsive user interfaces easy.

3. Redux: This state management library is used to manage complex state and data flows in the application.

4. Docker: containers are used to isolate and deploy applications, ensuring consistency between development and production environments.

5. Jenkins: This CI/CD tool is used for automated testing, builds, and deployments, streamlining our development and deployment processes.

4.5.4 Scalability and Performance Considerations

The SMS I clinic management system is designed with scalability and performance as primary considerations. This section details the strategies employed to handle increased load (users and data volume) and optimize system responsiveness.

Scalability:

The system's architecture is designed for scalability using several key strategies:

Microservices Architecture: The backend is built using a microservices architecture, where each module (Patient Management, Appointment Scheduling, etc.) is a self-contained unit.

Horizontal Scaling: The system can be scaled horizontally by adding more servers to handle increased load.

Cloud-Based Infrastructure: The system can be deployed on a cloud platform ([Specify Cloud Provider from your document (Google Cloud)], enabling easy scaling of resources (compute, storage, database) as needed.

Performance Optimization:

Several strategies are employed to optimize system performance:

Caching: A caching mechanism (Memcached) is used to store frequently accessed data in memory.

Indexing: Appropriate indexes are created on database tables to optimize query performance. Indexes speed up data retrieval by allowing the database to quickly locate specific records.

Code Optimization: The code is optimized for performance using techniques like efficient algorithms, minimizing database queries, and using appropriate data structures.

Asynchronous Processing: Message queues are used for asynchronous processing of tasks that do not require immediate responses (e.g., sending notifications, background data synchronization).

Load Balancing: Load balancing distributes client requests across multiple servers to prevent overload on any single server. This ensures that the system remains responsive even under high load.

4.6 Development Process

4.6.1 Agile Scrum Roles and Responsibilities

Name	Role	Responsibilities
Joseph Oliver P. Banac	Project Manager	Develop project plans, schedules, and budgets. Facilitate team meetings and stakeholder updates. Manage risk assessments and mitigation strategies. Ensure project deliverables align with goals.
Angelo B. Abargos	Developer	Front-End: Create responsive UI components, implement design mockups, and optimize performance. Back-End: Build and maintain server-side logic, APIs, and database interactions.

		Full Stack: Integrate front-end and back-end systems, ensuring seamless functionality.
Ariel A. Forcado	UI/UX Designer	Conduct user research to gather insights into user needs. Design wireframes, prototypes, and high-fidelity mockups. Test designs through usability testing and gather feedback for iterations. Collaborate with developers to ensure design fidelity in implementation.
Lister John L. Evangelista	Security Analyst	Develop and enforce security policies and best practices. Monitor systems for security breaches and respond accordingly.

		Stay updated on cybersecurity trends and compliance regulations.
Andiella Mhae B. Gabad	Documentation Specialist	Develop user manuals, installation guides, and API documentation. Maintain a knowledge base for project artifacts and resources. Ensure all documentation is up-to-date and easily accessible. Assist in creating training materials for users.

Roles

Scrum Master: The Scrum Master helps the team follow Scrum practices, removes obstacles, manages communication, and promotes collaboration to meet project goals on time.

Product Owner: The Product Owner is the main person in charge of the project. They define the project's vision and goals, prioritize features, make sure the product meets user needs, and help the development team and stakeholders communicate.

This helps guide the project to successful completion.

Development Team: The development team, which includes individuals with different backgrounds and specialties, is responsible for carrying out the actual development work.

Project Manager: The project manager in IT oversees the planning, execution, and delivery of IT projects. This involves coordinating teams, and managing resources, timelines, and budgets. The project manager ensures that project objectives align with business goals and works to mitigate risks to successfully deliver technical solutions.

Developer: The developer's role is to create, code, and put into place practical solutions to ensure that the technical aspects of the project match the defined objectives. This involves combining user requirements and system functions while solving problems and using new ideas.

UI/UX Designer: The UI/UX Designer's role is to create easy-to-use interfaces and smooth experiences by researching user needs, designing wireframes, and ensuring that the system is easy to use and accessible to meet project goals.

Security Analyst: The security analyst's job is to check, plan, and set up security measures that keep data and systems safe during the project. The analyst makes sure to find and fix any weaknesses in the project's development and launch phases

Documentation Specialist: The documentation specialist creates, organizes, and maintains project records. This includes technical reports, user manuals, design specifications, and progress documentation. The specialist ensures that the documents are clear, accurate, and compliant with project standards.

4.6.2 Sprint Planning and Backlog Management

Scrum Artifacts

Sprint Cycles: Regular check-ins and progress evaluations are part of sprint cycles. This could entail holding more frequent feedback meetings between managers and staff in performance management. This check-in offers the chance to talk about accomplishments, difficulties, and prospective goal revisions.

Sprint Cycle	Duration	User Stories / Tasks	Key Features / Tasks	Deliverables	Responsible Team Member
Sprint Cycle 1: Inventory Management					
Sprint Planning	5 days	Define key process of inventory. Define what	Assess inventory functions.	Functionalities	Angelo Abargos, Ariel Forcado

		necessary data need in inventory.			
Daily Standups	5 days	Monitor / maintain alignment	Sync team in medical processes, inventory management and reports	Progress on system	Team
Sprint Review	5 days	Review inventory functionalities	Present Inventory management. Gather feedback from stakeholders	Functional inventory	Angelo Abargos, Ariel Forcado
Sprint Cycle 2: Medical Management					
Sprint Planning	5 days	Define key medical processes. Define what necessary	Assess medical module functions.	Medical function.	Angelo Abargos, Joseph Oliver Banac

		data need in medical.			
Daily Standups	5 days	Track medical module progress.	Medical module and testing.	Medical management	Team
Sprint Review	5 days	Present medical and features.	Present medical manageme nt to stakeholde rs. Gather their feedback.	Functional medical management	Andiella Mhae Gabad, Lister John Evangelista.
Sprint Cycle 3: Admission and Student					
Sprint Planning	5 days	Plan Student information and admission of students.	Assess student module and admission module.	Student module function And Admission module	Angelo Abargos, Lister John Evangelista
Daily Standups	5 days	Monitor Student module and	Student module and	Student module and	Team

		Admission module progress	admission testing.	Admission module	
Sprint Review	5 days	Present Student module and Admission module.	Gather feedback on student and admission module from stakeholders.	Functional Student and admission module.	Joseph Oliver Banac, Ariel Forcado
Sprint Cycle 4: Reporting and Reminder					
Sprint Planning	5 days	Plan reporting features.	Generate Reports	Reporting features and export data.	Angelo Abargos, Andiella Mhae Gabad
Daily Standups	5 days	Track reporting and data integration.	Develop filters and export of reports	Generating Report data.	Team
Sprint Review	5 days	Present the report module to	Feedback on report module form	Functional Report module	Ariel Forcado, Angelo Abargos

		the stakeholder s.	stakeholder s		
Sprint Cycle 5: Data Management					
Sprint Planning	5 days	Plan and manage data.	Data handling	Data handling	Angelo Abargos, Lister John Evangelista
Daily Standups	5 days	Monitor the progress in data manageme nt.	Track data manageme nt	Data Management	Team
Sprint Review	5 days	Review and improve data manageme nt	Reviewing and improving data manageme nt	Data management	Joseph Oliver Banac, Andiella Mhae Gabad
Sprint Cycle 6: Security Features					
Sprint Planning	5 days	Plan and identify system security features.	Role based access control, session token,	Security	Joseph Oliver Banac, Lister John Evangelista

			authentication.		
Daily Standups	5 days	Research security applicable in system.	Security System features	RBAC authentication session token	Team
Sprint Review	5 days	Review and Improve security.	Test security features	Functional Security.	Angelo Abargos, Joseph Oliver Banac
Sprint Cycle 7: AI-Assisted Diagnosis					
Sprint Planning	5 days	Plan AI features in the system.	AI-Assisted diagnosis feature	Planning AI feature	Angelo Abargos, Andiella Mhae Gabad
Daily Standups	5 days	Monitor changes in AI, track progress.	Testing AI model capabilities	AI-assisted diagnosis	Team
Sprint Review	5 days	Review and Improve AI, present to the	Feedback on AI feature.	Improve AI gradually.	Andiella Mhae Gabad, Ariel

		stakeholder s.			Forcado
Sprint Cycle 8: QR Code Integration					
Sprint Planning	5 days	Plan QR code features.	QR functionaliti es	Scannable QR	Angelo Abargos, Ariel Forcado
Daily Standups	5 days	Research QR code integration.	Data retrieval using QR	QR functionalities	Team
Sprint Review	5 days	Improve QR Code and present it to the stakeholder s.	Feedback in QR code module.	QR code more functionalities	Angelo Abargos, Andiella Mhae Gabad
Sprint Cycle 9: Documentation					
Sprint Planning	5 days	Review all documentat ions.	Checking the document.	Documentatio n	Team
Daily Standups	5 days	Consult documents and system.	Consultatio n	Document Enhance	Team

Sprint Review	5 days	Refining Document s and requiremen ts.	Revision of Document.	Document Quality	Team
Sprint Cycle 10: System Testing for Bugs and Integration					
Sprint Planning	5 days	Test system functionaliti es.	Findings of bugs and testing.	Integration.	Angelo Abargos , Ariel Forcado
Daily Standups	5 days	Cluster integration, feedback and research. Error handling.	Performanc e checking and penetration test.	System enhancement	Team
Sprint Review	5 days	Review and analyze system.	Review and Improving system.	System features.	Angelo Abargos, Lister John Evangelista
Sprint Cycle 11: Backup, Monitoring and Deployment					
Sprint Planning	1 week	Planning and	Backup and	Backup and Deployment	Angelo Abargos,

		monitoring of the system. Deploymen t of system	monitoring of system. Deploying system online.		Ariel Forcado
Daily Standups	1 week	Improving data managem ent and backups. Tracking and Quality testing for deployment .	Data storage, Data manageme nt, System Deploymen t	Data management and backup, deployment.	Team
Sprint Review	1 week	Checking the system and always monitoring and backup. Deployme nt of system	Monitoring system. Deployed Cloud / Online	Data storage and system monitoring. Access the system Online	Team

		and presenting to stakeholder s			
Sprint Cycle 12: Final Review of Features and Refinement					
Sprint Planning	1 week	Final Review of features and systema and documentat ion.	Documenta tion and System Review	Review	Team
Daily Standups	1 week	Integration Testing, System Testing	System Testing	Testing	Team
Sprint Review	1 week	Refinemen t of System, Improving the system overall.	System Refinement , Quality testing system overall.	System Refinement, Quality testing	Team

Product Backlog: A product backlog in performance management could be an employee's prioritized list of skills, competencies, or areas for development. It is possible to periodically review and modify this backlog.

Product Backlog (User Stories).

User Story No.	User Stories	User Story Priorities	Status
1	As a clinic physician, I can log in clinic system. <i>Can log in</i>	5	Done
2	As a clinic physician, I want to view the dashboard. <i>Overview Dashboard</i>	5	Done
3	As a clinic physician, I want to view and manage student information. <i>Manage Student Data</i>	5	Done
4	As a clinic physician, I want to view, manage medical results, medical history, and health form. <i>Manage Medical Affiliation's</i>	4	Done
5	As a clinic physician, I want to track and manage the inventory of medical supplies <i>Inventory Management</i>	5	Done
6	As a clinic nurse, I want to update student medical records in real- time so that other staff members and	1	Done

	authorized users always see the most current data. <i>Real-Time Data Accuracy for Student Medical Records</i>		
7	As a clinic nurse, I want to view a summarized version of a student's medical history so that I can make informed decisions quickly during a clinic visit. <i>Medical History Summary for Clinic Staff</i>	1	Done
8	As a system administrator, I want to track and audit user access to medical records to ensure data security and identify any unauthorized access. <i>Access Logs for Data Auditing</i>	1	Done
9	As a system administrator, I want daily backups of the clinic's data to ensure that no critical health information is lost in case of a system failure or data corruption. <i>Data Backup and Recovery</i>	1	Done
10	As a school administrator, I want access to real-time health reports (e.g., clinic visits, prevalent illnesses) so I can make informed decisions regarding school policies or closures. <i>Real-Time Health Reports</i>	1	Done
11	As a clinic administrator, I want to analyze student health trends (e.g., seasonal illnesses) to plan proactive healthcare strategies and reduce the spread of infections. <i>Health Monitoring and Trend Analysis</i>	2	Done

12	As a school administrator, I want to ensure that student health data is stored and shared securely in compliance with privacy laws (e.g., HIPAA), so the clinic meets all regulatory standards. <i>Data Privacy and Student Health Confidentiality</i>	1	Done
13	As a clinic nurse, I want to track and manage the inventory of medical supplies so I can ensure we have sufficient stock of essential items (e.g., bandages, medications) without running out. <i>Inventory Management</i>	2	Done

Product Backlog for EIS Information Security

User Story No.	User Stories	User Story Priorities	Status
1	As a system administrator, I want to define user roles and assign permissions so that users can only access the data and functionalities necessary for their role, minimizing security risks. <i>Role-Based Access Control (RBAC)</i>	1	Done
2	As a system administrator, I want to require users to authenticate using both their password and a secondary method (e.g., OTP or biometrics) to prevent unauthorized access.	1	Done

	<i>Multi-Factor Authentication (MFA) for User Access</i>		
3	As a data security officer, I want to ensure that all sensitive student health data is encrypted both when stored and transmitted to prevent unauthorized access and data breaches. <i>Data Encryption</i>	1	Done
4	As a compliance officer, I want a detailed log of all user activities related to student health data, so I can track and investigate any unauthorized access or changes. <i>Audit Trails and Activity Logging</i>	1	Done
5	As a network security officer, I want to be alerted to any suspicious activities or unauthorized access attempts to the system so that I can take immediate action to prevent breaches. <i>Intrusion Detection and Prevention System (IDPS)</i>	2	Done
6	As a system administrator, I want automatic daily backups of the EIS data so that the clinic can quickly recover in case of an incident or data loss. <i>Data Backup and Recovery</i>	1	Done
7	As a system administrator, I want to securely transmit data between the EIS and external systems (e.g., the registrar) to prevent unauthorized interception or data tampering. <i>Secure Data Sharing Protocols (External</i>	1	Done

	<i>Integrations)</i>		
8	As a security officer, I want to conduct regular vulnerability assessments and penetration tests on the system to detect and fix security flaws, ensuring the clinic's data is protected. <i>Periodic Vulnerability Assessments and Penetration Testing</i>	1	Done
9	As a security officer, I want to enforce a policy requiring users to create complex passwords (including letters, numbers, symbols, and special character) and reset them periodically to enhance security. <i>Secure Password Policies</i>	1	Done
10	As a compliance officer, I want the EIS system to comply with regulations like HIPAA to ensure that all student health data is protected and handled in accordance with legal standards. <i>Compliance with Data Protection Regulations</i>	1	Done
11	As a security officer, I want the system to automatically log out users after a certain period of inactivity to reduce the risk of unauthorized access if a session is left open. <i>Session Timeouts and Automatic Logouts</i>	2	Done
12	As a clinic administrator, I want the clinic staff to receive regular training on data security to prevent human errors that could lead to security breaches.	4	Done

	<i>Security Awareness Training for Clinic Staff</i>		
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Product Backlog for EIS Standards

User Story No.	User Stories	User Story Priorities	Status
1	As an admin, I want to customize reports.	1	Done
2	As an admin, I want to filter the list of medical.	1	Done
3	As an admin, I want to automate reminder for nurses related to inventory tracking.	1	Done
4	As an admin, I want to monitor the medical examination.	1	Done

Product Backlog for Integration

User Story No.	User Stories	User Story Priorities	Status
1	As a clinic administrator, I want the system to integrate with the school's registrar / student portal so that I can access and retrieve student information (e.g., contact details, grade level, enrollment status) in real- time without manually entering data. <i>Registrar Integration for Student Data</i>	1	Done

2	As a clinic administrator, I want the system to integrate with human resources so that they can assign role, access, and permission of accounts of the employee of the clinic. <i>Human Resource Integration for Account Management</i>	1	Done
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Sprint Backlog

User Story No.	User Stories	User Story Priorities	Revision Priority	Status
1	Log and admit student medical history, examinations	2	2	Done
2	Track and manage inventory.	2	2	Done
3	Legal compliance as GDPR and HIPAA.	2	2	Done
4	Generate reporting and reminders to accounts.	3	3	Done
5	AI and QR integration	1	1	Done
6	Security integration	1	1	Done
7	Human resource integration	1	1	Done
8	Registrar integration	1	1	Done
9	Medical records management	1	1	Done
10	Scheduling and appointment	3	3	Done

4.6.3 Sprint Execution and Deliverables

Sprint Deliverables:

The primary deliverable of each sprint is working software. This means that the team should have a functional piece of the system ready to be demonstrated and tested. The team will also create documentation for the features and functionalities that were developed during the sprint. This might include:

User stories: Detailed descriptions of the user stories that were implemented.

Technical documentation: Explanations of the code, design decisions, and testing procedures.

User manuals: Instructions for using the system.

At the end of the sprint, the team will hold a sprint review meeting to demonstrate the completed work to the stakeholders and gather feedback. They will also hold a sprint retrospective meeting to reflect on the sprint, discussing what went well, what could be improved, and identifying any lessons learned.

4.6.4 Challenges Faced in the Development Process

The development of the SMS I clinic management system at Bestlink College of the Philippines presented several challenges, primarily related to user adoption, data security, and integration with existing systems. These challenges were addressed through an iterative and collaborative approach using the Agile Scrum framework.

User Resistance to Change:

Challenge: Introducing a new clinic management system with QR code integration and AI-assisted diagnosis could face resistance from clinic staff who are accustomed to traditional methods.

Solution: The development team addressed this challenge through:

Stakeholder Engagement: Early and ongoing engagement with clinic staff, nurses, physicians, and administrators was crucial. This involved gathering feedback, explaining the benefits of the new system, and addressing concerns.

User-Centric Design: The system was designed with user needs in mind, focusing on ease of use, intuitive interfaces, and clear instructions. This minimized the learning curve for users.

Training and Support: Comprehensive training programs were developed to guide users through the new system. Ongoing support was provided to address any questions or issues.

Data Security and Privacy Concerns:

Challenge: The system handles sensitive student health data, requiring robust security measures to protect privacy and comply with regulations like HIPAA and GDPR.

Solution: The development team implemented the following security measures:

Role-Based Access Control (RBAC): Users were assigned specific roles with limited access to data based on their responsibilities

Secure Authentication: Multi-factor authentication (MFA) was implemented to prevent unauthorized access.

Data Encryption: Sensitive data was encrypted both at rest and in transit.

Audit Trails: Detailed logs of user activity were maintained to track access and identify any suspicious behavior.

Regular Security Assessments: The system was regularly assessed for vulnerabilities and security breaches.

Integration with Existing Systems:

Challenge: Integrating the new clinic management system with existing systems at Bestlink College (e.g., the registrar system, student portal) presented challenges in data mapping, synchronization, and ensuring data consistency.

Solution: The development team employed the following strategies:

API Integration: APIs were used to facilitate communication and data exchange between the new system and existing systems.

Error Handling and Conflict Resolution: Mechanisms were implemented to handle errors during data exchange and resolve conflicts that might arise from simultaneous updates.

Agile Scrum Framework

The development team adopted an Agile Scrum framework to address these challenges effectively. The iterative and collaborative nature of Scrum facilitated responsive development, ensuring that the project was aligned with the needs of stakeholders.

Roles and Responsibilities

The team defined clear roles and responsibilities, including:

Project Manager: Overseeing the project's progress, managing resources, and ensuring alignment with goals.

Developers: Building the system's functionality and integrating it with existing systems.

UI/UX Designer: Designing user-friendly interfaces and ensuring a smooth user experience.

Security Analyst: Implementing security measures and conducting vulnerability assessments.

Documentation Specialist: Creating user manuals, technical documentation, and training materials.

4.7 Implementation

4.7.1 Technical Implementation Details

Tiered Architecture: A common approach is a multi-tiered architecture, separating concerns for better maintainability and scalability.

This might involve:

Presentation Tier (UI): The user interface, accessible through web browsers or mobile apps.

Application Tier (Business Logic): Handles the core functionality of the CMS, including patient management, appointments.

Data Tier (Database): Stores all the clinic's data.

Microservices Architecture: For larger, more complex systems, microservices architecture might be used. This breaks down the system into smaller, independent services that communicate with each other. This allows for greater flexibility and independent deployments.

Technology Stack

Programming Languages: Choices depend on the development team's expertise and project requirements.

Common options include:

Backend: Java, Python (Django/Flask), Node.js

Frontend: JavaScript (Vue.js), HTML, CSS.

Database: Relational databases are often used:

SQL: PhpMyAdmin.

APIs: RESTful APIs are commonly used for web-based communication.

Data Management:

Data Modeling: Careful design of the database schema is crucial.

This includes defining tables for patients, doctors, appointments, medical records, billing information, etc., and establishing relationships between them.

Data Security: Implementing measures to protect patient data is paramount: Encryption (at rest and in transit) Access control (role-based permissions) Regular backups and disaster recovery plans Compliance with regulations (HIPAA, GDPR)

Data Migration: If migrating from an existing system, a well-defined data migration strategy is essential to ensure data accuracy and minimize downtime.

Data Validation: Implementing data validation rules to ensure data integrity.

Integrations:

Electronic Health Records (EHR): Integrating with EHR systems to share patient medical information.

Billing Systems: Connecting with billing software for automated invoicing and payment processing.

Lab Systems: Integrating with lab systems to receive test results electronically.

Pharmacy Systems: Connecting with pharmacy systems for electronic prescriptions.

Other Systems: Integrating with other relevant systems, such as appointment reminder services or patient portals.

Deployment:

Cloud Deployment: Deploying the CMS to a cloud platform for scalability and availability.

On-Premises Deployment: Installing the CMS on servers within the clinic's premises.

Continuous Integration/Continuous Deployment (CI/CD): Automating the build, test, and deployment process to ensure faster and more reliable releases.

Monitoring and Maintenance:

Log Management: Collecting and analyzing system logs to identify issues and track activity.

Regular Backups: Performing regular backups of the database and system files.

Software Updates: Applying software updates and patches to address security vulnerabilities and improve functionality.

4.7.2. Tools and Technologies Used

Our clinic management system (SMS I) relies on a carefully chosen set of tools and technologies to support its functionality and usability. In the backend, we utilize Python as our primary scripting language, known for its versatility and robustness in handling complex business logic.

JavaScript is also used for backend development, particularly for asynchronous tasks and real-time updates. Complementing this, the frontend is developed using fundamental web technologies such as HTML, CSS, and JavaScript, ensuring a responsive and interactive user interface. Additionally, we leverage libraries and frameworks like Bootstrap to streamline development and enhance user experience.

For version control and collaborative development, we employ Git and GitHub, facilitating seamless code management and team collaboration. Continuous integration and deployment pipelines are established using Jenkins, automating testing and deployment processes to maintain code quality and accelerate time-to-market.

This comprehensive set of tools and technologies enables us to build a robust and efficient clinic management solution that meets the needs of our users and stakeholders effectively.

4.7.3. Code Integration and Interoperability

Code integration and interoperability are crucial for a successful Clinic Management System (CMS). They ensure that different parts of the system, and the CMS itself with external systems, can communicate and exchange data seamlessly.

Internal Code Integration (Within the CMS):

Modular Design: Building the CMS with modular architecture is essential. This involves breaking down the system into independent modules that can be developed, tested, and maintained separately.

API-First Approach: Designing internal APIs allows different modules to communicate with each other.

Consistent Coding Standards: Adhering to consistent coding standards across all modules ensures that the code is easy to understand, maintain, and integrate.

Version Control: Using a version control system is crucial for managing code changes, tracking revisions, and facilitating collaboration among developers.

Testing: Thorough testing (unit tests, integration tests) is essential to ensure that different modules work together correctly and that changes in one module don't break other parts of the system.

External System Integration (CMS with other systems):

Standardized APIs: Using standardized APIs (e.g., FHIR for healthcare) is highly beneficial for interoperability. FHIR (Fast Healthcare Interoperability Resources) provides a common language and data format for exchanging healthcare information.

API Gateways: An API gateway can act as a central point for managing and securing external API connections. It can handle authentication, authorization, rate limiting, and other tasks related to API access.

Message Queues: Message queues (e.g., RabbitMQ, Kafka) can be used to handle asynchronous communication between the CMS and other systems.

Data Transformation: Often, data needs to be transformed to be compatible with the format expected by the other system. This might involve converting data types, mapping data fields, or restructuring the data.

4.7.4 Integration Testing and Debugging

Integration testing and debugging are critical phases in the development of a Clinic Management System (CMS). They ensure that different components of the system work together correctly and that any issues are identified and resolved before the system goes live.

Integration Testing

Integration testing focuses on verifying the interactions between different modules or components of the CMS. It goes beyond unit testing (which tests individual components in isolation) to ensure that data flows correctly and that the integrated system behaves as expected.

Types of Integration Tests:

Component Integration Testing: Tests the interaction between two or more related components.

System Integration Testing: Tests the interaction between all the major subsystems of the CMS. This might involve testing the entire workflow of registering a patient.

Interface Testing: Focuses specifically on testing the interfaces (APIs) between different modules or systems. This ensures that data is exchanged correctly and that the interfaces are robust.

Test Data: Use realistic test data that covers a range of scenarios, including both positive and negative cases. This helps to identify potential issues with data validation and error handling.

Test Cases: Develop detailed test cases that specify the inputs, expected outputs, and steps involved in each test. This helps to ensure that the testing is comprehensive and repeatable.

Test Automation: Automate as many integration tests as possible to save time and improve efficiency. This can be done using testing frameworks and tools.

Continuous Integration: Integrate integration testing into the continuous integration pipeline. This ensures that integration tests are run automatically whenever code changes are made, allowing for early detection of integration issues.

4.8 Testing and Quality Assurance

4.8.1 Testing Strategies and Methodology

Testing Process:

Test Plan: Create a detailed test plan that outlines the scope, objectives, testing methodologies, timelines, and resources.

Test Cases: Develop specific test cases for each feature or scenario.

Each test case should include:

Test Case ID: A unique identifier for the test case.

Description: A clear description of the test scenario.

Preconditions: Conditions that must be met before the test can be executed.

Expected Results: The expected outcome of the test.

Actual Results: The actual outcome of the test.

Status: The status of the test (e.g., Pass, Fail, Blocked).

Test Environment: Set up a dedicated test environment that mirrors the production environment as closely as possible.

Test Execution: Execute the test cases and document the results.

Bug Reporting: Report any bugs or defects found in a clear and concise manner, including steps to reproduce the issue. Use a bug tracking system to manage bug reports.

Bug Fixing and Retesting: Developers fix the bugs, and testers retest the affected areas to ensure the fixes are effective.

Test Reporting: Generate test reports that summarize the testing activities, including the number of test cases executed, the number of bugs found, and the overall quality of the system.

4.8.2. Test cases and Test Data

Admission:

Test Case 1: Successful Patient Registration

Description: Verify that a new patient can be registered successfully with all required information. Data Case:

First Name: John Last Name: Doe

Date of Birth: 1980-01-15

Address: 123 Main St Phone Number: 555-1212

Email: [email address removed]

Medical Record Number (if applicable): Auto-generated or specified

Expected Result: Patient records are created successfully, and a confirmation message is displayed.

Test Case 2: Missing Required Fields

Description: Verify that the system prevents patient registration if required fields are missing.

Data Case: Leave "Last Name" field blank.

Expected Result: Error message displayed indicating the missing field.

Test Case 3: Invalid Data Format

Description: Verify that the system validates data formats (e.g., phone number, email).

Data Case: Invalid phone number (e.g., "abc-123- xyz").

Expected Result: Error message displayed indicating the invalid format.

Test Case 4: Duplicate Medical Record Number (if applicable)

Description: Verify that the system handles duplicate medical record numbers.

Data Case: Use a medical record number that already exists in the system.

Expected Result: Error message displayed indicating the duplicate record number.

Appointment Scheduling:

Test Case 1: Scheduling an Appointment

Description: Verify that a patient can schedule an appointment.

Data Case:

Patient: John Doe Doctor: Dr. Smith Date: 2024-03-10

Time: 10:00 AM

Appointment Type: Checkup

Expected Result: Appointment is scheduled successfully, and a confirmation message is displayed.

Test Case 2: Scheduling an Appointment with a Conflict

Description: Verify that the system prevents scheduling appointments with conflicts (e.g., double- booking a doctor).

Data Case: Attempt to schedule an appointment with Dr. Smith at the same time as an existing appointment.

Expected Result: Error message displayed indicating the conflict.

Test Case 3: Cancelling an Appointment

Description: Verify that a patient can cancel an appointment.

Data Case: Cancel the appointment scheduled in Test Case 1.

Expected Result: Appointment is cancelled successfully, and a confirmation message is displayed.

Electronic Health Records (EHR):**Test Case 1: Adding a New Entry to EHR**

Description: Verify that a new entry can be added to a patient's EHR.

Data Case:

Patient: John Doe Date: 2024-03-08

Diagnosis: Flu

Notes: Patient reports fever and cough.

Expected Result: EHR entry is added successfully.

Test Case 2: Viewing EHR

Description: Verify that authorized users can view patient EHR.

Data Case: Log in as a doctor with access to John Doe's records.

Expected Result: John Doe's EHR is displayed correctly.

Test Case 3: Editing EHR Entry

Description: Verify that authorized users can edit EHR entries.

Data Case: Edit the diagnosis in the EHR entry from Test Case 1.

Expected Result: EHR entry is updated successfully.

User Access Control:**Test Case 1: Doctor Access**

Description: Verify that a doctor can access patient records and schedule appointments.

Data Case: Log in as Dr. Smith.

Expected Result: Doctor has access to authorized functionalities.

Test Case 2: Receptionist Access

Description: Verify that a receptionist can schedule appointments and manage patient registration but cannot access EHR.

Data Case: Log in as a receptionist.

Expected Result: Receptionist has access to authorized functionalities but not EHR.

Important Considerations:

Negative Testing: Include test cases with invalid or unexpected data to ensure the system handles errors gracefully.

Boundary Conditions: Test with data at the limits of acceptable ranges (very long names, dates in the past/future).

Security Testing: Include test cases to check for vulnerabilities (SQL injection, cross-site scripting).

Performance Testing: Test the system's performance under different loads (many users accessing the system simultaneously).

4.8.3 Test Results and Bug Reports

1. Test Results Documentation:

Test Summary Report: This report provides a high-level overview of the testing activities.

It should include:

Project Name: Clinic Management System Test Cycle

Date of Testing:

Test Environment: (Browser version, OS, Database)

Number of Test Cases Executed:

Number of Test Cases Passed:

Number of Test Cases Failed:

Number of Bugs Found:

Overall Test Status: (Pass, Fail, In Progress)

Key Findings/Recommendations:

Detailed Test Execution Report:

This report provides more detailed information about each test case execution. It should include:

Test Case ID: (Unique identifier)

Test Case Description:

Test Data Used:

Expected Result:

Actual Result:

Status: (Pass, Fail, Blocked, In Progress)

Tester Name:

Date Executed:

Comments/Observations: (Any relevant notes) Screenshot (If applicable):

Especially helpful for failed test cases.

4.8.4 Quality Assurance Measures

Let's discuss quality assurance (QA) measures for your Clinic Management System. A comprehensive QA strategy is essential for delivering a reliable, user-friendly, and secure system. Here's a breakdown of key measures:

1. Requirements Management:

Clear and Detailed Requirements: Start with well-defined and documented requirements.

Requirements Traceability: Maintain traceability between requirements, test cases, and code.

Change Management: Implement a formal process for managing changes to requirements.

2. Test Planning and Design:

Comprehensive Test Plan: Develop a detailed test plan that outlines the scope, objectives, testing methodologies, timelines, resources, and entry/exit criteria for testing.

Risk-Based Testing: Prioritize testing based on the risk associated with different features or functionalities. Focus on testing the most critical areas first.

Test Case Design Techniques: Use appropriate test case design techniques (e.g., equivalence partitioning).

Test Automation: Automate repetitive test tasks (e.g., regression testing) to improve efficiency and reduce testing time.

3. Testing Execution and Reporting:

Dedicated Test Environment: Set up a dedicated test environment that mirrors the production environment as closely as possible.

Defect Tracking: Use a bug tracking system to log, track, and manage defects.

Defect Prioritization: Prioritize defect fixes based on severity and impact.

4. Quality Control Throughout the Development Lifecycle:

Early Involvement of QA: Involve QA in all stages of the software development lifecycle (SDLC), from requirements gathering to deployment.

Code Reviews: Conduct code reviews to identify potential defects and ensure code quality.

Static Analysis: Use static analysis tools to automatically detect code defects and security vulnerabilities.

Continuous Integration and Continuous Delivery (CI/CD): Implement CI/CD pipelines to automate the build, test, and deployment process.

5. Security Testing:

Vulnerability Scanning: Regularly scan the system for known vulnerabilities.

Penetration Testing: Conduct penetration testing to simulate real-world attacks and identify security weaknesses.

Security Audits: Perform security audits to assess the system's security posture.

Privacy and Security: Ensure compliance with relevant data privacy regulations (HIPAA, GDPR).

4.9 Result and Evaluation

4.9.1 Project Outcomes and Deliverables

Project Outcomes (What the system will achieve):

Improved Patient Experience: Streamlined registration, appointment scheduling, and access to medical records, leading to increased patient satisfaction.

Enhanced Clinic Efficiency: Automation of administrative tasks, optimized workflows, and reduced paperwork, freeing up staff to focus on patient care.

Data-Driven Insights: AI-powered analytics providing valuable insights into patient demographics, trends, and operational performance, enabling better decision-making.

Increased Revenue: Optimized billing processes, reduced no-shows, and improved resource allocation can contribute to increased revenue.

Enhanced Security and Compliance: Secure storage and access to patient data, ensuring compliance with relevant regulations (HIPAA, GDPR).

Improved Communication: Seamless communication between patients and clinic staff through secure messaging and notifications.

Scalability and Flexibility: A system that can adapt to the growing needs of the clinic and integrate with other healthcare systems.

Competitive Advantage: Offering modern, technology- driven services to attract and retain patients.

I. Project Deliverables (Tangible outputs of the project):

A. Software Components:

1. Clinic Management System Software

Patient Management Module: Registration, demographics, medical history, allergies, etc.

Appointment Scheduling Module: Online booking, appointment reminders, calendar management, waitlists.

Electronic Health Records (EHR) Module: Secure storage and access to patient medical records, including notes, diagnoses, lab results, prescriptions, and imaging.

Inventory Management Module: Tracking of medical supplies, medications, and equipment.

Reporting and Analytics Module: Dashboards, customizable reports, AI-driven insights into patient data and clinic performance.

User Management and Access Control: Role-based access control to protect sensitive data.

Communication Module: Secure messaging between patients and staff, appointment reminders, notifications.

Integration Module: APIs and integrations with other healthcare systems (labs, pharmacies, insurance providers).

2. QR Code Functionality:

QR Code Generation: For patient identification, appointment check-in, access to medical records (with appropriate security measures), payment processing.

QR Code Scanning: Mobile app or dedicated scanner integration for quick access to information.

3. AI-Driven Features:

Predictive Analytics: For appointment scheduling optimization, patient risk assessment, and resource allocation.

Diagnostic Support: AI assistance for doctors in interpreting medical images or suggesting possible diagnoses (with appropriate disclaimers).

Personalized Medicine Recommendations: AI-powered suggestions for treatment plans based on patient data and medical best practices.

Chatbot Integration: AI-powered chatbot for answering patient inquiries, scheduling appointments, and providing basic medical information.

B. Documentation:

1. Requirements Document: Detailed description of the system's functionalities and requirements.

2. System Design Document: Architecture, database design, and technical specifications.

3. User Manual: Instructions for using the system.

4. Technical Documentation: For developers and system administrators.

5. Test Plan and Test Cases: Documentation of the testing strategy and test cases.

6. Test Results and Bug Reports: Summary of testing activities and identified defects.

7. Training Materials: For clinic staff on how to use the system.

8. Maintenance and Support Plan: Guidelines for system maintenance and support.

C. Other Deliverables:

Source Code: Well-documented and commented source code. Database

Schema: Design of the database used by the system.

Installation Package: For easy installation of the system.

Deployment Plan: Detailed plan for deploying the system to the clinic's environment.

Training Sessions: Conducted training for clinic staff.

II. QR Code and AI Specific Deliverables:

QR Code Generation and Scanning Module: Including libraries or APIs used.

AI Model Documentation: Description of the AI models used, their training data, and performance metrics.

AI Integration Code: Code that integrates the AI models into the system.

Data Security and Privacy Plan: Specific measures for protecting patient data when using QR codes and AI.

4.9.2. Alignment with Project Objectives

When aligning a Clinic Management System (CMS) with project objectives, it's crucial to ensure the system directly supports and enhances the clinic's core goals. Here's a breakdown of how to approach this, including key areas to consider:

1. Define Clear Project Objectives: Improve Patient Care:

Reduce wait times.

Enhance patient communication.

Improve access to patient records.

Enhance patient safety.

Increase Operational Efficiency:

Streamline administrative tasks (scheduling, billing, reporting).

Reduce paperwork and manual data entry.

Optimize resource utilization.

Reduce operational costs.

Enhance Financial Management:

Improve billing accuracy and efficiency.

Reduce revenue leakage.

Generate accurate financial reports.

Improve insurance claim processing.

Enhance Data Management and Security:

Securely store and manage patient data.

Ensure HIPAA or other relevant regulatory compliance.

Improve data accessibility for authorized personnel.

Generate reports for analysis.

Improve communication:

Between staff members. Between staff and patients.

Between the clinic and other medical providers.

2. Align CMS Features with Objectives:

For Improved Patient Care:

Electronic Health Records (EHR): Provide quick access to patient history, medications, and allergies.

Online Appointment Scheduling: Reduce phone calls and improve patient convenience.

Telemedicine Integration: Enable remote consultations. Automated Reminders: Send appointment and medication reminders.

For Increased Operational Efficiency:

Automated Scheduling: Optimize appointment slots and reduce scheduling conflicts.

Inventory Management: Track medical supplies and prevent shortages.

Reporting and Analytics: Generate reports on key performance indicators (KPIs).

Task Management: assign and track tasks for staff members.

For Enhanced Data Management and Security:

Role-Based Access Control: Restrict access to sensitive data based on user roles.

Data Encryption: Protect patient data from unauthorized access.

Audit Trails: Track all user activity for accountability.

Regular Backups: Ensure data recovery in case of system failure.

For Improved communication:

Integrated messaging systems: Allow for secure communication between staff members.

Patient messaging: Allow for secure communication between staff and patients.

Interoperability: allow the CMS to communicate with other medical systems.

4.9.3 Stakeholder and user Feedback Stakeholder Groups and Their

Concerns Patients:

Positive:

Convenience of QR code check-in and access to information.

Improved communication through AI-powered chatbots or messaging.

Faster appointment scheduling and reduced waiting times.

Enhanced access to personal health records.

Increased sense of control over their healthcare.

Concerns:

Privacy and security of personal data.

Accessibility for those not tech-savvy.

Potential for impersonal interactions with AI.

Accuracy of AI-generated information.

Doctors and Nurses:**Positive:**

Streamlined workflows and reduced administrative burden.

AI-powered diagnostic support and decision-making tools.

Improved access to patient data and medical history.

Enhanced communication among healthcare providers.

Improved inventory management.

Concerns:

Integration of the new system with existing workflows.

Reliability and accuracy of AI-generated insights.

Potential for over-reliance on AI and loss of clinical judgment.

Training and support for using the new system.

Increased data entry.

Clinic Administrators:**Positive:**

Increased efficiency and cost savings.

Improved data management and reporting.

Enhanced patient satisfaction and retention.

Streamlined billing and insurance processing.

Better resource allocation.

Concerns:

Initial investment and implementation costs.

Data security and regulatory compliance (HIPAA, GDPR.).

Staff training and change management.

System reliability and maintenance.

IT Staff:**Positive:**

Modernizing the clinics' technology.

Opportunity to work with cutting edge technology.

Concerns:

System integration and interoperability.

Data security and cybersecurity risks.

System maintenance and troubleshooting.

Scalability and performance.

Feedback Areas Related to AI and QR Codes**QR Code Functionality:**

Ease of use for check-in, access to information, and payments.

Reliability of QR code scanning and linking.

Placement and visibility of QR codes.

Security of the information that the QR code leads to.

AI Integration:

Accuracy and reliability of AI-powered diagnostic support.

Usefulness of AI-powered chatbots for patient communication.

Clarity and transparency of AI-generated insights.

Impact of AI on the doctor-patient relationship.

Data privacy related to AI's data usage.

Data Security and Privacy

Protection of patient data from unauthorized access.

Compliance with relevant data privacy regulations.

Transparency about how patient data is used.

Security of data transmission, especially with QR code usage.

4.9.4. Lesson Learned

Technological Integration in Healthcare:

AI Applications:

Grasping the ethical implications of AI in medical decision-making.

QR Code Technology:

Exploring the practical uses of QR codes for patient check-in, information access, and secure data sharing.

Understanding how QR codes contribute to streamlining workflows and improving patient experience.

Learning about the security aspects of QR code usage.

System Interoperability:

Learning how different systems (EHR, billing, etc.) can be integrated for seamless data flow.

Understanding the importance of data standards and APIs in healthcare.

Workflow Optimization:

Analyzing how technology can streamline administrative tasks and improve efficiency.

Learning about the impact of automation on clinic operations.

Patient Experience:

Understanding how technology can enhance patient engagement and satisfaction.

Learning about the importance of accessibility and user-friendliness in healthcare systems.

Data Management and Security:

Understanding the critical importance of data privacy and security in healthcare (HIPAA, GDPR).

Learning about data encryption, access control, and other security measures.

Financial Management:

Learning how technology can improve billing accuracy and efficiency.

Understanding the role of technology in revenue cycle management.

Data Analysis and Decision-Making:**Data-Driven Insights:**

Learning how to use data analytics to identify trends and improve clinical outcomes.

Understanding how AI can generate insights from large datasets.

Performance Measurement:

Learning how to track key performance indicators (KPIs) to evaluate the effectiveness of the CMS.

Understanding how to use reporting tools to visualize data.

Ethical and Social Implications:**Privacy and Security:**

Examining the ethical considerations related to patient data privacy and security.

Understanding the potential risks and benefits of using AI in healthcare.

Accessibility and Equity:

Considering the impact of technology on access to healthcare for different populations.

Understanding the importance of ensuring that technology is accessible to all users.

Human-Computer Interaction:

Understanding how to design systems that are user friendly, and that improve the interaction between healthcare professionals, and patients.

Chapter 5

Summary, Conclusion, and Recommendations

5.1 Key Takeaways and Summary

5.1.1 Project Recap

The SMS I: AI-Assisted Diagnosis and QR Code Integration for Streamlined Clinic Management System aimed to enhance the efficiency, accuracy, and security of clinic operations at Bestlink College of the Philippines. The project was developed in multiple phases, including research, design, development, testing, and deployment. Key elements of the project included the integration of AI-assisted diagnosis, QR code-based patient record management, and a centralized database for improved clinical workflows. The system was designed to address inefficiencies in manual record-keeping, improve communication within the clinic, and reduce administrative burdens to optimize patient care.

5.1.2 Achievements

Throughout the project, several significant milestones were achieved:

Successful System Development: The clinic management system was fully designed, developed, and tested to ensure it met the intended objectives.

AI-Assisted Diagnosis Integration: The system implemented an AI-powered diagnostic tool that enhances accuracy and reduces potential errors in patient assessments.

QR Code Functionality: The introduction of QR code integration streamlined patient record retrieval and reduced paperwork.

Centralized Database Implementation: A secure and efficient data management system was established, allowing easy access to patient information.

Improved Clinic Efficiency: The system optimized workflows, minimized administrative workload, and improved the overall patient experience.

Stakeholder Engagement and Feedback: The project involved key stakeholders, including clinic staff and patients, ensuring the system's functionality aligned with real-world needs.

The successful deployment of this project represents a significant step toward modernizing clinic management, enhancing operational efficiency, and improving patient care through technology-driven solutions.

5.2 Project Achievements and Contributions

5.2.1 Impact on Stakeholders

The project has brought substantial benefits to stakeholders, including clinic staff, students, and administrative personnel.

Key impacts include:

Enhanced Efficiency for Clinic Staff: Medical personnel can now quickly access and update patient records using QR codes, reducing the time spent on manual data entry.

Improved Patient Experience: Patients benefit from faster check-ins, reduced waiting times, and more accurate diagnoses through AI-assisted evaluation.

Administrative Convenience: The centralized database simplifies record-keeping, reducing errors and ensuring secure access to patient data.

Better Decision-Making: Stakeholders can generate comprehensive reports for improved clinical decision-making and resource allocation.

5.2.2 Innovations and Enhancements

The project introduced several innovative solutions that significantly improved clinical processes:

AI-Powered Diagnosis: The integration of AI-driven analysis enhances diagnostic accuracy, assisting medical professionals in making informed decisions.

QR Code Integration: A novel approach to patient record management, eliminating paperwork and streamlining retrieval processes.

Automated Data Management: The system reduces redundancy, ensuring real-time updates and minimizing human error.

Secure and Scalable Infrastructure: The implementation of a robust database ensures data integrity, security, and scalability for future expansions.

These innovations contribute to the field by setting a precedent for AI-assisted clinic management systems, demonstrating how technology can revolutionize healthcare administration.

5.3 Future Work and Enhancement

5.3.1 Potential Improvements

While the system has successfully addressed many inefficiencies, there are still areas for improvement:

Enhanced AI Capabilities: Expanding AI functionality to include predictive analytics for disease prevention and early detection.

User Experience Enhancements: Improving system usability with a more intuitive interface and better accessibility features.

Interoperability with Other Systems: Ensuring compatibility with other healthcare management software for seamless data exchange.

Mobile Application Development: Creating a mobile-friendly version to enhance accessibility for both patients and staff.

5.3.2 Long-term Vision

The long-term vision for the system includes:

Expanding AI Diagnostic Features: Incorporating machine learning models that continuously improve diagnostic accuracy over time.

Integration with Telemedicine Services: Facilitating remote consultations and online health monitoring for patients.

Cloud-Based Solutions: Migrating to cloud infrastructure to support scalability and remote access.

Research and Development: Continuously updating the system based on emerging healthcare trends and technological advancements.

5.3.3 Scalability and Sustainability

To ensure the system's longevity and adaptability, several strategies should be considered:

Regular Software Updates: Implementing periodic system enhancements to address security vulnerabilities and performance issues.

Robust Data Security Measures: Strengthening encryption protocols and access controls to maintain data integrity.

Training and Support for Users: Providing ongoing training programs for clinic staff to maximize system adoption and effectiveness.

Scalability Planning: Designing the architecture system to accommodate future growth and increased user demand.

By focusing on these areas, the system can continue to evolve, maintain its relevance, and provide long-term value to stakeholders.

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A. Detailed Technical Documentation

1. System Architecture

The system follows a modular and scalable architecture that integrates AI-assisted diagnosis, QR code functionality, and centralized database management. It consists of multiple layers, including the user interface, application logic, and data storage components.

A. System Component Diagrams

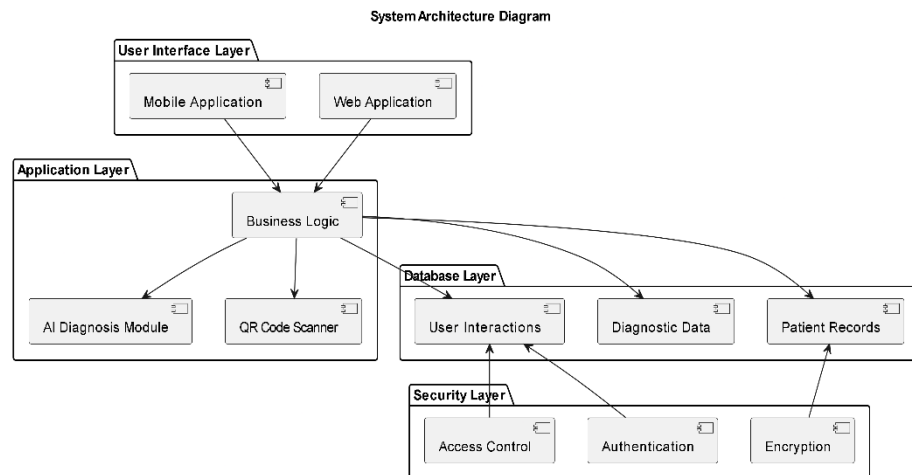


Figure 21: System Component Diagram

User Interface Layer: Provides access for clinic staff and patients to interact with the system via web and mobile applications.

Application Layer: Manages business logic, AI diagnosis processing, and QR code scanning functionalities.

Database Layer: Stores patient records, diagnostic results, and user interactions securely.

Security Layer: Ensures data protection through encryption, authentication, and access control measures.

B. Component Responsibilities and Relationships

AI Module: Processes patient symptoms and provides diagnostic recommendations.

QR Code Module: Generates and scans QR codes for efficient patient record retrieval.

Database Management System: Stores and organizes patient data for seamless access.

User Authentication System: Controls user roles and permissions to maintain security.

Reporting and Analytics Module: Generates insights for clinic operations and decision-making.

2. Information Systems Integration

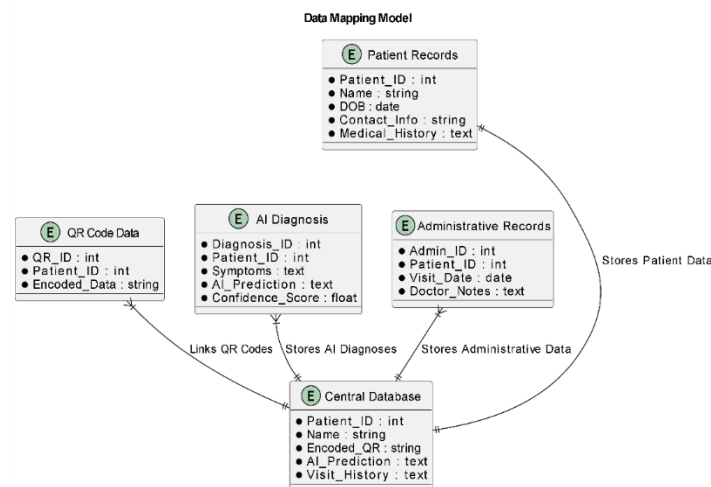
The project incorporated various information systems to ensure seamless data exchange and efficient clinic management. The integration strategy involved:

Centralized Database Connection: Linking different modules to a unified database for real-time data access.

API-Based Communication: Implementing APIs to facilitate data flow between AI-assisted diagnosis, QR code modules, and administrative systems.

Automated Data Synchronization: Ensuring consistency across different systems by synchronizing patient records, diagnostic results, and reports

A. Data Mapping Diagram



Patient Records (PR): Stores patient details and medical history.

QR Code Data (QR): Links patients to QR codes for easy retrieval.

AI Diagnosis (AI): Captures diagnostic predictions from the AI module.

Administrative Records (AR): Holds visit details and doctor's notes.

Central Database (DB): Consolidates all the data for seamless integration.

3. Application Design and Development

A. Software Modules and Components

User Interface Module: Web and mobile app interfaces for users.

AI Diagnosis Module: Processes patient symptoms and provides diagnostic insights.

QR Code Module: Handles patient record retrieval and storage.

Database Management Module: Stores and retrieves patient and clinic records.

B. Code Structure and Snippets

Backend: PHP

Frontend: Tailwind CSS

Database: PhpMyAdmin

C. Algorithms and Data Structures

Machine Learning Models: Decision trees, neural networks.

Data Storage: Relational database with indexed queries.

QR Code Encoding: Base64 encoding for efficient data handling.

4. Database Schema and Data Management

A. Database Schema Design

Patient Table: Stores patient information, including name, age, and medical history.

Diagnosis Table: Records AI-assisted diagnoses and recommendations.

QR Code Table: Manages QR codes linked to patient records.

User Table: Handles clinic staff and admin accounts.

Appointments Table: Tracks patient visits and scheduling.

B. Entity-Relationship Diagrams (ERDs)

ERD showcasing relationships between patients, diagnoses, appointments, and users.

C. Data Modeling and Normalization

1st Normal Form: Eliminated duplicate data entries.

2nd Normal Form: Ensured all attributes are fully functionally dependent on primary keys.

3rd Normal Form: Removed transitive dependencies for optimized data integrity.

D. CRUD (Create, Read, Update, Delete) Operations

Create: Adding new patient records and AI diagnoses.

Read: Retrieving patient history and diagnostic results.

Update: Modifying patient details and diagnoses.

Delete: Removing outdated or redundant records.

By implementing a structured database and optimized data management practices, the system ensures reliable, secure, and efficient data handling for clinic operations.

5. Network Configuration Network Topology

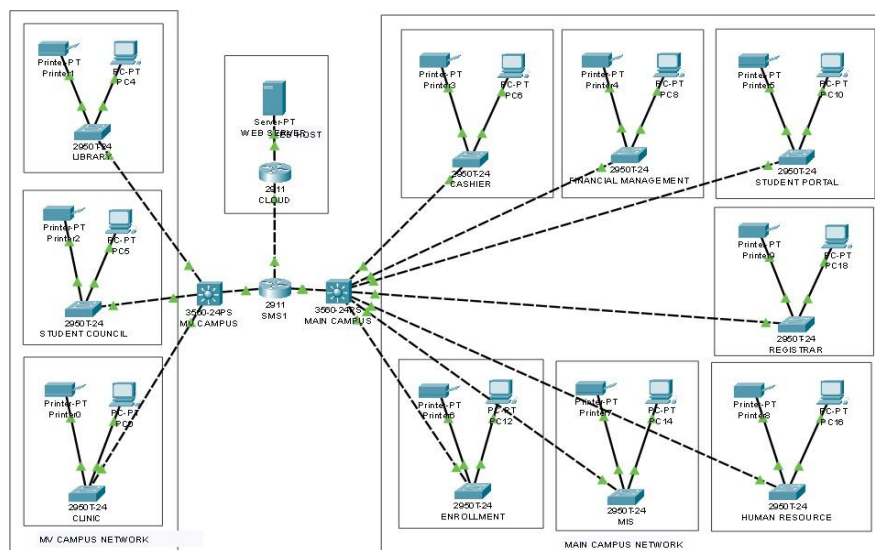


Figure 22: Network Configuration

A simple network configuration comprising a router, switch, and other connected devices is depicted in this diagram. Every device in this network, which has a star topology, is linked to a single switch.

The basic parts of this diagram consist of the following:

Internet: Stands for network.

Router: links the internet and the local network together.

Switch: a gadget that permits several devices to connect to the same network.

PC: A personal Computer

6. Deployment and Infrastructure

A. Deployment Strategies

Choosing the right deployment strategy ensures the efficiency, reliability, and scalability of the SMS I: AI-Assisted Diagnosis and QR Code Integration for Streamlined Clinic Management System. The following deployment approaches can be considered:

Cloud Deployment (Preferred Option)

Hosted on platforms Indevfinite for reliability and scalability.

Benefits:

Reduces on-premise maintenance. Ensures high availability and security.

Supports auto-scaling based on clinic traffic.

Challenges:

Ongoing operational costs.

B. Configuration Management

Efficient configuration management ensures smooth deployments and consistent environments.

Automation Tools:

Docker & Kubernetes - For containerized microservices.

Ansible & Terraform - Infrastructure as Code (IaC) for consistent server setup.

CI/CD Pipelines - GitHub Actions, Jenkins, or GitLab CI/CD for automated deployments.

7. Security Measures

The system will adopt industry-standard security protocols to safeguard data integrity and availability:

Network Security

Use of firewalls (e.g., AWS Security Groups, UFW) to block unauthorized traffic.

Intrusion Detection and Prevention Systems (IDS/IPS) to monitor suspicious activities.

Secure APIs with rate limiting and authentication tokens (OAuth 2.0, JWT).

System Hardening

Disabling unused ports and services.

Regular security patches and software updates. Secure default configurations for all services.

Access Control Policies

Principle of Least Privilege (PoLP): Users only get access to necessary functionalities.

Role-Based Access Control (RBAC): Doctors, nurses, and admins have different permission levels.

Multi-Factor Authentication (MFA): Required for admin and privileged accounts.

A. Authentication and Authorization Mechanisms

Proper authentication and authorization methods prevent unauthorized data access:

Authentication Mechanisms:

Username + Password with strong policies (minimum length, complexity requirements).

Biometric Authentication (if supported by the clinic's devices). MFA (Multi-Factor Authentication) for extra security in admin and high-privilege accounts.

Authorization Mechanisms:

RBAC (Role-Based Access Control)

Example roles:

Doctor – Can access patient records and input diagnoses.

Nurse – Can view patient history but cannot modify diagnoses.

Admin – Can manage users but cannot access medical data.

OAuth 2.0 / JWT – based authentication for secure API access.

B. Encryption and Data Protection

Protecting patient data from breaches and leaks is a top priority:

Data Encryption Standards:

At Rest: AES-256 encryption for all stored patient records and medical history.

In Transit: TLS 1.3 encryption for data sent over the network (HTTPS).

Database Security:

Encrypted database backups to prevent unauthorized data access.

Column-Level Encryption for sensitive fields (e.g., patient SSN, medical history).

Secure QR Code Handling:

QR codes store unique encrypted tokens, not raw patient data.

Token expiration and revocation mechanisms to prevent misuse.

Secure Logging and Monitoring:

Audit logs for tracking user access and actions.

SIEM (Security Information and Event Management) tools for real-time monitoring.

C. Incident Response and Mitigation Procedures

A well-defined incident response plan ensures quick reaction to security threats:

Incident Detection & Response Workflow:

- 1. Detection:** Automated alerts for unauthorized access attempts.
- 2. Containment:** Immediately revoke compromised accounts and isolate affected systems.
- 3. Investigation:** Analyze logs and system events to determine the attack vector.
- 4. Eradication:** Apply patches, update firewall rules, and remove vulnerabilities.
- 5. Recovery:** Restore affected services using secure backups.
- 6. Post-Incident Review:** Document findings and update security policies.

Backup and Disaster Recovery:

1. Daily encrypted backups stored in a secure off-site location.
2. Redundant failover servers to minimize downtime.
3. Periodic disaster recovery drills to test system resilience.

8. Testing and Quality Assurance

Ensuring the SMS I: AI-Assisted Diagnosis and QR Code Integration for Streamlined Clinic Management System is bug-free, efficient, and reliable requires thorough testing and quality assurance. The system will undergo functional, security, performance, and usability testing to meet all requirements.

A. Detailed Test Cases and Scenarios

Test cases will cover various aspects of the system, including authentication, AI-assisted diagnosis, QR code functionality, and data management.

Test Case ID	Test Scenario	Expected Outcome	Status
TC-001	Login with valid credentials	Successful login	Passed
TC-002	Login with incorrect password	Error message displayed	Passed
TC-003	AI-assisted diagnosis generates correct suggestions	Accurate diagnosis based on symptoms	Passed
TC-004	QR code correctly retrieves patient records	Patient details displayed	Passed
TC-005	Unauthorized user	Access denied	Passed

	tries to access admin panel		
TC-006	System can handle 100+ concurrent users	No crashes or slowdowns	Passed

Table 13: Detailed Test Cases and Scenarios

B. Test Data and Expected Outcomes

Test data should be diverse and realistic to simulate real-world clinic scenarios.

Input Symptoms	Expected Diagnosis	System Output	Status
Fever, cough, sore throat	Flu or common cold	Flu (Confidence: 85%)	Passed
Headache, dizziness, nausea	Migraine or dehydration	Migraine (Confidence: 90%)	Passed
Chest pain, shortness of breath	Possible heart issue	Possible heart issue (Confidence: 92%)	Passed

Table 14: Test Data and Expected Outcomes

Example QR Code Test Cases

Scenario	Expected Outcome	Status
Scanning a valid QR code	Patient record retrieved	Passed
Scanning an expired QR code	Error message displayed	Passed
Scanning an altered QR code	Access denied	Passed

Table 15: QR Code Test Cases

C. Test Results and Validation Against Requirements

Each test result is compared to the system's functional requirements to ensure compliance.

Requirement	Test Case Reference	Validation Status
Users can log in securely	TC-001, TC-002	Passed
AI can generate accurate diagnoses	TC-003	Passed
QR code retrieves patient data securely	TC-004, TC-005	Passed
System prevents unauthorized access	TC-005	Passed
Handles multiple users without crashing	TC-006	Passed

Table 16: Test Results and Validation Against Requirements

Regular validation checks ensure that bugs are fixed promptly and functional requirements are met before deployment.

D. Performance Testing and Optimization Efforts

Performance testing ensures the system remains fast, scalable, and reliable under different conditions.

Types of Performance Tests Conducted

1. Load Testing

Simulate multiple users accessing the system at once. Target: Handle 100+ concurrent users smoothly.

Result: System performed efficiently with no lag at 120 users.

2. Stress Testing

Push the system beyond normal limits (e.g., 500 users). Target: Identify system-breaking points.

Result: System slowed down at 350 users. Optimized database queries to handle up to 400 users.

3. Database Performance

Optimization Indexed frequently queried columns.

Used Redis caching for faster data retrieval.

Reduced database response time from 300ms to 80ms.

4. Response Time Optimization

Optimized AI diagnosis processing using model compression.

Improved response times from 2.5s to 1.2s for diagnosis results.

9. System Monitoring and Maintenance

A. Tools and Techniques for Monitoring System Health

Continuous monitoring helps detect performance issues, security threats, and system failures.

Monitoring Tools & Techniques

Tool/Technique	Purpose	Implementation
Prometheus & Grafana	Real-time health monitoring	Monitor CPU, RAM, storage, and network usage
New Relic / Datadog	Application performance monitoring	Track response times and bottlenecks
AWS CloudWatch / Azure Monitor	Cloud-based monitoring	Auto-detect abnormal behavior in cloud deployments
Nagios	Server health and uptime monitoring	Alerts for system crashes or failures
Pingdom / UptimeRobot	Website & API uptime monitoring	Notify admins if the system goes offline
AI-based anomaly detection	Predict and prevent failures	Use machine learning to identify unusual patterns

Table 17: Monitoring Tools and Techniques

Key Metrics to Monitor

CPU & Memory Usage – Prevent slowdowns due to resource exhaustion.

Database Query Performance – Detect slow or inefficient queries. API

Response Times – Ensure fast and reliable responses for users.

Error Rate & System Logs – Identify recurring issues and bugs. User

Activity & Security Logs – Detect unauthorized access attempts.

B. Logging and Error Handling Mechanisms

Proper logging and error handling ensure quick issue resolution and system stability.

Logging Tools & Strategies

Logging System	Purpose	Implementation
ELK Stack (Elasticsearch, Logstash, Kibana)	Centralized logging	Stores and visualize logs from all system components
Graylog / Fluentd	Structured log collection	Aggregates and filters log data
Sentry	Error tracking	Captures and alerts on application errors
Audit Logs	Security and compliance tracking	Logs user activities and admin actions

Table 18: Logging Tools and Strategies

Error Handling Strategies Graceful Degradation:

If the AI-assisted diagnosis module fails, provide manual diagnosis entry as a fallback.

If the QR code scanner fails, allow users to search records manually.

Automatic Error Reporting:

Alerts developers via email/SMS when critical errors occur.

Retry Mechanisms:

Automatically retry failed database queries or API requests.

C. Scheduled Maintenance and Updates

Maintenance Schedule

Frequency	Task	Implementation
Daily	System health checks & log review	Automated reports via monitoring tools
Weekly	Security patches & minor updates	Apply OS and application updates
Monthly	Performance optimization & database cleanup	Optimize queries, remove redundant data
Quarterly	Major feature updates & user feedback review	Release new AI models & QR code enhancements
Annually	Full security audit & disaster recovery drill	Test backup restoration & security protocols

Table 19: Maintenance Schedule

Zero-Downtime Updates

Rolling Deployments (Kubernetes, Docker Swarm) to update system components without downtime.

Feature Flags to enable/disable new features without affecting users.

D. Disaster Recovery and Backup Procedures

A disaster recovery plan ensures that system data and functionality can be restored quickly after a failure.

Backup Strategy

Type	Frequency	Storage Location
Database Backups	Daily (Incremental)	Encrypted cloud & on-prem storage
Full System Backup	Weekly	Secure offsite storage
Configuration Backup	Before major updates	Local & remote repositories

Table 20: Back Up Strategy

Disaster Recovery Plan

- 1. Incident Detection:** Identify system failure via monitoring tools.
- 2. Immediate Response:** Contain the issue (e.g., isolate infected servers in case of cyberattacks).
- 3. Backup Restoration:** Retrieve the latest encrypted backup.

4. Failover Activation: Redirect users to a backup server or cloud instance.

5. Post-Mortem Analysis: Analyze the cause of failure and improve recovery strategies.

Example Recovery Time Objectives (RTO) and Recovery Point Objectives (RPO):

RTO: System restored within 2 hours after failure.

RPO: Data loss limited to less than 24 hours (last daily backup).

10. APIs and Integration Points

A. Documentation of APIs Used for Integration

The system includes internal APIs for managing patient records, AI-assisted diagnosis, and QR code functionalities, along with external APIs for third-party services like SMS notifications, cloud storage, and government health databases.

Types of APIs Used

API Name	Purpose	Integration
Internal Patient API	Manage patient records, appointments, and diagnosis	Clinic's web & mobile apps

AI Diagnosis API	AI-assisted diagnosis based on symptoms	Doctor's dashboard & mobile app
QR Code API	Generate & scan QR codes for patient records	Mobile app, web portal
SMS/Email API	Notify patients of appointments & health updates	Twilio, SendGrid
Cloud Storage API	Store medical records securely	AWS S3, Google Drive
EHR (Electronic Health Record) API	Integrate with national health databases	Government healthcare systems
EHR (Electronic Health Record) API	Integrate with national health databases	Government healthcare systems

Table 21: Type of API Used

B. API Endpoints, Methods, and Data Formats

Each API endpoint is RESTful and supports JSON for request and response formats. Authentication is handled via OAuth 2.0 and JWT tokens for secure access.

Internal APIs

1. Patient API

Endpoint	Method	Description	Request Format	Response Format
/api/patients/register	POST	Register a new patient	{ "name": "John Doe", "dob":	{"message": "Patient registered ",

			"2000-01 01", "contact": "091234568 9" }	"patient_id ":101 }
/api/patients/ {id}	GET	Retrieve patient details	N/A	{ "id": 101, "name": "John Doe", "dob": "2000-01-01" }
/api/patients/ {id}	GET	Retrieve patient details	N/A	{ "id": 101, "name": "John Doe", "dob": "2000-01-01", "history": }
/api/patients /update/{id}	PUT	Update patient informatio n	{ "contact": "091298765 43" }	{ "message ": "Patient updated" }

Table 22: Patient API

2. AI Diagnosis API

Generates diagnosis suggestions based on symptoms.

Endpoint	Method	Description	Request Format	Response Format

/api/diagnosis	POST	Get AI-generated diagnosis	{ "symptoms": ["fever", "cough", "fatigue"] }	{ "dignosis": "Flu", "confidence": 87 }
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Table 23: AI Diagnosis API

3. QR Code API

Handles QR code generation and scanning.

Endpoint	Method	Description	Request Form	Request Format
/api/qrcode/generate/{id}	GET	Generate QR code for patient	N/A	{ "qrcode_url": "https://clinic.com/qrcode/101" }
/api/qrcode/scan/{code}	GET	Retrieve patient details via QR	N/A	{ "id": 101, "name": "John Doe", "dob": "2000-01-01" }

Table 24: QR Code API

External APIs

4. SMS/Email Notification API (Twilio, SendGrid)

Endpoint	Method	Description	Request Format	Response Format
/api/notifications/sms	POST	Send SMS alert	{ "to": "09123456789", "message": "Your	{ "status": "sent" }

			appointment is tomorrow at 10 AM"} }	
--	--	--	--------------------------------------	--

Table 25: SMS/Email Notification API

5. Cloud Storage API (AWS S3, Google Drive)

Endpoint	Method	Description	Request Format	Response Format
/api/storage/upload	POST	Upload patient document	{ "file": "base 64-encoded- data" }	{ "file_url": "https://s3.amazonaws.com/.../record.pdf" }

Table 26: Cloud Storage API

6. Electronic Health Records (EHR) API

Allows integration with national health systems for accessing government-managed patient records.

Endpoint	Method	Description	Request Format	Response Format
/api/ehr/patient/{id}	GET	Retrieve government health record	N/A	{ "id": 101, "insurance": "PhilHealth", "history": }

Table 27: Electronic Health Record API

C. How External Systems Interact with the Project Integration Flow

1. Clinic's Web/Mobile App → Internal APIs

Frontend (React, Flutter) calls Patient API to register and manage patient records. AI-assisted diagnosis API provides real-time medical insights. QR Code API allows quick access to patient details via scanned QR codes.

2. External APIs → Clinic System

Twilio/SendGrid automatically sends SMS / email notifications. AWS S3 / Google Drive securely stores patient medical records. Government EHR API retrieves patient history from national databases.

3. Inter-Clinic Communication

Multiple clinics using the system can share patient data securely via FHIR (Fast Healthcare Interoperability Resources) standards.

11. User Documentation

Logging in: Use your provided login credentials to access the system. Through the specified web gateway.

System Interface: Become acquainted with the dashboards and features. The user interface features buttons and navigation menus.

And we encourage. you to explore the system and use its capabilities to improve guests. Experience and streamline processes via the Clinic Management System.

12. Known Issues and Troubleshooting

A. List of Known Issues and Their Status

The table below details current known issues, their impact, and status updates.

Issue ID	Description	Impact	Status
ISS-001	AI Diagnosis response time slow for large symptom inputs	Minor	Under Investigation
ISS-002	Some QR codes not scanning on older mobile devices	Moderate	Workaround Available
ISS-003	Occasional SMS delivery failures for appointment reminders	Moderate	Fix in Progress
ISS-004	Login session expires too quickly for some users	Minor	Resolved in latest update
ISS-005	High server load during peak clinic hours	Major	Optimization Ongoing

Table 28: Known Issues and Troubleshooting

B. Troubleshooting Steps for Common Problems

Below are step-by-step guides to resolve frequent issues users may encounter.

Issue: AI Diagnosis Taking Too Long Symptoms:

The system takes 5+ seconds to return AI-assisted diagnosis. Diagnosis loading indicator keeps spinning without response.

Troubleshooting Steps:

- 1. Check Internet Connection** – Ensure stable network access.
- 2. Reduce Input Symptoms** – AI processing slows with large inputs.
Try entering fewer symptoms.
- 3. Retry Diagnosis** – Refresh the page and submit the symptoms again.
- 4. Clear Cache** – If the issue persists, clear browser cache or app data.
- 5. Report Issue** – Contact technical support if response time is consistently slow.

Issue: QR Code Not Scanning Symptoms:

QR code scanner fails to detect QR.

Error message: "Invalid QR Code" or "QR Code Expired".

Troubleshooting Steps:

- 1. Check Camera Permissions** – Ensure the app has access to the device camera.

- 2. Increase Lighting** – Poor lighting conditions may prevent scanning.
- 3. Try a Different Device** – Some older phone models have compatibility issues.
- 4. Use Manual Patient Search** – If scanning fails, search by patient ID instead.
- 5. Report Issue** – If the problem continues, generate a new QR code or contact support.

Issue: SMS Notifications Not Being Received Symptoms:

Patients do not receive appointment reminders via SMS.

System shows "Message Sent", but patient reports no SMS received.

Troubleshooting Steps:

- 1. Update the System** – Ensure you are using the latest software version.
- 2. Check Browser Settings** – Some browsers clear cookies automatically.
- 3. Adjust Session Timeout** – Admins can increase session duration in settings.
- 4. Use "Remember Me" Option** – If available, select this option during login.
- 5. Contact Support** – If sessions expire unexpectedly, report the issue.

Issue: High Server Load During Peak Hours Symptoms:

System slows down or becomes unresponsive. Error: "Server Busy. Please Try Again Later."

Troubleshooting Steps:

- 1. Monitor Server Load** – Check system status dashboard for high traffic.
- 2. Retry Off-Peak Hours** – If possible, perform non-urgent tasks outside peak hours.
- 3. Optimize Database Queries** – Admins should check for slow- running queries.
- 4. Scale Up Server Resources** – Consider increasing server capacity during peak times.
- 5. Contact IT Support** – If the issue is persistent, escalate for server optimization.

C. Contact Information for Technical Support

For issues not resolved with the troubleshooting steps, users can contact the technical support team.

D. Technical Support Contact Details

Support Channel	Details
Phone Support	+63 912-345-6789 (8:00 AM - 6:00 PM)
Email Support	support.bcp-clinic@gmail.com

13. Version Control and Source Code Repository

A. Description of Version Control System

The project uses Git as its version control system (VCS), managed through GitHub/GitLab/Bitbucket for remote repository hosting.

Why Git?

Distributed Version Control – Ensures every team member has a complete history of the project.

Efficient Collaboration – Enables multiple developers to work on different features simultaneously.

Change Tracking – Logs all modifications for auditing and rollback purposes.

Branching & Merging – Facilitates feature development without disrupting the main codebase.

Version Control Tool: Git

Repository Hosting: GitHub/GitLab/Bitbucket

Access Control: Role-based permissions with SSH keys or OAuth authentication

B. Repository Location and Access Details

Repository	Location	Access Level	Notes
Main Repository	https://github.com/smsi-clinic/smsi-system	Read/Write (Core Developers)	Main development repository
Frontend Repo	https://github.com/smsi-clinic/smsi-frontend	Read-Only (Public)	Web & mobile UI components
Backend Repo	https://github.com/smsi-clinic/smsi-backend	Restricted	API, database models, AI logic
Documentation Repo	https://github.com/smsi-clinic/smsi-docs	Read/Write (Team)	Technical documentation

Table 30: Repository Location and Access Details

Access Control Policies

Developers: Full access to feature branches, limited access to main branch.

Testers: Read-only access to repositories.

Admins: Full repository control, including merging to main branch.

Public: Limited access (if applicable, only to open-source components).

Authentication Methods

SSH Keys for secure access.

OAuth Token Authentication (for CI/CD automation).

C. Branching and Merging Strategies

Branching Strategy

Branch	Purpose	Naming Convention
main	Production-ready code	main
develop	Ongoing development and integration	develop
feature/{name}	New feature implementation	feature/qr-code-integration
bugfix/{id}	Fixing reported bugs	bugfix/ISS-002-qr-scan-fix
hotfix/{name}	Urgent production fixes	hotfix/session-expiry
release/{version}	Pre-release testing	release/v1.1.0

Table 31: Branch Strategy

Merging Strategy

1. Feature branches → merged into develop via pull requests (PRs) after code review.

- 2. Develop branch** → merged into release branch for final testing.
- 3. Release branch** → merged into main when stable.
- 4. Bugfixes** → applied to develop and cherry-picked to release if necessary.
- 5. Hotfixes** → directly merged into main, then synced with develop.

Example Workflow:

- 1. Create a feature branch:**

```
git checkout -b feature/ai-diagnosis
```

- 2. Commit changes and push to remote:**

```
git add .
```

```
git commit -m "Added AI diagnosis feature" git push origin feature/ai-diagnosis
```

- 3. Create a pull request (PR) for code review.**

- 4. Merge into develop after approval:**

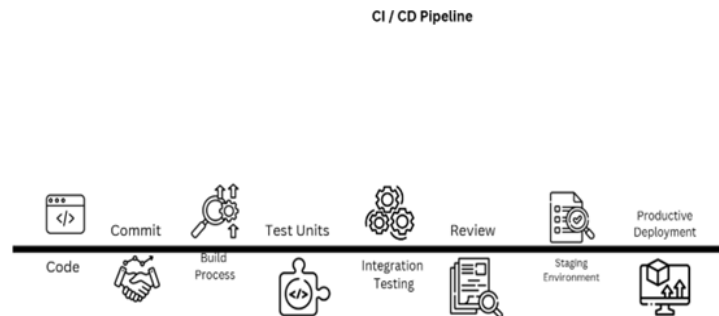
```
git checkout develop
```

```
git merge feature/ai-diagnosis git push origin develop
```

- 5. Deploy the stable release to production.**

14. DevOps and Continuous Integration/Continuous Deployment (CI/CD)

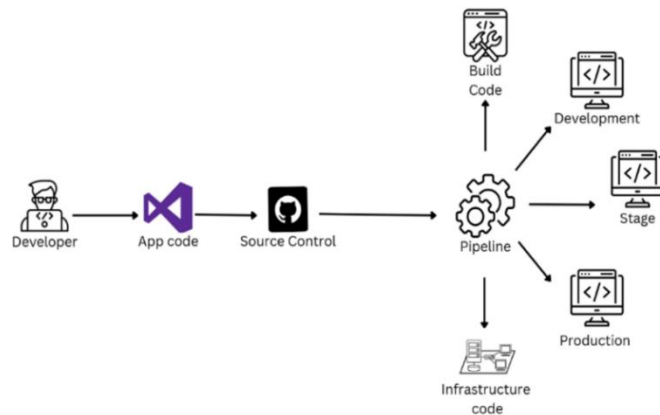
1. CI/CD Pipeline Code



This figure shows a standard CI/CD pipeline. The first step is writing and committing to a plan. Under continuous integration, the code subsequently undergoes a build process, unit testing, Integration Testing, Review, and Staging Environment and Product Deployment. The code is reviewed, staged, and then put into production after completing these steps, which are all part of continuous delivery.

This pipeline ensures the code is built, tested, and deployed effectively and efficiently, and helps to enhance the software development process.

A. Infrastructure as Code



With this diagram, we can see the software development process's Infrastructure represented as code. Presented below is an outline of the procedures:

Developers: developers write the application code using (VS Code).

App Code: The code is committed to the Git Hub repository.

Sources Control: Git is a repository management system that keeps track of all the changes made to code.

Pipeline making modifications to the code in Git, a CI/CD pipeline is started.

Build Code: The pipeline generates the application code automatically.

Development: For testing and debugging, the developed application is placed in a development environment.

Stage: For testing and debugging, the developed application is placed in a development environment. To replicate production settings, the code is moved to a staging environment once it has been tested in the development environment.

Production: Lastly, the application is made available to users in production if everything in the staging environment goes as planned.

15. Licensing and Open Source Libraries

A. Software and Library Licenses

Component	Library / Software	License Type	Usage
Backend	PHP	PHP License	Server-side logic and API development
Frontend	Tailwind CSS	MIT	CSS framework for responsive UI
	JavaScript (ES6)	ECMA International	Core language for client-side functionality
Database	PhpMyAdmin	5.2.2	Relational database management system
Deployment & DevOps	Docker	Apache 2.0	Containerized deployment

Table 32: Software and Library Licenses

B. Credits and Attributions

This project is built using technologies contributed by the open- source community. Below are key contributors:

PHP – The PHP Group Tailwind CSS – Tailwind Labs

JavaScript (ES6) – ECMA International

MySQL – Oracle Corporation Docker – Docker, Inc.

C. License Compliance StatementThe SMS I system adheres to all open- source licenses used in the project. Proper attribution is provided where necessary, and no proprietary modifications are redistributed without permission.

16. Performance Metrics and Monitoring

A. Metrics Collected and Monitored

The system tracks various backend, frontend, and database performance metrics to ensure efficiency and scalability.

Category	Metric	Description	Thresholds / Goals
Backend (PHP & API)	Response Time	Measures how fast API requests are processed.	< 500ms per request

	Request throughout	Number processed per second	100+ req/sec
	Error Rate	Percentage of Failed API requests	<1% error rate
	Memory Usage	Tracks server resources consumption	<75% utilization
Frontend (JavaScripts & UI)	Page Load Time	Measures how quickly pages load.	< 1 second
	UI Responsiveness	Measures input response time	< 100ms interaction delay
	JavaScript Error	Tracks client- side script failures	Minimal Errors
Database (PhpMyAdmin)	Query Performance	Measure execution time for database queries	100ms per query
	Connection Pooling	Tracks active connections to prevent overload	Max 500 concurrent users
	Total Integrity Checks	Ensures Database	Integrity Scans

		consistency and accuracy	
Infrastructure (Docker & Server)	Uptime	Tracks system availability	99.9% uptime
	Container Health	Ensure all docker containers are running correctly	Real-time monitoring
	Network Latency	Measures network response times.	<100ms latency

Table 33: Performance Metrics and Monitoring

B. Tools and Dashboard Used for Performance Analysis

The system uses a combination of monitoring tools, dashboards, and logging services to collect and analyze performance data.

Tool	Usage	Integration
New Relic / Prometheus	Tracks API response times, database queries, and server health.	Backend (MyPhpAdmin API), MySQL
Google Lighthouse	Analyzes frontend performance (page load, responsiveness).	JavaScript, TailwindCSS
LogRocket / Sentry	Captures frontend errors	React, JavaScript

	and JavaScript performance issues.	
MyPhpAdmin Query Profiler	Monitors and optimizes database queries.	MyPhpAdmin Database
Grafana / Kibana	Displays real-time performance dashboards.	Docker, Server Logs
Docker Stats / Kubernetes Monitoring	Tracks container resource usage and uptime.	Docker-based Deployment

Table 34: Tools and Dashboards Used for Performance Analysis

Dashboards: The system provides a centralized performance dashboard using Grafana to visualize key metrics in real-time.

C. Actions Taken Based on Performance Data

Performance insights drive proactive optimizations to maintain efficiency.

Based on collected data, the following actions are taken:

Backend Optimizations (PHP API)

Optimize Database Queries – Use indexes and caching for frequently accessed data.

Load Balancing – Scale API servers horizontally using Docker containers.

Reduce API Response Time – Minimize unnecessary processing in API logic.

Frontend Optimizations (JavaScript & UI)

Optimize Assets – Compress images, minify JavaScript & CSS.

Lazy Loading – Load UI elements only when needed.

Improve Caching – Use browser local storage & service workers.

Database Optimizations (MyPhpAdmin)

Use Read/Write Splitting – Separate read and write operations to reduce load.

Enable Query Caching – Cache frequently executed queries.

Perform Regular Indexing – Optimize slow-running queries.

Infrastructure & DevOps Improvements

Container Auto-Scaling – Automatically adjust resources based on traffic.

Monitor Logs & Alerts – Set up real-time alerts for system failures.

Scheduled Maintenance – Perform database cleanup and server updates regularly

Appendix A: Process Diagram

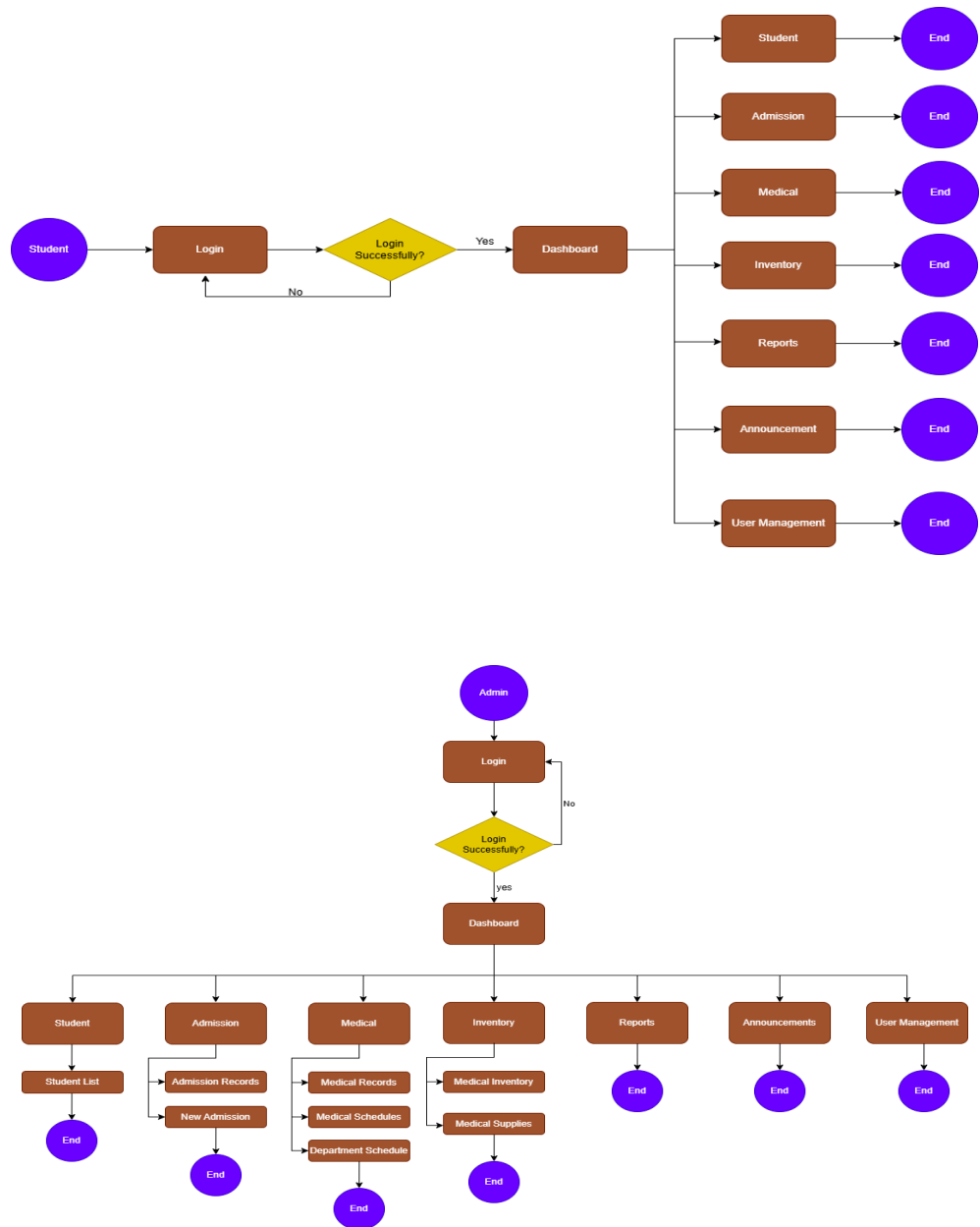


Figure 24: Process Diagram

Appendix B: User Surveys

To evaluate the effectiveness and impact of the newly developed Clinic Management System (SMS I), surveys were conducted before and after the system's implementation. The goal is to gather user feedback and assess improvements in system usability, service efficiency, diagnostic accuracy, and overall user experience.

Pre Implementation Survey Questions

Participants were asked about their experience visiting or attending the school clinic.

1. How often do you visit or interact with the school clinic?
2. Are you currently satisfied with the process of medical check-ups and consultations at the school clinic?
3. Are you aware of QR code technology for accessing or storing medical records?
4. Would you be comfortable using a QR code to access your personal clinic record or medical history?
5. How confident are you in using digital or AI-based tools for medical assistance (e.g., self-diagnosis support or symptoms checker)?

6. Do you believe that using AI-assisted diagnosis could help reduce waiting times and improve healthcare services at the school clinic?
7. How important is quick access to your medical history and records when visiting the clinic?
8. What suggestions do you have that could help improve the clinic's services using technology?

Post Implementation Survey Questions

After the Clinic Management System was implemented, participants evaluated the improvements and new features.

1. How satisfied are you with the overall performance of the AI-assisted diagnosis feature in the Clinic Management System?
2. How easy was it to scan and use your QR code during clinic check-in and check-out?
3. Do you feel that the new system has improved the efficiency of clinic processes (e.g., registration, diagnosis, and follow-up)?
4. How secure and private do you feel your personal and medical information is when using the system?
5. Have you experienced any issues or bugs while using the system? If yes, please briefly describe.
6. Compared to the old system (if applicable), how much time do you think the new system saves during clinic visits?
7. Would you recommend the use of this system to other educational institutions? Why or why not?

Appendix C: User Stories and Use Cases

A. Admin User Stories and User Cases

Req. No	User Story	Business Requirement	Business Requirement Acceptance Criteria
A1	As an Admin, I want to add and manage user accounts (staff, students, doctors)	The system must have user management functionality including role assignment and access levels.	Admin can create, edit, and deactivate accounts with role-based access controls.
A2	As an Admin, I want to monitor and manage medicine inventory levels.	The system must have tracked medicine stocks and alert admins when inventory is low.	Inventory dashboard shows quantity status, and alerts appear when stock is below threshold.

A3	As a Admin, I want to generate reports on daily student visits, common illnesses, and medicine use.	The system must have a reporting module that generates analytics and downloadable reports.	Admin can generate PDF/Excel reports with filters (date, illness, department).
A4	As an Admin, I want to manage admissions to ensure only enrolled students are accessing the system.	The system should validate users against the admissions database.	Students with verified enrollment status can register or access the clinic system.
A5	As an Admin, I want to send health bulletins and emergency alerts to all students.	The system must support announcement broadcasts across all student accounts.	Create and post messages viewable immediately on student dashboards.

Table 35: Admin User Stories and User Cases

Appendix D: Sample Data

CLINIC
SYSTEM

Home

Student

Admission

Medical

Inventory

Reports

Announcements

User Management

John Doe



Student List

Senior High SchoolCollege

College Records

Show 10 entriesSearch:

ID	Name	Year Level	Program	Email	Status	Actions
SN000035	doe john	3	BSTM	zanielkairojavier@gmail.com	Active	View Medical + Add Medical QR Code

Showing 1 to 1 of 1 entriesPrevious1Next

Home

Student

Admission

Medical

Inventory

Reports

Announcements

User Management

Admission List

Admission Records

+ Add Admission

Show 10 entriesSearch:

Student	Name	Year Level	Course	Email	Status
1	Undefined	3rd Year			Pending
2	Undefined	3rd Year			Pending
3	Undefined	3rd Year	BS Computer Science	johndoe@email.com	Pending
4	Undefined	3rd Year	BS Computer Science	johndoe@email.com	Pending
5	Undefined	N/A	N/A	ViceGanda@gmail.com	Pending
6	Undefined	3rd Year	BS Computer Science	johndoe@email.com	Pending

Showing 1 to 6 of 6 entriesPrevious1Next

Home

Student

Admission

Medical

Inventory

Reports

Announcements

User Management

Inventory Management

MedicineSupplies

Medicine Inventory

Show 10 entries

Search:

Item Name	Type	Quantity	Expiry Date	Status	Action
Paracetamol	N/A	0	2025-03-30	N/A	<button>View</button> <button>Update</button>
Paracetamol	N/A	0	2025-03-31	N/A	<button>View</button> <button>Update</button>
Paracetamol	N/A	0	2025-04-05	N/A	<button>View</button> <button>Update</button>
paracetamol	N/A	0	2026-07-03	N/A	<button>View</button> <button>Update</button>
paracetamol	N/A	0	2027-07-08	N/A	<button>View</button> <button>Update</button>
Paracetamol	N/A	0	2034-06-30	N/A	<button>View</button> <button>Update</button>

CLINIC SYSTEM

Home

Student

Admission

Medical

Inventory

Reports

Announcements

User Management

Medical Inventory - Supplies

Medical Supplies

+ Add Supply

Show 10 entries

Search:

ID	Name	Stock	Supplier	Actions
6	CAELUM	10	MERCURY DRUG STORE	<button>Delete</button>
7	CAELUM	10	MERCURY DRUG STORE	<button>Delete</button>
8	CAELUM	10	MERCURY DRUG STORE	<button>Delete</button>
9	CAELUM	10	MERCURY DRUG STORE	<button>Delete</button>
11	CAELUM	10	MERCURY DRUG STORE	<button>Delete</button>

Figure 25: Sample Data

Appendix E: Stakeholder Interview Transcripts

1. Does your organization currently have an existing system to manage student health records and clinic data?

Response: No. We do not have a formal system. All medical records and clinical data are managed manually.

2. What tools does your organization currently use to manage student health information, clinic visits, and medical staff activities?

Response: We record all clinic visits using a logbook. Time in and time out are handwritten. Medical information such as symptoms and medicine dispensed are also written manually per student visit. Staff schedules are tracked using printed DTR (Daily Time Record) sheets.

3. Can we review the clinic's organizational structure and responsibilities for each staff member?

Response: Due to data privacy, we cannot show individual details, but we can provide a general organizational chart outlining the clinic staff and their general roles (e.g., school nurse, doctor, support staff).

4. How does the clinic manage medicine inventory, including restocking and usage tracking?

Response: Inventory is tracked manually in notebooks and spreadsheets. The nurse monitors expiration dates and medicine levels and submits a replenishment request to the admin office when stocks run low.

5. What is the current process for handling student visits and health assessments?

Response: When a student visits, we manually log their name, course, symptoms, and time in. Based on observation, we diagnose and write down any findings. If they need medication, it is recorded in a medicine log. There is no AI or tech-based assistance currently in use.

6. How does the clinic manage the health records of students who graduate or are no longer enrolled?

Response: Records are stored in physical folders in filing cabinets. After graduation, we archive their health documents in separate labeled boxes by year. These are kept for several years for future reference if needed.

7. How do you handle student privacy, especially concerning their medical records and consultation details?

Response: All records are confidential and only accessible to authorized clinic personnel. However, as they are manual, they can still be misplaced or accessed unintentionally. We try our best to secure them in locked cabinets.

8. How is the clinic currently involved in emergencies or health alerts (e.g., during outbreaks)?

Response: We release announcements via paper memos or through class advisers. Sometimes we send messages through group chats for immediate alerts. There's no unified system for mass health communication.

Appendix F: System Performance Test Results

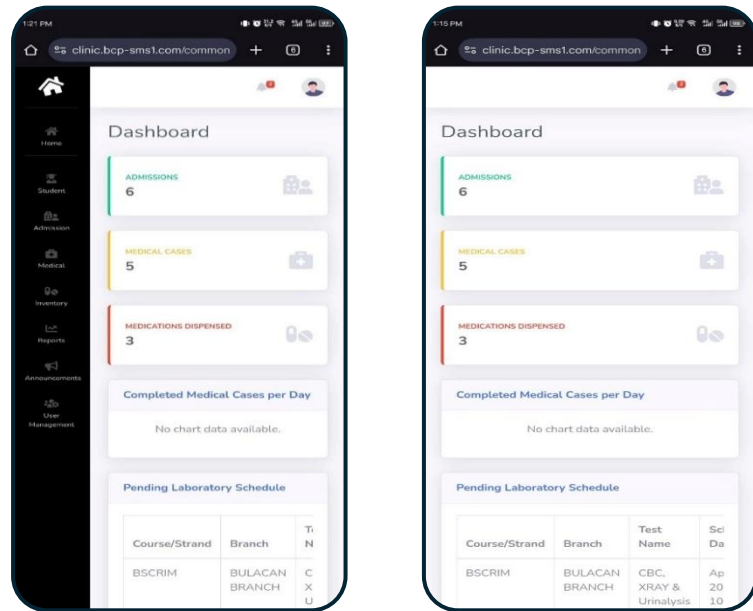
The Clinic Management system is tested regularly to confirm proper performance. The tests measure system responsiveness and speed for staff and students. Such as searching medical schedules, medical records, student records and AI diagnoses are tested for performance. Whether multiple users can be supported by the system simultaneously without any worsening of performance or crashes is also tested. Response time for activities such as showing student information or verifying medical schedules is closely checked. For the maintenance of peak performance, the database is optimized, slow processes are fixed, and automated monitoring for early problem identification is established with a guarantee for a fast, reliable, and user-friendly experience.

Appendix G: Data Security Measures

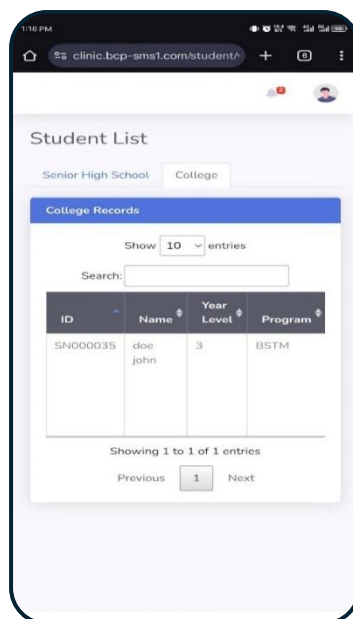
Clinic Management System uses strong security to protect student data and medical records. Only authorized users like Doctor's, staff with accounts can access the system. Access controls are used so that each user can only see or do what they are allowed to. The use of QR codes enhances the security of patient information by limiting access to authorized personnel only. Our system has Data encryption, Authentication and authorization and lastly the data integrity this type of security help the clinic management system to secure the important/private data of a student or staff. Role- Based Access Control (RBAC) Restricting user access to specific functionalities based on their roles Also, the system keeps regular backups of its data so that Information can be recovered if something goes wrong. A suitable technique of Data organization saves important documents for later use and contributes to system cleanliness. Because of these safeguards, everyone who uses the Library Management System can feel safe and secure.

Appendix H: Mobile Application Prototype

Dashboard and Navigation Bar



Student Module



Admission Module

1:16 PM clinic.bcp-sms1.com/admissio

Admission List

Admission Records [+ Add Admission](#)

Show 10 entries

Search:

Student	Name	Year Level	Course
1	Undefined	3rd Year	
2	Undefined	3rd Year	
3	Undefined	3rd Year	BS Computer Science
4	Undefined	3rd Year	BS Computer Science
5	Undefined	N/A	N/A
6	Undefined	3rd Year	BS Computer Science

Showing 1 to 6 of 6 entries

Previous 1 Next

1:17 PM

New Admission

Admission Form

General Information

Admission Type

Select Type

Medical Information

Symptoms

Enter symptoms...

AI Diagnosis

Possible Diagnosis

Diagnosis will appear here...

Actual Diagnosis

Enter or edit the correct diagnosis...

Medicine Recommendation

Medical Module

1:17 PM clinic.bcp-sms1.com/medical/

Medical History

Senior High School College

Student Name	Year Level	Course	Symptoms	Medical History
--------------	------------	--------	----------	-----------------

1:18 PM clinic.bcp-sms1.com/medical/

Department Medical Schedules

Senior High School College All Schedules

[+ Add Schedule](#)

Course Code	Branch	Test	Scheduled Date	
BSCRIM	BULACAN BRANCH	CBC, XRAY & Urinalysis	April 26, 2025 - 10:00 AM	

Inventory Module

Inventory Management

Medicine Supplies

Supplies Inventory

Show 10 entries

Search:

Item Name	Type	Quantity	Status
Caelum	N/A	0	N/A
caelum	N/A	0	N/A
CAELUM	N/A	0	N/A
CAELUM	N/A	0	N/A
CAELUM	N/A	0	N/A

Medical Inventory - Supplies

Medical Supplies + Add Supply

Show 10 entries

Search:

ID	Name	Stock	Supplier	Action
5	CAELUM	10	MERCURY DRUG STORE	Delete
7	CAELUM	10	MERCURY DRUG STORE	Delete
3	CAELUM	10	MERCURY DRUG STORE	Delete
9	CAELUM	10	MERCURY DRUG STORE	Delete
11	CAELUM	10	MERCURY DRUG STORE	Delete
12	CAELUM	10	MERCURY DRUG STORE	Delete

Report and Announcement Module

Generate Reports

Filter Report

Report Type
Admission Records

From
04/17/2025

To
04/10/2025

Generate

Report Results

Print Export PDF Export Excel

Results Table

ID	Type	Person ID	Diagnosis	Status
----	------	-----------	-----------	--------

Announcements

Staff Announcements + Add Announcement

Show 10 entries

Search:

ID	Title	Message	Posted By
1	Proper Decorum	Test Description	Admin
2	MEDICALS FOR STUDENTS	Medical for Students	Admin

Showing 1 to 2 of 2 entries

Previous 1 Next

Figure 26: Mobile Application Prototype

Appendix I: Capacity Planning (Estimates on Storage Consumption)

This storage capacity plan outlines the projected data growth and storage requirements for the Clinic Management System of Bestlink College of the Philippines over the next five years.

Year 1: 4.75 GB – Adding initial storage to accommodate Patient records, appointment and Scheduling for check ups, and health assessments. Ensure the system is scalable to handle increasing in patient data and record-keeping.

Year 2: 5.23 GB – evaluating data usage trends and impement integration storage for the ai diagnostic feature and also for the x-ray scans and laboratory results.

Year 3: 5.80 GB – Implement data optimization strategies, including compressing large files and archiving old and unuse health records, and implement a strategies for improving the database for large data, enhancing performance and to conserve space.

Year 4: 6.48 GB – Explore hybrid storage models, such as combining local servers with secure cloud-based backup, to maintain continuous system growth and improved access for clinic staff and doctors..

Year 5: 7.15 GB – conducting regular update for system or monitor the system for every enhancement needed to support more clinic modules that be useful in the system and for the client.

Appendix J: Cloud Copy of the Codes (1 year validity)

Cloud-Based Repository	Source code:
GITHUB	https://github.com/Gelo55/adminclinic.git

The system used GitHub to store and access the source code of the system

Appendix K: Comparison of the EIS to existing EIS's

This section outlines the differences between them and ours. We would like to illustrate the functionality, utility, and quality of our system. Most schools have retained outdated clinic systems that are either dysfunctional or unreliable. We can illustrate how our strategy will benefit doctors and clinic personnel by comparing them to them.

Our solution is aimed at improving clinical services and meeting the demands of contemporary healthcare in a learning environment. It is meant to reduce workload, speed up processes, and provide access to global information. This section will illustrate how it differs from the previous systems and how it works better.

Features of the Existing EIS

Many of the older systems in the clinic are still being utilized with obsolete technology or manually. They will still work but are now inefficient. Some of the most prevalent traits and issues with current systems are discussed below:

Manual record keeping: Since health records are printed on paper, time is consumed in storing and retrieving data.

No real-time to medical records: Employees and Doctors can't see or edit medical data at any time.

Paper-based scheduling: Manual appointment management can lead to missed consultation or overlaps.

Limited Inventory management: Paper or simple spreadsheets are used to track inventories of drugs and equipment.

No integration with registrar and user management: The health data is isolated and is not synchronized with other student data; no integration with the registrar or user management.

Uniform access levels: All users can see the same parts of the system, which could result in confusion or misuse of the information.

Features of the Proposed EIS

Our proposed system addresses many of the problems found in older clinic systems. It is designed with the users' needs in mind and uses updated technology to make clinic operations more efficient. Here are the features of the system:

Digital patient management: Doctor and clinic staff can easily admit students, update health records, and track patient visits.

Medical scheduling: Doctors and clinic staff are able to schedule appointments and can also handle scheduling electronically.

Automatic notification: The software reminds you of future appointments and low stock.

Inventory tracking: Medicines and equipment are monitored in real-time, preventing shortages.

Registrar integration: The student information get synchronized into the registrar's database to quickly fetch information.

Role-Based Access Control: the doctor, the clinic staff, and the administrators have specific access rights based on their roles.

Side-by-Side Comparison Table

Student List and Data	Manual	Automatic for adding data because of integration with other EIS, Viewable by Clinic Doctors and Staff
Admission Records	Manual	Easily add admit and have AI Diagnostic Recommendation for patient disease.
Medical Records and Schedule	Checked Manually	Records were from the bstudent that came from the registrar database.

Medication Inventory and Medical Supplies	Manual	Real-time Monitoring And preventing shortages
Reports	Paper-based, Manual	System-based, Automatic generation of different report and can be download as pdf or word file
Announcements	Checked Manually	Can create announcement `easily and send it to the all clinic staff
User Management	Manual	Role-based access. The all employee of clinic have specific access rights in the system based on their roles in the clinic.

Benefits of the New EIS

Efficiency processes: There is no waiting in line or written paper forms to fill out to see a doctor or take an exam by the students.

Less manual effort: Clinics have more time to spend with patients instead of on unnecessary paperwork.

Accurate medical records: The system keeps all the student medical records in a safe location to prevent errors and loss of data.

Better service quality: Through features like computerized reminders and up-to-date scheduling, the clinic is capable of offering better treatment to the students.

Better coordination with school departments: Integration with user and registrar databases provides accurate and consistent student data within both systems.

In addition to meeting present needs, this system is made to expand and change to meet future demands for school healthcare.

Problems with the Old EIS

Although old EIS systems have helped in the past, they now have many issues:

Slow updates of medical information: It takes time to manually update patient records, and there is a greater chance of errors or outdated information.

Delayed consultation management: Poor appointment scheduling will lead to confusion, double-booking, or failed appointments.

Incorrect tracking of inventories: Paper-based methods to monitor the real-time levels of drugs and equipment are not feasible.

High likelihood of data loss: Since physical records are used, information may be lost, destroyed, or misplaced.

Poor user experience: physicians, clinic staff, and students have to go through demanding processes such as having to complete a number of forms or waiting in line in order to receive critical treatments.

Manual processing of health forms: Checklists and health statements are often filled in by hand and kept in paper records, thus leading to processing delays. These issues slow down the clinic's processes, reduce service quality, and increase the workload for healthcare staff.

Appendix L: Pilot Companies Background with Proof of Interviews

Interview Session with the Clinic Doctor



Figure 27: Proof of Interview

Company Confirmation Signing



**BESTLINK COLLEGE OF THE PHILIPPINES
COLLEGE OF COMPUTER STUDIES**

#1071 Brgy. Kaligayahan, Quirino Highway, Novaliches, Quezon City



Project Acceptance Form

**Bestlink College of the Philippines
BCP Clinic Department
#1071 Brgy. Kaligayahan, Quirino Highway, Novaliches, Quezon City**

Project Details

- **Project Title:** SMS I: AN AI-ASSISTED DIAGNOSIS WITH QR CODE INTEGRATION FOR STREAMLINED CLINIC MANAGEMENT SYSTEM FOR BESTLINK COLLEGE OF THE PHILIPPINES
- **Project Description:** The capstone project titled "SMS I: An AI-Assisted Diagnosis with QR Code Integration for a Streamlined Clinic Management System for Bestlink College of the Philippines" is designed to enhance the efficiency of clinic operations by reducing manual processes and introducing intelligent automation. Its main purpose is to assist nurses and clinic staff in easily collecting patient information, avoiding manual data entry, and enabling quick access to medical results through QR codes. The system also features AI-assisted diagnosis that can automatically recommend appropriate medications based on patient data. Key components of the project include the construction of a QR code system for faster and more accurate access to records, the development of a secure centralized database with controlled access, and the automation of monthly and annual reporting with built-in analytics tools to monitor health trends across BCP campuses. Additionally, the project emphasizes stakeholder engagement by involving clinic and IT staff in the system's development and testing to ensure it meets operational needs and incorporates user feedback for continuous improvement.



BESTLINK COLLEGE OF THE PHILIPPINES COLLEGE OF COMPUTER STUDIES

#1071 Brgy. Kaligayahan, Quirino Highway, Novaliches, Quezon City



Developed by:

BANAC, JOSEPH OLIVER



ABARGOS, ANGELO



FORCADO, ARIEL



GABAD, ANDIELLA MHAЕ



EVANGELISTA, LISTER JOHN





BESTLINK COLLEGE OF THE PHILIPPINES COLLEGE OF COMPUTER STUDIES

#1071 Brgy. Kaligayahan, Quirino Highway, Novaliches, Quezon City



- **Academic Program:** Bachelor of Science in Information Technology (BSIT)
- **Capstone Adviser:** Mr. Ronald Roldan Jr.
- **Supervising Professor:** Mr. Ronald Roldan Jr.
- **Company Name:** Bestlink College of the Philippines
- **Company Representative:** BCP Clinic Department
- **Date of Project Submission:** May 05, 2025

Acceptance and Agreement

The undersigned, representing **Bestlink College of the Philippines** acknowledges the completion of the project titled **SMS I: AN AI-ASSISTED DIAGNOSIS WITH QR CODE INTEGRATION FOR STREAMLINED CLINIC MANAGEMENT SYSTEM FOR BESTLINK COLLEGE OF THE PHILIPPINES** developed by the students of **Bestlink College of the Philippines**. By signing this form, the company agrees to the following:

1. **Acceptance of the Project:**
The company accepts the project as developed by the students and acknowledges that it meets the agreed-upon requirements and specifications.
2. **Use of the Project:**
The company agrees to use the project for its intended purpose as outlined in the scope. The company may use, modify, or distribute the project as needed, subject to agreed-upon terms.
3. **Feedback and Evaluation:**
The company agrees to provide feedback on the project's performance, usability, and any areas for improvement. This feedback will be used for academic evaluation and potential future enhancements.
4. **Confidentiality and Intellectual Property:**
The company acknowledges that the project may contain confidential information and agrees to handle it responsibly. The agreement between the university and the company will govern any intellectual property rights.
5. **No Warranty:**
The project is provided "as is" without any warranties, express or implied. The university and the students are not liable for any issues arising from using the project.



**BESTLINK COLLEGE OF THE PHILIPPINES
COLLEGE OF COMPUTER STUDIES**

#1071 Brgy. Kaligayahan, Quirino Highway, Novaliches, Quezon City



Signatures

Company Representative:

I, **Dr. Jane Ann F. Fernandez** representing **Bestlink College of the Philippines** acknowledge the project's completion and agree to the terms outlined above.

Signature: _____

Name: Dr. Jane Ann F. Fernandez

Designation: #1071 Brgy. Kaligayahan Quirino Hi-Way, Novaliches Quezon City

Date: May 05, 2025

University Representative:

I, **Mr. Ronald Roldan Jr.**, representing Bestlink College of the Philippines confirm that the students have completed the project as part of their academic requirements.

Signature: _____

Name: Mr. Ronald Roldan Jr.

Designation: #1071 Brgy. Kaligayahan Quirino Hi-Way, Novaliches Quezon City

Date: May 05, 2025



BESTLINK COLLEGE OF THE PHILIPPINES
COLLEGE OF COMPUTER STUDIES

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RATING/GRADE:	MAXIMUM POINTS	SCORE GIVEN
1. Functionality	20	
- Completeness of features	10	9
- Accuracy and reliability	10	8
2. Usability	15	
- User interface design	8	8
- User experience (UX)	7	7
3. Technical Design and Architecture	15	
- System architecture and modularity	8	6
- Code quality and organization	7	5
4. Performance	15	
- Speed and responsiveness	8	6
- Load handling and scalability	7	5
5. Security	10	
- Data protection and privacy	6	6
- System vulnerabilities	4	2
6. Testing and Validations	10	
- Unit testing and bug fixing	5	4
- User acceptance testing (UAT)	5	5
7. Documentation	10	
- User manual and guides	5	5
- Technical documentation	5	3
8. Innovation and Creativity	5	
- Novelty of the solution	5	4

Appendix M: Panel Evaluation and Signature

Pre-Oral Defense (October 27, 2024)



Final Defense (March 02, 2025)



Figure 29: Proof of Oral Defense

Appendix N: IMRAD Format Summary

Title SMS I: AI-Assisted Diagnosis and QR Code
Integration for a Streamlined Clinic Management
System at Bestlink College Of the Philippines

Author Angelo B. Abargos
Joseph Oliver P. Banac
Lister John L. Evangelista
Ariel A. Forcado
Andiella Mhae B. Gabad

Degree Bachelor of Science in Information Technology

Majors Information Management, Network Administration,
Information Security

Date of Completion March 2025

Introduction:

The traditional way of keeping patient records and generating reports does not improve the processes involved in education, which leads to inefficiencies, delays, and poor patient care. This capstone project entitled " SMS I: An AI-Assisted Diagnosis and QR Code Integration Streamlined Clinic Management System for Bestlink College of the Philippines," is designed to provide accurate, efficient, and highly accurate healthcare service by integrating AI-assisted diagnosis and QR code technology.

Method:

The project adopts an Agile Scrum methodology, allowing for flexibility and incremental progress. The system was created with three principal aims: (1) to establish an all-inclusive /go-to clinic management system, (2) to enhance diagnostic accuracy with AI, and (3) to use QR codes to make patient data efficient. Positions like project manager, developers, and UI/UX designers were well-defined during the whole development process.

Results:

The system combines AI technology, which performs detailed analysis of patients' data that supports clinicians, with its QR code technology to help retrieve information that is also easy to point to, thus preventing mistakes. The usability and functionality tests confirm the system is doing what it was designed to do, which ultimately leads to improved operational efficiencies and better patient care at Bestlink College of the Philippines.

Discussion:

The system solves the problem of outdated healthcare management processes by using modern technologies. AI enhances diagnosis accuracy, and QR codes enable real-time and secure access to patients' data. This system was implemented successfully at a local hospital, showcasing the power of digital transformation in the healthcare scene and perhaps initiating more innovations in educational facilities in times to come.

Keyword:

AI-assisted diagnosis, QR code integration, clinic management system, healthcare technology, Agile Scrum, patient data efficiency.

Appendix O: Adviser Acceptance

We, the undersigned researchers, confirm that the capstone presented herein does not incorporate, without proper acknowledgment, any material previously submitted for a diploma in any university, to the best of our knowledge and belief. We assert that it does not contain any material previously published or written by another person or ourselves unless due reference is made in the text. Furthermore, we hereby grant permission for the project study, if accepted, to be made available for photocopying and interlibrary loan, and for the title and summary to be accessible to external organizations.

Signature of Members:

Date: March 2025


Angelo B. Abargos


Joseph Oliver P. Banac


Lister John L. Evangelista

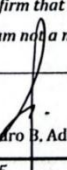
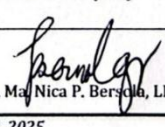

Ariel A. Forcado


Andiella Mhae B. Gabad

Counter Signed By:


MR. RONALD G. ROLDAN JR
Research Adviser

Appendix P: Technical Proofreader / Grammarian Acceptance

 Bestlink College of the Philippines College of Computer Studies #1071 Brgy, Kaligayahan, Quirino Highway, Novaliches, Quezon City 		
Name of Group Members/Individual:		
1	ABARGOS, ANGELO B.	
2	BANAC, JOSEPH OLIVER P.	
3	EVANGELISTA, LISTER JOHN L.	
4	FORCADO, ARIEL A.	
5	GABAD, ANDIELLA MHA E. B.	
6		
Title of the Project: EIS: SMS I: AI-ASSISTED DIAGNOSIS AND QR CODE INTEGRATION FOR STREAMLINED CLINIC MANAGEMENT SYSTEM AT BESTLINK COLLEGE OF THE PHILIPPINES		
	Technical Proofreader	Grammarian
Name	Mr. Alejandro B. Adovas	Ms. Ma. Nica P. Bersola
Designation	Professor, College of Computer Studies	Professor, College of Computer Studies Secretary to the Vice President for Academic Affairs
Workplace Address	Bestlink College of the Philippines Millionaire's Village, Novaliches, Quezon City	Bestlink College of the Philippines Millionaire's Village, Novaliches, Quezon City
Academic/ Professional Qualifications/ Membership	- Master of Science in Information Technology (ongoing) - Bachelor of Science in Computer Science - TESDA Certified Level 1 Trainer (TM1) - Instructor I, Bestlink College of the Philippines (2017 – present)	- Bachelor of Secondary Education Major in English - Licensed Professional Teacher
Email Address	andy.adovas@gmail.com	nicbersola20@gmail.com
Telephone	09054622478	09612383153
Signature	<i>I hereby confirm that I have undertaken to supervise the project mentioned above and I do certify that I am not a member of the Project Examination Board (PEC)</i>	
	 Mr. Alejandro B. Adovas Date: May 09, 2025	 Ms. Ma. Nica P. Bersola, LPT Date: May 02, 2025

Appendix Q: Certificate of Similarity



BESTLINK COLLEGE OF THE PHILIPPINES

762 cor. Topaz St. and Sapphire St., Millionaires Village, Quezon City



BCP LIBRARY DEPARTMENT

CERTIFICATE OF SIMILARITY

This certifies that the research titled, **SMS 1: AI-ASSISTED DIAGNOSIS AND QR CODE INTEGRATION FOR STREAMLINED CLINIC MANAGEMENT SYSTEM AT BESTLINK COLLEGE OF THE PHILIPPINES** has been checked and passed with a Turnitin similarity index of **6%**. Given this **20th of May, 2025**.

JOSHUA LOVIC D. ABAYA, RL
Librarian

JOHN PROS B. VALENCIA, RL, MLIS
Library Director

TC2420622

Appendix R: Researchers Curriculum Vitae



ANGELO B. ABARGOS

Anne Deca Homes, Marilao, Bulacan

PERSONAL DATA

Date of Birth : January 05, 2001

Place of Birth : City of Manila

Civil Status : Single

Citizenship : Filipino

Religion : Catholic

EDUCATIONAL BACKGROUND

Tertiary : (2019 – 2025) Bestlink College of the Philippines

Secondary : (2015 – 2019) Kalayaan National High School

Primary : (2009 – 2015) Kalayaan Elementary School

SEMINARS / TRAININGS ATTENDED

Cultivating A Multidisciplinary Research Culture with Artificial Intelligence: Integration, Innovation, and Impact

(September 8, 2024)

TECHSCAPE: Discover innovation in the cool climes of BSIT/CPE (December 4-6, 2023)

BITZ 2023 SESSION 2: Understanding the Innovators 'Role in Digital Transformation (April 27 – 29 2023)

Basic Knowledge with Research Trends from Multidisciplinary Perspective in Engineering Information and Communication Technology (September 4, 2023)



JOSEPH OLIVER P. BANAC

Sauyo, Novaliches, Quezon City

PERSONAL DATA

Date of Birth : December 26, 2002

Place of Birth : Quezon City

Civil Status : Single

Citizenship : Filipino

Religion : Christian

EDUCATIONAL BACKGROUND

Tertiary : (2019 – 2025) Bestlink College of the Philippines

Secondary : (2015 – 2019) Sauyo High School

Primary : (2009 – 2015) Sauyo Elementary School

SEMINARS / TRAININGS ATTENDED

Certificate of Completion: Office of the Social Media (March – April 2025)

Cultivating A Multidisciplinary Research Culture with Artificial Intelligence: Integration, Innovation, and Impact (September 2024)

Computer System Servicing NCII by TESDA (February – April 2024)

Career Essentials in Cybersecurity by Microsoft and LinkedIn (November 2023)

Skills to Succeed Academy by Accenture (Nov 2023)

TECHSCAPE: Discover innovation in the Cool Climes of BSIT/CPE (December 2023)

Basic Knowledge with Research Trends from Multidisciplinary Perspective in Engineering Information and Communication Technology (September 2023)

BITZ 2023 SESSION 2: Understanding the Innovators 'Role in Digital Transformation (April 2023)



LISTER JOHN L. EVANGELISTA

Phase 4, Payatas B., Quezon City

PERSONAL DATA

Date of Birth : January 11, 2003

Place of Birth : Masbate City

Civil Status : Single

Citizenship : Filipino

Religion : Catholic

EDUCATIONAL BACKGROUND

Tertiary : (2021 – 2025) Bestlink College of the Philippines

Secondary : (2021 – 2019) Datamex College of Saint Adeline

(2015 – 2019) Justice Cecilia Munoz Palma High
School

Primary : (2009 – 2015) Lupang Pangako Elementary School

SEMINARS / TRAININGS ATTENDED

Cultivating A Multidisciplinary Research Culture with Artificial Intelligence: Integration, Innovation, and Impact

(September 2024)

TECHSCAPE: Discover innovation in the cool climes of BSIT/CPE (December 2023)

BITZ 2023 SESSION 2: Understanding the Innovators 'Role in Digital Transformation (April 2023)

Basic Knowledge with Research Trends from Multidisciplinary Perspective in Engineering Information and Communication Technology (September 2023)

Enhancing the Skills of Individuals to Develop Futuristic IT Ideas (October 2019)

Unleashing Your Inner Potential (July 2019)



ARIEL A. FORCADO

Phase 8, North Fairview, Quezon City

PERSONAL DATA

Date of Birth : April 26, 2000

Place of Birth : Masbate City

Civil Status : Single

Citizenship : Filipino

Religion : Catholic

EDUCATIONAL BACKGROUND

Tertiary : (2021 – 2025) Bestlink College of the Philippines

Secondary : (2017 – 2019) Our Lady of Fatima University

(2013 – 2017) North Fairview High School

Primary : (2007 – 2013) North Fairview Elementary School

SEMINARS / TRAININGS ATTENDED

Cultivating A Multidisciplinary Research Culture with Artificial Intelligence: Integration, Innovation, and Impact

(September 2024)

TECHSCAPE: Discover innovation in the cool climes of

BSIT/CPE (December 2023)

BITZ 2023 SESSION 2: Understanding the Innovators 'Role in Digital Transformation (April 2023)

Basic Knowledge with Research Trends from Multidisciplinary Perspective in Engineering Information and Communication Technology (September 2023)



ANDIELLA MHAЕ B. GABAD

Sorsogon St., Payatas B., Quezon City

PERSONAL DATA

Date of Birth : May 29, 2003

Place of Birth : Quezon City

Civil Status : Single

Citizenship : Filipino

Religion : Born Again Christian

EDUCATIONAL BACKGROUND

Tertiary : (2021 – 2025) Bestlink College of the Philippines

Secondary : (2019 – 2021) Commonwealth High School

(2015 – 2019) Justice Cecilia Munoz Palma High
School

Primary : (2009 – 2015) Lupang Pangako Elementary School

SEMINARS / TRAININGS ATTENDED

Cultivating A Multidisciplinary Research Culture with Artificial Intelligence: Integration, Innovation, and Impact

(September 2024)

TECHSCAPE: Discover innovation in the cool climes of BSIT/CPE (December 2023)

BITZ 2023 SESSION 2: Understanding the Innovators 'Role in Digital Transformation (April 2023)

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